

2025-10-05 00:08:12,839 - root - INFO - Dataset loaded with 42879 samples  
2025-10-05 00:08:12,854 - root - INFO - Using device: mps  
2025-10-05 00:08:12,874 - root - INFO -  
2025-10-05 00:08:12,874 - root - INFO - N  
2025-10-05 00:08:12,874 - root - INFO - E  
2025-10-05 00:08:12,874 - root - INFO - W  
2025-10-05 00:08:12,874 - root - INFO - STARTING TRAINING FLOW :  
CSILSTMNet:('csi\_input\_size', 'meta\_input\_size', 'window\_size', 'num\_classes',  
'lr', 'epochs'):([99], [12], [128], [31], [0.001, 0.0005], [100])  
2025-10-05 00:08:13,123 - root - INFO - START RUN MLFLOW :  
CSILSTMNet:{'csi\_input\_size': 99, 'meta\_input\_size': 12, 'window\_size': 128,  
'num\_classes': 31, 'lr': 0.001, 'epochs': 100, 'model\_name': 'CSILSTMNet'}  
2025-10-05 00:08:13,136 - root - INFO - START T-N-E CSILSTMNet  
2025-10-05 00:08:13,624 - root - INFO - CSILSTMNet USING LEARNING RATE 0.001  
2025-10-05 00:08:48,097 - root - INFO - Epoch 1/100, Model CSILSTMNet, Train  
Loss: 1.1840, Train Acc: 0.6213, Val Loss: 0.8900, Val Acc: 0.6934, Epoch  
Duration: 34.47 sec  
2025-10-05 00:09:23,433 - root - INFO - Epoch 2/100, Model CSILSTMNet, Train  
Loss: 0.8803, Train Acc: 0.6990, Val Loss: 1.1249, Val Acc: 0.6051, Epoch  
Duration: 33.22 sec  
2025-10-05 00:09:57,233 - root - INFO - Epoch 3/100, Model CSILSTMNet, Train  
Loss: 0.8209, Train Acc: 0.7191, Val Loss: 0.7436, Val Acc: 0.7481, Epoch  
Duration: 33.80 sec  
2025-10-05 00:10:32,755 - root - INFO - Epoch 4/100, Model CSILSTMNet, Train  
Loss: 0.7616, Train Acc: 0.7383, Val Loss: 0.7237, Val Acc: 0.7411, Epoch  
Duration: 33.45 sec  
2025-10-05 00:11:05,921 - root - INFO - Epoch 5/100, Model CSILSTMNet, Train  
Loss: 0.6752, Train Acc: 0.7635, Val Loss: 0.7224, Val Acc: 0.7587, Epoch  
Duration: 33.16 sec  
2025-10-05 00:11:41,379 - root - INFO - Epoch 6/100, Model CSILSTMNet, Train  
Loss: 0.6157, Train Acc: 0.7873, Val Loss: 0.5273, Val Acc: 0.8175, Epoch  
Duration: 33.51 sec  
2025-10-05 00:12:16,513 - root - INFO - Epoch 7/100, Model CSILSTMNet, Train  
Loss: 0.5334, Train Acc: 0.8153, Val Loss: 0.5271, Val Acc: 0.8123, Epoch  
Duration: 33.16 sec  
2025-10-05 00:12:49,724 - root - INFO - Epoch 8/100, Model CSILSTMNet, Train  
Loss: 0.5199, Train Acc: 0.8212, Val Loss: 0.4688, Val Acc: 0.8418, Epoch  
Duration: 33.21 sec  
2025-10-05 00:13:24,825 - root - INFO - Epoch 9/100, Model CSILSTMNet, Train  
Loss: 0.5181, Train Acc: 0.8217, Val Loss: 0.5432, Val Acc: 0.8156, Epoch  
Duration: 33.16 sec  
2025-10-05 00:13:57,695 - root - INFO - Epoch 10/100, Model CSILSTMNet, Train  
Loss: 0.4404, Train Acc: 0.8482, Val Loss: 0.4183, Val Acc: 0.8587, Epoch  
Duration: 32.87 sec  
2025-10-05 00:14:32,782 - root - INFO - Epoch 11/100, Model CSILSTMNet, Train  
Loss: 0.3801, Train Acc: 0.8689, Val Loss: 0.3961, Val Acc: 0.8633, Epoch  
Duration: 33.23 sec  
2025-10-05 00:15:07,626 - root - INFO - Epoch 12/100, Model CSILSTMNet, Train  
Loss: 0.3646, Train Acc: 0.8729, Val Loss: 0.3648, Val Acc: 0.8723, Epoch  
Duration: 32.95 sec  
2025-10-05 00:15:43,332 - root - INFO - Epoch 13/100, Model CSILSTMNet, Train  
Loss: 0.3074, Train Acc: 0.8924, Val Loss: 0.3204, Val Acc: 0.8872, Epoch  
Duration: 33.77 sec

2025-10-05 00:16:18,408 - root - INFO - Epoch 14/100, Model CSILSTMNet, Train Loss: 0.2690, Train Acc: 0.9061, Val Loss: 0.2779, Val Acc: 0.9066, Epoch Duration: 33.12 sec

2025-10-05 00:16:53,205 - root - INFO - Epoch 15/100, Model CSILSTMNet, Train Loss: 0.2488, Train Acc: 0.9149, Val Loss: 0.2731, Val Acc: 0.9100, Epoch Duration: 33.12 sec

2025-10-05 00:17:28,540 - root - INFO - Epoch 16/100, Model CSILSTMNet, Train Loss: 0.2081, Train Acc: 0.9282, Val Loss: 0.2070, Val Acc: 0.9281, Epoch Duration: 33.51 sec

2025-10-05 00:18:03,636 - root - INFO - Epoch 17/100, Model CSILSTMNet, Train Loss: 0.1896, Train Acc: 0.9350, Val Loss: 0.1946, Val Acc: 0.9318, Epoch Duration: 33.16 sec

2025-10-05 00:18:38,702 - root - INFO - Epoch 18/100, Model CSILSTMNet, Train Loss: 0.1824, Train Acc: 0.9371, Val Loss: 0.2017, Val Acc: 0.9295, Epoch Duration: 33.02 sec

2025-10-05 00:19:11,593 - root - INFO - Epoch 19/100, Model CSILSTMNet, Train Loss: 0.1527, Train Acc: 0.9470, Val Loss: 0.1834, Val Acc: 0.9362, Epoch Duration: 32.89 sec

2025-10-05 00:19:46,814 - root - INFO - Epoch 20/100, Model CSILSTMNet, Train Loss: 0.1346, Train Acc: 0.9539, Val Loss: 0.1642, Val Acc: 0.9466, Epoch Duration: 33.24 sec

2025-10-05 00:20:21,799 - root - INFO - Epoch 21/100, Model CSILSTMNet, Train Loss: 0.1204, Train Acc: 0.9584, Val Loss: 0.1502, Val Acc: 0.9508, Epoch Duration: 32.95 sec

2025-10-05 00:20:56,708 - root - INFO - Epoch 22/100, Model CSILSTMNet, Train Loss: 0.1033, Train Acc: 0.9640, Val Loss: 0.1478, Val Acc: 0.9489, Epoch Duration: 32.87 sec

2025-10-05 00:21:29,405 - root - INFO - Epoch 23/100, Model CSILSTMNet, Train Loss: 0.1008, Train Acc: 0.9648, Val Loss: 0.1364, Val Acc: 0.9534, Epoch Duration: 32.70 sec

2025-10-05 00:22:04,334 - root - INFO - Epoch 24/100, Model CSILSTMNet, Train Loss: 0.0906, Train Acc: 0.9691, Val Loss: 0.1325, Val Acc: 0.9530, Epoch Duration: 32.90 sec

2025-10-05 00:22:37,070 - root - INFO - Epoch 25/100, Model CSILSTMNet, Train Loss: 0.0832, Train Acc: 0.9710, Val Loss: 0.1291, Val Acc: 0.9577, Epoch Duration: 32.73 sec

2025-10-05 00:23:11,759 - root - INFO - Epoch 26/100, Model CSILSTMNet, Train Loss: 0.0767, Train Acc: 0.9734, Val Loss: 0.1316, Val Acc: 0.9564, Epoch Duration: 32.72 sec

2025-10-05 00:23:44,711 - root - INFO - Epoch 27/100, Model CSILSTMNet, Train Loss: 0.0698, Train Acc: 0.9758, Val Loss: 0.1124, Val Acc: 0.9621, Epoch Duration: 32.95 sec

2025-10-05 00:24:19,670 - root - INFO - Epoch 28/100, Model CSILSTMNet, Train Loss: 0.0600, Train Acc: 0.9798, Val Loss: 0.1094, Val Acc: 0.9628, Epoch Duration: 32.98 sec

2025-10-05 00:24:54,386 - root - INFO - Epoch 29/100, Model CSILSTMNet, Train Loss: 0.0674, Train Acc: 0.9779, Val Loss: 0.1022, Val Acc: 0.9674, Epoch Duration: 32.77 sec

2025-10-05 00:25:29,259 - root - INFO - Epoch 30/100, Model CSILSTMNet, Train Loss: 0.0509, Train Acc: 0.9833, Val Loss: 0.0948, Val Acc: 0.9685, Epoch Duration: 32.84 sec

2025-10-05 00:26:04,126 - root - INFO - Epoch 31/100, Model CSILSTMNet, Train Loss: 0.0515, Train Acc: 0.9819, Val Loss: 0.0901, Val Acc: 0.9704, Epoch Duration: 32.87 sec

2025-10-05 00:26:40,369 - root - INFO - Epoch 32/100, Model CSILSTMNet, Train Loss: 0.0487, Train Acc: 0.9832, Val Loss: 0.1047, Val Acc: 0.9661, Epoch Duration: 34.13 sec

2025-10-05 00:27:14,219 - root - INFO - Epoch 33/100, Model CSILSTMNet, Train Loss: 0.0430, Train Acc: 0.9852, Val Loss: 0.0997, Val Acc: 0.9693, Epoch Duration: 33.85 sec

2025-10-05 00:27:47,855 - root - INFO - Epoch 34/100, Model CSILSTMNet, Train Loss: 0.0420, Train Acc: 0.9855, Val Loss: 0.1019, Val Acc: 0.9675, Epoch Duration: 33.63 sec

2025-10-05 00:28:21,627 - root - INFO - Epoch 35/100, Model CSILSTMNet, Train Loss: 0.0393, Train Acc: 0.9870, Val Loss: 0.1129, Val Acc: 0.9640, Epoch Duration: 33.77 sec

2025-10-05 00:28:55,414 - root - INFO - Epoch 36/100, Model CSILSTMNet, Train Loss: 0.0473, Train Acc: 0.9845, Val Loss: 0.1063, Val Acc: 0.9675, Epoch Duration: 33.79 sec

2025-10-05 00:29:29,013 - root - INFO - Epoch 37/100, Model CSILSTMNet, Train Loss: 0.0366, Train Acc: 0.9879, Val Loss: 0.0829, Val Acc: 0.9738, Epoch Duration: 33.60 sec

2025-10-05 00:30:04,470 - root - INFO - Epoch 38/100, Model CSILSTMNet, Train Loss: 0.0303, Train Acc: 0.9901, Val Loss: 0.0841, Val Acc: 0.9754, Epoch Duration: 33.50 sec

2025-10-05 00:30:40,383 - root - INFO - Epoch 39/100, Model CSILSTMNet, Train Loss: 0.0333, Train Acc: 0.9889, Val Loss: 0.1103, Val Acc: 0.9674, Epoch Duration: 33.94 sec

2025-10-05 00:31:13,918 - root - INFO - Epoch 40/100, Model CSILSTMNet, Train Loss: 0.0334, Train Acc: 0.9888, Val Loss: 0.0998, Val Acc: 0.9712, Epoch Duration: 33.53 sec

2025-10-05 00:31:47,886 - root - INFO - Epoch 41/100, Model CSILSTMNet, Train Loss: 0.0328, Train Acc: 0.9897, Val Loss: 0.1023, Val Acc: 0.9675, Epoch Duration: 33.97 sec

2025-10-05 00:32:21,745 - root - INFO - Epoch 42/100, Model CSILSTMNet, Train Loss: 0.0359, Train Acc: 0.9880, Val Loss: 0.0888, Val Acc: 0.9731, Epoch Duration: 33.86 sec

2025-10-05 00:32:55,387 - root - INFO - Epoch 43/100, Model CSILSTMNet, Train Loss: 0.0272, Train Acc: 0.9910, Val Loss: 0.0795, Val Acc: 0.9755, Epoch Duration: 33.64 sec

2025-10-05 00:33:31,516 - root - INFO - Epoch 44/100, Model CSILSTMNet, Train Loss: 0.0269, Train Acc: 0.9912, Val Loss: 0.0894, Val Acc: 0.9721, Epoch Duration: 34.11 sec

2025-10-05 00:34:05,418 - root - INFO - Epoch 45/100, Model CSILSTMNet, Train Loss: 0.0320, Train Acc: 0.9899, Val Loss: 0.0935, Val Acc: 0.9718, Epoch Duration: 33.90 sec

2025-10-05 00:34:38,967 - root - INFO - Epoch 46/100, Model CSILSTMNet, Train Loss: 0.0272, Train Acc: 0.9914, Val Loss: 0.0912, Val Acc: 0.9719, Epoch Duration: 33.55 sec

2025-10-05 00:35:12,702 - root - INFO - Epoch 47/100, Model CSILSTMNet, Train Loss: 0.0279, Train Acc: 0.9910, Val Loss: 0.0847, Val Acc: 0.9735, Epoch Duration: 33.73 sec

2025-10-05 00:35:46,688 - root - INFO - Epoch 48/100, Model CSILSTMNet, Train Loss: 0.0278, Train Acc: 0.9908, Val Loss: 0.0927, Val Acc: 0.9733, Epoch Duration: 33.98 sec

2025-10-05 00:36:20,196 - root - INFO - Epoch 49/100, Model CSILSTMNet, Train Loss: 0.0233, Train Acc: 0.9924, Val Loss: 0.0779, Val Acc: 0.9785, Epoch Duration: 33.51 sec

2025-10-05 00:36:56,111 - root - INFO - Epoch 50/100, Model CSILSTMNet, Train Loss: 0.0268, Train Acc: 0.9916, Val Loss: 0.0920, Val Acc: 0.9734, Epoch Duration: 33.83 sec

2025-10-05 00:37:29,580 - root - INFO - Epoch 51/100, Model CSILSTMNet, Train Loss: 0.0232, Train Acc: 0.9922, Val Loss: 0.0781, Val Acc: 0.9776, Epoch Duration: 33.47 sec

2025-10-05 00:38:03,022 - root - INFO - Epoch 52/100, Model CSILSTMNet, Train Loss: 0.0255, Train Acc: 0.9918, Val Loss: 0.0805, Val Acc: 0.9757, Epoch Duration: 33.44 sec

2025-10-05 00:38:36,540 - root - INFO - Epoch 53/100, Model CSILSTMNet, Train Loss: 0.0200, Train Acc: 0.9935, Val Loss: 0.0804, Val Acc: 0.9778, Epoch Duration: 33.52 sec

2025-10-05 00:39:10,007 - root - INFO - Epoch 54/100, Model CSILSTMNet, Train Loss: 0.0207, Train Acc: 0.9938, Val Loss: 0.0874, Val Acc: 0.9754, Epoch Duration: 33.47 sec

2025-10-05 00:39:43,718 - root - INFO - Epoch 55/100, Model CSILSTMNet, Train Loss: 0.0169, Train Acc: 0.9945, Val Loss: 0.0692, Val Acc: 0.9804, Epoch Duration: 33.71 sec

2025-10-05 00:40:19,753 - root - INFO - Epoch 56/100, Model CSILSTMNet, Train Loss: 0.0256, Train Acc: 0.9917, Val Loss: 0.0839, Val Acc: 0.9753, Epoch Duration: 33.93 sec

2025-10-05 00:40:53,393 - root - INFO - Epoch 57/100, Model CSILSTMNet, Train Loss: 0.0185, Train Acc: 0.9942, Val Loss: 0.0839, Val Acc: 0.9766, Epoch Duration: 33.64 sec

2025-10-05 00:41:26,990 - root - INFO - Epoch 58/100, Model CSILSTMNet, Train Loss: 0.0209, Train Acc: 0.9933, Val Loss: 0.0751, Val Acc: 0.9791, Epoch Duration: 33.60 sec

2025-10-05 00:42:00,473 - root - INFO - Epoch 59/100, Model CSILSTMNet, Train Loss: 0.0170, Train Acc: 0.9947, Val Loss: 0.0778, Val Acc: 0.9790, Epoch Duration: 33.48 sec

2025-10-05 00:42:34,735 - root - INFO - Epoch 60/100, Model CSILSTMNet, Train Loss: 0.0160, Train Acc: 0.9945, Val Loss: 0.0796, Val Acc: 0.9787, Epoch Duration: 34.26 sec

2025-10-05 00:43:08,218 - root - INFO - Epoch 61/100, Model CSILSTMNet, Train Loss: 0.0181, Train Acc: 0.9945, Val Loss: 0.0816, Val Acc: 0.9787, Epoch Duration: 33.48 sec

2025-10-05 00:43:41,649 - root - INFO - Epoch 62/100, Model CSILSTMNet, Train Loss: 0.0191, Train Acc: 0.9940, Val Loss: 0.0975, Val Acc: 0.9748, Epoch Duration: 33.43 sec

2025-10-05 00:44:14,969 - root - INFO - Epoch 63/100, Model CSILSTMNet, Train Loss: 0.0145, Train Acc: 0.9955, Val Loss: 0.0804, Val Acc: 0.9802, Epoch Duration: 33.32 sec

2025-10-05 00:44:48,474 - root - INFO - Epoch 64/100, Model CSILSTMNet, Train Loss: 0.0173, Train Acc: 0.9948, Val Loss: 0.0708, Val Acc: 0.9806, Epoch Duration: 33.50 sec

2025-10-05 00:45:24,158 - root - INFO - Epoch 65/100, Model CSILSTMNet, Train Loss: 0.0149, Train Acc: 0.9951, Val Loss: 0.0740, Val Acc: 0.9808, Epoch Duration: 33.56 sec

2025-10-05 00:45:59,549 - root - INFO - Epoch 66/100, Model CSILSTMNet, Train Loss: 0.0154, Train Acc: 0.9952, Val Loss: 0.0806, Val Acc: 0.9777, Epoch Duration: 33.45 sec

2025-10-05 00:46:33,018 - root - INFO - Epoch 67/100, Model CSILSTMNet, Train Loss: 0.0177, Train Acc: 0.9944, Val Loss: 0.0852, Val Acc: 0.9773, Epoch Duration: 33.47 sec

2025-10-05 00:47:06,354 - root - INFO - Epoch 68/100, Model CSILSTMNet, Train Loss: 0.0116, Train Acc: 0.9962, Val Loss: 0.0794, Val Acc: 0.9788, Epoch Duration: 33.33 sec

2025-10-05 00:47:39,863 - root - INFO - Epoch 69/100, Model CSILSTMNet, Train Loss: 0.0202, Train Acc: 0.9934, Val Loss: 0.0972, Val Acc: 0.9735, Epoch Duration: 33.51 sec

2025-10-05 00:48:13,792 - root - INFO - Epoch 70/100, Model CSILSTMNet, Train Loss: 0.0148, Train Acc: 0.9959, Val Loss: 0.0940, Val Acc: 0.9736, Epoch Duration: 33.93 sec

2025-10-05 00:48:47,354 - root - INFO - Epoch 71/100, Model CSILSTMNet, Train Loss: 0.0148, Train Acc: 0.9955, Val Loss: 0.0732, Val Acc: 0.9796, Epoch Duration: 33.56 sec

2025-10-05 00:49:20,669 - root - INFO - Epoch 72/100, Model CSILSTMNet, Train Loss: 0.0138, Train Acc: 0.9957, Val Loss: 0.0778, Val Acc: 0.9803, Epoch Duration: 33.31 sec

2025-10-05 00:49:54,115 - root - INFO - Epoch 73/100, Model CSILSTMNet, Train Loss: 0.0152, Train Acc: 0.9953, Val Loss: 0.0895, Val Acc: 0.9757, Epoch Duration: 33.44 sec

2025-10-05 00:50:27,522 - root - INFO - Epoch 74/100, Model CSILSTMNet, Train Loss: 0.0122, Train Acc: 0.9958, Val Loss: 0.0806, Val Acc: 0.9789, Epoch Duration: 33.41 sec

2025-10-05 00:51:00,858 - root - INFO - Epoch 75/100, Model CSILSTMNet, Train Loss: 0.0116, Train Acc: 0.9966, Val Loss: 0.0820, Val Acc: 0.9791, Epoch Duration: 33.33 sec

2025-10-05 00:51:34,348 - root - INFO - Epoch 76/100, Model CSILSTMNet, Train Loss: 0.0147, Train Acc: 0.9956, Val Loss: 0.0725, Val Acc: 0.9834, Epoch Duration: 33.49 sec

2025-10-05 00:52:09,842 - root - INFO - Epoch 77/100, Model CSILSTMNet, Train Loss: 0.0151, Train Acc: 0.9957, Val Loss: 0.0781, Val Acc: 0.9816, Epoch Duration: 33.57 sec

2025-10-05 00:52:43,331 - root - INFO - Epoch 78/100, Model CSILSTMNet, Train Loss: 0.0096, Train Acc: 0.9971, Val Loss: 0.0699, Val Acc: 0.9811, Epoch Duration: 33.49 sec

2025-10-05 00:53:16,788 - root - INFO - Epoch 79/100, Model CSILSTMNet, Train Loss: 0.0137, Train Acc: 0.9958, Val Loss: 0.1233, Val Acc: 0.9706, Epoch Duration: 33.45 sec

2025-10-05 00:53:50,347 - root - INFO - Epoch 80/100, Model CSILSTMNet, Train Loss: 0.0148, Train Acc: 0.9955, Val Loss: 0.0758, Val Acc: 0.9809, Epoch Duration: 33.56 sec

2025-10-05 00:54:23,800 - root - INFO - Epoch 81/100, Model CSILSTMNet, Train Loss: 0.0127, Train Acc: 0.9962, Val Loss: 0.1201, Val Acc: 0.9711, Epoch Duration: 33.45 sec

2025-10-05 00:54:57,251 - root - INFO - Epoch 82/100, Model CSILSTMNet, Train Loss: 0.0145, Train Acc: 0.9952, Val Loss: 0.0856, Val Acc: 0.9799, Epoch Duration: 33.45 sec

2025-10-05 00:55:31,162 - root - INFO - Epoch 83/100, Model CSILSTMNet, Train Loss: 0.0151, Train Acc: 0.9954, Val Loss: 0.0823, Val Acc: 0.9795, Epoch Duration: 33.91 sec

2025-10-05 00:56:04,731 - root - INFO - Epoch 84/100, Model CSILSTMNet, Train Loss: 0.0103, Train Acc: 0.9969, Val Loss: 0.0878, Val Acc: 0.9771, Epoch Duration: 33.57 sec

2025-10-05 00:56:38,313 - root - INFO - Epoch 85/100, Model CSILSTMNet, Train Loss: 0.0095, Train Acc: 0.9971, Val Loss: 0.0700, Val Acc: 0.9805, Epoch Duration: 33.58 sec

2025-10-05 00:57:11,854 - root - INFO - Epoch 86/100, Model CSILSTMNet, Train Loss: 0.0094, Train Acc: 0.9973, Val Loss: 0.0704, Val Acc: 0.9811, Epoch Duration: 33.54 sec

2025-10-05 00:57:45,476 - root - INFO - Epoch 87/100, Model CSILSTMNet, Train Loss: 0.0140, Train Acc: 0.9957, Val Loss: 0.1025, Val Acc: 0.9753, Epoch Duration: 33.62 sec

2025-10-05 00:58:18,961 - root - INFO - Epoch 88/100, Model CSILSTMNet, Train Loss: 0.0138, Train Acc: 0.9956, Val Loss: 0.0827, Val Acc: 0.9789, Epoch Duration: 33.48 sec

2025-10-05 00:58:52,880 - root - INFO - Epoch 89/100, Model CSILSTMNet, Train Loss: 0.0107, Train Acc: 0.9965, Val Loss: 0.1186, Val Acc: 0.9722, Epoch Duration: 33.92 sec

2025-10-05 00:59:26,550 - root - INFO - Epoch 90/100, Model CSILSTMNet, Train Loss: 0.0102, Train Acc: 0.9969, Val Loss: 0.0862, Val Acc: 0.9782, Epoch Duration: 33.67 sec

2025-10-05 01:00:00,105 - root - INFO - Epoch 91/100, Model CSILSTMNet, Train Loss: 0.0117, Train Acc: 0.9965, Val Loss: 0.0753, Val Acc: 0.9791, Epoch Duration: 33.55 sec

2025-10-05 01:00:33,759 - root - INFO - Epoch 92/100, Model CSILSTMNet, Train Loss: 0.0118, Train Acc: 0.9961, Val Loss: 0.0794, Val Acc: 0.9790, Epoch Duration: 33.65 sec

2025-10-05 01:01:07,437 - root - INFO - Epoch 93/100, Model CSILSTMNet, Train Loss: 0.0112, Train Acc: 0.9966, Val Loss: 0.0845, Val Acc: 0.9804, Epoch Duration: 33.68 sec

2025-10-05 01:01:40,903 - root - INFO - Epoch 94/100, Model CSILSTMNet, Train Loss: 0.0060, Train Acc: 0.9981, Val Loss: 0.0844, Val Acc: 0.9785, Epoch Duration: 33.46 sec

2025-10-05 01:02:14,415 - root - INFO - Epoch 95/100, Model CSILSTMNet, Train Loss: 0.0156, Train Acc: 0.9954, Val Loss: 0.0905, Val Acc: 0.9777, Epoch Duration: 33.51 sec

2025-10-05 01:02:47,950 - root - INFO - Epoch 96/100, Model CSILSTMNet, Train Loss: 0.0099, Train Acc: 0.9966, Val Loss: 0.0894, Val Acc: 0.9804, Epoch Duration: 33.53 sec

2025-10-05 01:03:21,373 - root - INFO - Epoch 97/100, Model CSILSTMNet, Train Loss: 0.0088, Train Acc: 0.9971, Val Loss: 0.0860, Val Acc: 0.9809, Epoch Duration: 33.42 sec

2025-10-05 01:03:55,031 - root - INFO - Epoch 98/100, Model CSILSTMNet, Train Loss: 0.0085, Train Acc: 0.9974, Val Loss: 0.0763, Val Acc: 0.9806, Epoch Duration: 33.66 sec

2025-10-05 01:04:28,475 - root - INFO - Epoch 99/100, Model CSILSTMNet, Train Loss: 0.0108, Train Acc: 0.9965, Val Loss: 0.0785, Val Acc: 0.9810, Epoch Duration: 33.44 sec

2025-10-05 01:05:01,909 - root - INFO - Epoch 100/100, Model CSILSTMNet, Train Loss: 0.0107, Train Acc: 0.9968, Val Loss: 0.0736, Val Acc: 0.9817, Epoch Duration: 33.43 sec

2025-10-05 01:05:01,909 - root - CRITICAL - Training complete. Best Val Acc: 0.9834 at epoch 76.

2025-10-05 01:05:01,913 - root - CRITICAL - SAVED BEST MODEL /Users/sanjeev/VNIT/FINAL\_PRJ\_PHASE2/experiments/20251005\_000538/checkpoints/best\_model\_CSILSTMNet\_epoch76.pt USING LEARNING RATE 0.001

2025-10-05 01:05:01,913 - root - CRITICAL - BEST VALIDATION ACCURACY: 0.9834 at epoch 76.

2025-10-05 01:05:02,222 - root - INFO - Saved stats plot to /Users/sanjeev/VNIT/FINAL\_PRJ\_PHASE2/experiments/20251005\_000538/plots/CSILSTMNe

t\_{'csi\_input\_size': 99, 'meta\_input\_size': 12, 'window\_size': 128,  
'num\_classes': 31, 'lr': 0.001, 'epochs': 100, 'model\_name':  
'CSILSTMNet'}\_stats.png

2025-10-05 01:05:05,432 - root - INFO - Confusion matrix:

```
[[277  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
  0  0  0  0  0  0  0  0  0  0  0  0]
 [ 0 270  0  0  7  5  0  1  0  0  0  0  0  0  0  0  0
  0  0  0  0  0  0  0  0  0  0  0  0]
 [ 3  0 285  0  3  1  0  0  0  0  0  0  0  0  0  0  0
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 [ 0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  1  0  2  0
  0  0  0  0  0  6 299 0  0  0  0]
```

```

[ 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0
 0 0 0 0 0 0 0 262 0 0 0 0]
[ 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
 0 0 1 0 0 0 0 0 286 0 0 0]
[ 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
 0 0 0 0 0 0 0 0 293 0 0]
[ 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0
 0 0 0 5 0 0 0 0 1 0 276 11]
[ 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
 0 0 4 2 0 0 0 0 0 0 5 254]]

```

2025-10-05 01:05:05,432 - root - INFO - Classification report:  
precision recall f1-score support

|    |      |      |      |     |
|----|------|------|------|-----|
| 1  | 0.98 | 1.00 | 0.99 | 277 |
| 2  | 0.97 | 0.95 | 0.96 | 283 |
| 3  | 0.99 | 0.98 | 0.98 | 292 |
| 4  | 1.00 | 0.99 | 0.99 | 288 |
| 5  | 0.96 | 0.97 | 0.97 | 272 |
| 6  | 0.97 | 0.99 | 0.98 | 295 |
| 7  | 1.00 | 1.00 | 1.00 | 290 |
| 8  | 1.00 | 1.00 | 1.00 | 267 |
| 9  | 0.99 | 0.99 | 0.99 | 267 |
| 10 | 1.00 | 0.98 | 0.99 | 272 |
| 11 | 0.99 | 1.00 | 0.99 | 270 |
| 12 | 1.00 | 1.00 | 1.00 | 272 |
| 13 | 0.97 | 0.95 | 0.96 | 332 |
| 14 | 1.00 | 1.00 | 1.00 | 301 |
| 15 | 0.99 | 1.00 | 0.99 | 253 |
| 16 | 0.99 | 1.00 | 0.99 | 307 |
| 17 | 0.95 | 0.96 | 0.95 | 290 |
| 18 | 0.96 | 0.95 | 0.96 | 306 |
| 19 | 0.96 | 0.96 | 0.96 | 282 |
| 20 | 1.00 | 0.99 | 0.99 | 285 |
| 21 | 0.98 | 0.99 | 0.99 | 289 |
| 22 | 0.97 | 0.97 | 0.97 | 316 |
| 23 | 1.00 | 0.99 | 0.99 | 288 |
| 24 | 0.97 | 0.99 | 0.98 | 272 |
| 25 | 0.99 | 0.97 | 0.98 | 308 |
| 26 | 0.98 | 1.00 | 0.99 | 263 |
| 27 | 0.99 | 1.00 | 0.99 | 287 |
| 28 | 1.00 | 1.00 | 1.00 | 293 |
| 29 | 0.97 | 0.94 | 0.96 | 294 |
| 30 | 0.93 | 0.96 | 0.94 | 265 |

|              |      |      |      |      |
|--------------|------|------|------|------|
| accuracy     |      |      | 0.98 | 8576 |
| macro avg    | 0.98 | 0.98 | 0.98 | 8576 |
| weighted avg | 0.98 | 0.98 | 0.98 | 8576 |

2025-10-05 01:05:05,435 - root - CRITICAL - 99. SAVED BEST MODEL IN CSILSTMNet.  
BEST VALIDATION ACCURACY: 0.9834 at epoch 76.

2025-10-05 01:05:05,436 - root - CRITICAL - 100. LOGGED BEST MODEL IN MLFLOW  
/Users/sanjeev/VNIT/FINAL\_PRJ\_PHASE2/experiments/20251005\_000538/checkpoints/best\_model\_CSILSTMNet\_epoch76.pt. BEST VALIDATION ACCURACY: 0.9834 at epoch 76.

2025-10-05 01:05:05,437 - root - INFO - START RUN MLFLOW :



```
CSILSTMNet: {'csi_input_size': 99, 'meta_input_size': 12, 'window_size': 128,
'num_classes': 31, 'lr': 0.0005, 'epochs': 100, 'model_name': 'CSILSTMNet'}
2025-10-05 01:05:05,444 - root - INFO - START T-N-E CSILSTMNet
2025-10-05 01:05:05,444 - root - INFO - CSILSTMNet USING LEARNING RATE 0.0005
2025-10-05 01:05:38,966 - root - INFO - Epoch 1/100, Model CSILSTMNet, Train
Loss: 1.1632, Train Acc: 0.6408, Val Loss: 0.9701, Val Acc: 0.6624, Epoch
Duration: 33.52 sec
2025-10-05 01:06:14,999 - root - INFO - Epoch 2/100, Model CSILSTMNet, Train
Loss: 0.8579, Train Acc: 0.7071, Val Loss: 0.8479, Val Acc: 0.7060, Epoch
Duration: 33.80 sec
2025-10-05 01:06:50,678 - root - INFO - Epoch 3/100, Model CSILSTMNet, Train
Loss: 0.7128, Train Acc: 0.7536, Val Loss: 0.6757, Val Acc: 0.7671, Epoch
Duration: 33.68 sec
2025-10-05 01:07:26,376 - root - INFO - Epoch 4/100, Model CSILSTMNet, Train
Loss: 0.6562, Train Acc: 0.7702, Val Loss: 0.6326, Val Acc: 0.7739, Epoch
Duration: 33.75 sec
2025-10-05 01:08:01,898 - root - INFO - Epoch 5/100, Model CSILSTMNet, Train
Loss: 0.5762, Train Acc: 0.7978, Val Loss: 0.6075, Val Acc: 0.7858, Epoch
Duration: 33.60 sec
2025-10-05 01:08:37,583 - root - INFO - Epoch 6/100, Model CSILSTMNet, Train
Loss: 0.5960, Train Acc: 0.7911, Val Loss: 0.5734, Val Acc: 0.8027, Epoch
Duration: 33.70 sec
2025-10-05 01:09:13,273 - root - INFO - Epoch 7/100, Model CSILSTMNet, Train
Loss: 0.5212, Train Acc: 0.8198, Val Loss: 0.5197, Val Acc: 0.8161, Epoch
Duration: 33.66 sec
2025-10-05 01:09:49,342 - root - INFO - Epoch 8/100, Model CSILSTMNet, Train
Loss: 0.4843, Train Acc: 0.8314, Val Loss: 0.4635, Val Acc: 0.8383, Epoch
Duration: 34.10 sec
2025-10-05 01:10:25,006 - root - INFO - Epoch 9/100, Model CSILSTMNet, Train
Loss: 0.4373, Train Acc: 0.8462, Val Loss: 0.4061, Val Acc: 0.8587, Epoch
Duration: 33.71 sec
2025-10-05 01:11:00,699 - root - INFO - Epoch 10/100, Model CSILSTMNet, Train
Loss: 0.4087, Train Acc: 0.8580, Val Loss: 0.3754, Val Acc: 0.8684, Epoch
Duration: 33.57 sec
2025-10-05 01:11:36,382 - root - INFO - Epoch 11/100, Model CSILSTMNet, Train
Loss: 0.3367, Train Acc: 0.8845, Val Loss: 0.4486, Val Acc: 0.8428, Epoch
Duration: 33.75 sec
2025-10-05 01:12:09,809 - root - INFO - Epoch 12/100, Model CSILSTMNet, Train
Loss: 0.3693, Train Acc: 0.8743, Val Loss: 0.6228, Val Acc: 0.8113, Epoch
Duration: 33.43 sec
2025-10-05 01:12:44,175 - root - INFO - Epoch 13/100, Model CSILSTMNet, Train
Loss: 0.3785, Train Acc: 0.8711, Val Loss: 0.3919, Val Acc: 0.8645, Epoch
Duration: 34.36 sec
2025-10-05 01:13:17,459 - root - INFO - Epoch 14/100, Model CSILSTMNet, Train
Loss: 0.3360, Train Acc: 0.8855, Val Loss: 0.5173, Val Acc: 0.8230, Epoch
Duration: 33.28 sec
2025-10-05 01:13:50,825 - root - INFO - Epoch 15/100, Model CSILSTMNet, Train
Loss: 0.3364, Train Acc: 0.8850, Val Loss: 0.3662, Val Acc: 0.8758, Epoch
Duration: 33.36 sec
2025-10-05 01:14:26,288 - root - INFO - Epoch 16/100, Model CSILSTMNet, Train
Loss: 0.3047, Train Acc: 0.8947, Val Loss: 0.3217, Val Acc: 0.8912, Epoch
Duration: 33.44 sec
2025-10-05 01:15:01,772 - root - INFO - Epoch 17/100, Model CSILSTMNet, Train
Loss: 0.2904, Train Acc: 0.8993, Val Loss: 0.2700, Val Acc: 0.9102, Epoch
```

Duration: 33.49 sec  
2025-10-05 01:15:37,185 - root - INFO - Epoch 18/100, Model CSILSTMNet, Train Loss: 0.2594, Train Acc: 0.9105, Val Loss: 0.2475, Val Acc: 0.9199, Epoch  
Duration: 33.49 sec  
2025-10-05 01:16:12,424 - root - INFO - Epoch 19/100, Model CSILSTMNet, Train Loss: 0.2348, Train Acc: 0.9188, Val Loss: 0.2593, Val Acc: 0.9078, Epoch  
Duration: 33.38 sec  
2025-10-05 01:16:45,826 - root - INFO - Epoch 20/100, Model CSILSTMNet, Train Loss: 0.2365, Train Acc: 0.9195, Val Loss: 0.2192, Val Acc: 0.9289, Epoch  
Duration: 33.40 sec  
2025-10-05 01:17:21,127 - root - INFO - Epoch 21/100, Model CSILSTMNet, Train Loss: 0.1972, Train Acc: 0.9330, Val Loss: 0.2164, Val Acc: 0.9289, Epoch  
Duration: 33.43 sec  
2025-10-05 01:17:54,539 - root - INFO - Epoch 22/100, Model CSILSTMNet, Train Loss: 0.1825, Train Acc: 0.9371, Val Loss: 0.2276, Val Acc: 0.9228, Epoch  
Duration: 33.41 sec  
2025-10-05 01:18:27,841 - root - INFO - Epoch 23/100, Model CSILSTMNet, Train Loss: 0.1634, Train Acc: 0.9445, Val Loss: 0.2150, Val Acc: 0.9275, Epoch  
Duration: 33.30 sec  
2025-10-05 01:19:01,277 - root - INFO - Epoch 24/100, Model CSILSTMNet, Train Loss: 0.1431, Train Acc: 0.9505, Val Loss: 0.1593, Val Acc: 0.9447, Epoch  
Duration: 33.43 sec  
2025-10-05 01:19:36,838 - root - INFO - Epoch 25/100, Model CSILSTMNet, Train Loss: 0.1468, Train Acc: 0.9510, Val Loss: 0.1531, Val Acc: 0.9480, Epoch  
Duration: 33.60 sec  
2025-10-05 01:20:11,999 - root - INFO - Epoch 26/100, Model CSILSTMNet, Train Loss: 0.1160, Train Acc: 0.9611, Val Loss: 0.1548, Val Acc: 0.9450, Epoch  
Duration: 33.26 sec  
2025-10-05 01:20:45,324 - root - INFO - Epoch 27/100, Model CSILSTMNet, Train Loss: 0.1116, Train Acc: 0.9623, Val Loss: 0.1442, Val Acc: 0.9493, Epoch  
Duration: 33.32 sec  
2025-10-05 01:21:21,943 - root - INFO - Epoch 28/100, Model CSILSTMNet, Train Loss: 0.0979, Train Acc: 0.9666, Val Loss: 0.1518, Val Acc: 0.9496, Epoch  
Duration: 34.71 sec  
2025-10-05 01:21:57,894 - root - INFO - Epoch 29/100, Model CSILSTMNet, Train Loss: 0.1100, Train Acc: 0.9634, Val Loss: 0.1339, Val Acc: 0.9563, Epoch  
Duration: 34.01 sec  
2025-10-05 01:22:33,852 - root - INFO - Epoch 30/100, Model CSILSTMNet, Train Loss: 0.0861, Train Acc: 0.9715, Val Loss: 0.1269, Val Acc: 0.9567, Epoch  
Duration: 34.04 sec  
2025-10-05 01:23:09,687 - root - INFO - Epoch 31/100, Model CSILSTMNet, Train Loss: 0.0790, Train Acc: 0.9725, Val Loss: 0.1103, Val Acc: 0.9643, Epoch  
Duration: 33.81 sec  
2025-10-05 01:23:45,335 - root - INFO - Epoch 32/100, Model CSILSTMNet, Train Loss: 0.0763, Train Acc: 0.9739, Val Loss: 0.1096, Val Acc: 0.9626, Epoch  
Duration: 33.74 sec  
2025-10-05 01:24:19,098 - root - INFO - Epoch 33/100, Model CSILSTMNet, Train Loss: 0.0712, Train Acc: 0.9754, Val Loss: 0.1008, Val Acc: 0.9647, Epoch  
Duration: 33.76 sec  
2025-10-05 01:24:54,886 - root - INFO - Epoch 34/100, Model CSILSTMNet, Train Loss: 0.0647, Train Acc: 0.9774, Val Loss: 0.0991, Val Acc: 0.9658, Epoch  
Duration: 33.88 sec  
2025-10-05 01:25:30,625 - root - INFO - Epoch 35/100, Model CSILSTMNet, Train Loss: 0.0599, Train Acc: 0.9793, Val Loss: 0.1006, Val Acc: 0.9668, Epoch

Duration: 33.73 sec  
2025-10-05 01:26:06,483 - root - INFO - Epoch 36/100, Model CSILSTMNet, Train Loss: 0.0581, Train Acc: 0.9802, Val Loss: 0.0900, Val Acc: 0.9706, Epoch  
Duration: 33.88 sec  
2025-10-05 01:26:42,381 - root - INFO - Epoch 37/100, Model CSILSTMNet, Train Loss: 0.0566, Train Acc: 0.9800, Val Loss: 0.1005, Val Acc: 0.9657, Epoch  
Duration: 33.86 sec  
2025-10-05 01:27:16,001 - root - INFO - Epoch 38/100, Model CSILSTMNet, Train Loss: 0.0462, Train Acc: 0.9844, Val Loss: 0.0987, Val Acc: 0.9688, Epoch  
Duration: 33.62 sec  
2025-10-05 01:27:49,784 - root - INFO - Epoch 39/100, Model CSILSTMNet, Train Loss: 0.0506, Train Acc: 0.9826, Val Loss: 0.1068, Val Acc: 0.9660, Epoch  
Duration: 33.78 sec  
2025-10-05 01:28:23,437 - root - INFO - Epoch 40/100, Model CSILSTMNet, Train Loss: 0.0412, Train Acc: 0.9865, Val Loss: 0.0970, Val Acc: 0.9683, Epoch  
Duration: 33.65 sec  
2025-10-05 01:28:57,324 - root - INFO - Epoch 41/100, Model CSILSTMNet, Train Loss: 0.0431, Train Acc: 0.9852, Val Loss: 0.0848, Val Acc: 0.9741, Epoch  
Duration: 33.89 sec  
2025-10-05 01:29:33,046 - root - INFO - Epoch 42/100, Model CSILSTMNet, Train Loss: 0.0440, Train Acc: 0.9848, Val Loss: 0.0835, Val Acc: 0.9736, Epoch  
Duration: 33.62 sec  
2025-10-05 01:30:06,554 - root - INFO - Epoch 43/100, Model CSILSTMNet, Train Loss: 0.0394, Train Acc: 0.9871, Val Loss: 0.0906, Val Acc: 0.9704, Epoch  
Duration: 33.51 sec  
2025-10-05 01:30:40,146 - root - INFO - Epoch 44/100, Model CSILSTMNet, Train Loss: 0.0345, Train Acc: 0.9882, Val Loss: 0.0915, Val Acc: 0.9699, Epoch  
Duration: 33.59 sec  
2025-10-05 01:31:13,907 - root - INFO - Epoch 45/100, Model CSILSTMNet, Train Loss: 0.0338, Train Acc: 0.9885, Val Loss: 0.0798, Val Acc: 0.9757, Epoch  
Duration: 33.76 sec  
2025-10-05 01:31:50,178 - root - INFO - Epoch 46/100, Model CSILSTMNet, Train Loss: 0.0317, Train Acc: 0.9894, Val Loss: 0.0919, Val Acc: 0.9728, Epoch  
Duration: 33.94 sec  
2025-10-05 01:32:23,621 - root - INFO - Epoch 47/100, Model CSILSTMNet, Train Loss: 0.0330, Train Acc: 0.9893, Val Loss: 0.0804, Val Acc: 0.9753, Epoch  
Duration: 33.44 sec  
2025-10-05 01:32:56,974 - root - INFO - Epoch 48/100, Model CSILSTMNet, Train Loss: 0.0278, Train Acc: 0.9907, Val Loss: 0.0780, Val Acc: 0.9753, Epoch  
Duration: 33.35 sec  
2025-10-05 01:33:30,330 - root - INFO - Epoch 49/100, Model CSILSTMNet, Train Loss: 0.0330, Train Acc: 0.9889, Val Loss: 0.0750, Val Acc: 0.9788, Epoch  
Duration: 33.35 sec  
2025-10-05 01:34:05,873 - root - INFO - Epoch 50/100, Model CSILSTMNet, Train Loss: 0.0266, Train Acc: 0.9915, Val Loss: 0.1526, Val Acc: 0.9628, Epoch  
Duration: 33.46 sec  
2025-10-05 01:34:39,257 - root - INFO - Epoch 51/100, Model CSILSTMNet, Train Loss: 0.0294, Train Acc: 0.9909, Val Loss: 0.1585, Val Acc: 0.9573, Epoch  
Duration: 33.38 sec  
2025-10-05 01:35:12,599 - root - INFO - Epoch 52/100, Model CSILSTMNet, Train Loss: 0.0284, Train Acc: 0.9913, Val Loss: 0.0758, Val Acc: 0.9774, Epoch  
Duration: 33.34 sec  
2025-10-05 01:35:46,112 - root - INFO - Epoch 53/100, Model CSILSTMNet, Train Loss: 0.0242, Train Acc: 0.9921, Val Loss: 0.0760, Val Acc: 0.9782, Epoch

Duration: 33.51 sec  
2025-10-05 01:36:19,491 - root - INFO - Epoch 54/100, Model CSILSTMNet, Train Loss: 0.0273, Train Acc: 0.9917, Val Loss: 0.0704, Val Acc: 0.9789, Epoch  
Duration: 33.38 sec  
2025-10-05 01:36:55,025 - root - INFO - Epoch 55/100, Model CSILSTMNet, Train Loss: 0.0179, Train Acc: 0.9945, Val Loss: 0.0667, Val Acc: 0.9797, Epoch  
Duration: 33.48 sec  
2025-10-05 01:37:30,415 - root - INFO - Epoch 56/100, Model CSILSTMNet, Train Loss: 0.0192, Train Acc: 0.9939, Val Loss: 0.0766, Val Acc: 0.9776, Epoch  
Duration: 33.46 sec  
2025-10-05 01:38:03,787 - root - INFO - Epoch 57/100, Model CSILSTMNet, Train Loss: 0.0218, Train Acc: 0.9928, Val Loss: 0.0908, Val Acc: 0.9747, Epoch  
Duration: 33.37 sec  
2025-10-05 01:38:37,177 - root - INFO - Epoch 58/100, Model CSILSTMNet, Train Loss: 0.0190, Train Acc: 0.9940, Val Loss: 0.0767, Val Acc: 0.9790, Epoch  
Duration: 33.39 sec  
2025-10-05 01:39:10,580 - root - INFO - Epoch 59/100, Model CSILSTMNet, Train Loss: 0.0190, Train Acc: 0.9938, Val Loss: 0.0838, Val Acc: 0.9762, Epoch  
Duration: 33.40 sec  
2025-10-05 01:39:44,192 - root - INFO - Epoch 60/100, Model CSILSTMNet, Train Loss: 0.0188, Train Acc: 0.9943, Val Loss: 0.0735, Val Acc: 0.9795, Epoch  
Duration: 33.61 sec  
2025-10-05 01:40:17,488 - root - INFO - Epoch 61/100, Model CSILSTMNet, Train Loss: 0.0186, Train Acc: 0.9942, Val Loss: 0.0643, Val Acc: 0.9816, Epoch  
Duration: 33.29 sec  
2025-10-05 01:40:52,987 - root - INFO - Epoch 62/100, Model CSILSTMNet, Train Loss: 0.0190, Train Acc: 0.9937, Val Loss: 0.0890, Val Acc: 0.9754, Epoch  
Duration: 33.39 sec  
2025-10-05 01:41:26,326 - root - INFO - Epoch 63/100, Model CSILSTMNet, Train Loss: 0.0166, Train Acc: 0.9949, Val Loss: 0.0761, Val Acc: 0.9795, Epoch  
Duration: 33.34 sec  
2025-10-05 01:41:59,701 - root - INFO - Epoch 64/100, Model CSILSTMNet, Train Loss: 0.0226, Train Acc: 0.9932, Val Loss: 0.0682, Val Acc: 0.9803, Epoch  
Duration: 33.37 sec  
2025-10-05 01:42:33,032 - root - INFO - Epoch 65/100, Model CSILSTMNet, Train Loss: 0.0162, Train Acc: 0.9952, Val Loss: 0.0676, Val Acc: 0.9806, Epoch  
Duration: 33.33 sec  
2025-10-05 01:43:06,323 - root - INFO - Epoch 66/100, Model CSILSTMNet, Train Loss: 0.0154, Train Acc: 0.9955, Val Loss: 0.0749, Val Acc: 0.9775, Epoch  
Duration: 33.29 sec  
2025-10-05 01:43:39,745 - root - INFO - Epoch 67/100, Model CSILSTMNet, Train Loss: 0.0178, Train Acc: 0.9946, Val Loss: 0.0861, Val Acc: 0.9775, Epoch  
Duration: 33.42 sec  
2025-10-05 01:44:13,017 - root - INFO - Epoch 68/100, Model CSILSTMNet, Train Loss: 0.0202, Train Acc: 0.9936, Val Loss: 0.0652, Val Acc: 0.9803, Epoch  
Duration: 33.27 sec  
2025-10-05 01:44:46,465 - root - INFO - Epoch 69/100, Model CSILSTMNet, Train Loss: 0.0116, Train Acc: 0.9965, Val Loss: 0.0658, Val Acc: 0.9810, Epoch  
Duration: 33.45 sec  
2025-10-05 01:45:19,849 - root - INFO - Epoch 70/100, Model CSILSTMNet, Train Loss: 0.0163, Train Acc: 0.9944, Val Loss: 0.0726, Val Acc: 0.9816, Epoch  
Duration: 33.38 sec  
2025-10-05 01:45:53,141 - root - INFO - Epoch 71/100, Model CSILSTMNet, Train Loss: 0.0126, Train Acc: 0.9959, Val Loss: 0.0640, Val Acc: 0.9844, Epoch

Duration: 33.29 sec  
2025-10-05 01:46:28,822 - root - INFO - Epoch 72/100, Model CSILSTMNet, Train Loss: 0.0140, Train Acc: 0.9955, Val Loss: 0.0699, Val Acc: 0.9804, Epoch  
Duration: 33.58 sec  
2025-10-05 01:47:02,028 - root - INFO - Epoch 73/100, Model CSILSTMNet, Train Loss: 0.0122, Train Acc: 0.9963, Val Loss: 0.0676, Val Acc: 0.9818, Epoch  
Duration: 33.20 sec  
2025-10-05 01:47:35,303 - root - INFO - Epoch 74/100, Model CSILSTMNet, Train Loss: 0.0138, Train Acc: 0.9957, Val Loss: 0.0586, Val Acc: 0.9846, Epoch  
Duration: 33.27 sec  
2025-10-05 01:48:10,589 - root - INFO - Epoch 75/100, Model CSILSTMNet, Train Loss: 0.0156, Train Acc: 0.9959, Val Loss: 0.0628, Val Acc: 0.9840, Epoch  
Duration: 33.34 sec  
2025-10-05 01:48:43,893 - root - INFO - Epoch 76/100, Model CSILSTMNet, Train Loss: 0.0107, Train Acc: 0.9969, Val Loss: 0.0673, Val Acc: 0.9818, Epoch  
Duration: 33.30 sec  
2025-10-05 01:49:17,100 - root - INFO - Epoch 77/100, Model CSILSTMNet, Train Loss: 0.0115, Train Acc: 0.9970, Val Loss: 0.0710, Val Acc: 0.9812, Epoch  
Duration: 33.21 sec  
2025-10-05 01:49:50,592 - root - INFO - Epoch 78/100, Model CSILSTMNet, Train Loss: 0.0147, Train Acc: 0.9953, Val Loss: 0.0641, Val Acc: 0.9829, Epoch  
Duration: 33.49 sec  
2025-10-05 01:50:23,979 - root - INFO - Epoch 79/100, Model CSILSTMNet, Train Loss: 0.0116, Train Acc: 0.9963, Val Loss: 0.0786, Val Acc: 0.9788, Epoch  
Duration: 33.38 sec  
2025-10-05 01:50:57,369 - root - INFO - Epoch 80/100, Model CSILSTMNet, Train Loss: 0.0097, Train Acc: 0.9969, Val Loss: 0.0699, Val Acc: 0.9811, Epoch  
Duration: 33.39 sec  
2025-10-05 01:51:31,001 - root - INFO - Epoch 81/100, Model CSILSTMNet, Train Loss: 0.0170, Train Acc: 0.9950, Val Loss: 0.0821, Val Acc: 0.9788, Epoch  
Duration: 33.63 sec  
2025-10-05 01:52:04,331 - root - INFO - Epoch 82/100, Model CSILSTMNet, Train Loss: 0.0100, Train Acc: 0.9969, Val Loss: 0.0762, Val Acc: 0.9810, Epoch  
Duration: 33.33 sec  
2025-10-05 01:52:37,739 - root - INFO - Epoch 83/100, Model CSILSTMNet, Train Loss: 0.0101, Train Acc: 0.9971, Val Loss: 0.0856, Val Acc: 0.9782, Epoch  
Duration: 33.41 sec  
2025-10-05 01:53:11,220 - root - INFO - Epoch 84/100, Model CSILSTMNet, Train Loss: 0.0130, Train Acc: 0.9960, Val Loss: 0.0643, Val Acc: 0.9839, Epoch  
Duration: 33.48 sec  
2025-10-05 01:53:44,591 - root - INFO - Epoch 85/100, Model CSILSTMNet, Train Loss: 0.0114, Train Acc: 0.9965, Val Loss: 0.0694, Val Acc: 0.9816, Epoch  
Duration: 33.37 sec  
2025-10-05 01:54:17,858 - root - INFO - Epoch 86/100, Model CSILSTMNet, Train Loss: 0.0097, Train Acc: 0.9969, Val Loss: 0.0711, Val Acc: 0.9813, Epoch  
Duration: 33.27 sec  
2025-10-05 01:54:51,143 - root - INFO - Epoch 87/100, Model CSILSTMNet, Train Loss: 0.0115, Train Acc: 0.9966, Val Loss: 0.0743, Val Acc: 0.9804, Epoch  
Duration: 33.28 sec  
2025-10-05 01:55:24,481 - root - INFO - Epoch 88/100, Model CSILSTMNet, Train Loss: 0.0099, Train Acc: 0.9967, Val Loss: 0.0821, Val Acc: 0.9783, Epoch  
Duration: 33.34 sec  
2025-10-05 01:55:57,932 - root - INFO - Epoch 89/100, Model CSILSTMNet, Train Loss: 0.0097, Train Acc: 0.9967, Val Loss: 0.0685, Val Acc: 0.9827, Epoch

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Duration: 33.45 sec
2025-10-05 01:56:31,301 - root - INFO - Epoch 90/100, Model CSILSTMNet, Train
Loss: 0.0089, Train Acc: 0.9969, Val Loss: 0.0848, Val Acc: 0.9796, Epoch
Duration: 33.37 sec
2025-10-05 01:57:04,677 - root - INFO - Epoch 91/100, Model CSILSTMNet, Train
Loss: 0.0131, Train Acc: 0.9956, Val Loss: 0.0765, Val Acc: 0.9794, Epoch
Duration: 33.37 sec
2025-10-05 01:57:37,986 - root - INFO - Epoch 92/100, Model CSILSTMNet, Train
Loss: 0.0050, Train Acc: 0.9987, Val Loss: 0.0671, Val Acc: 0.9815, Epoch
Duration: 33.31 sec
2025-10-05 01:58:11,476 - root - INFO - Epoch 93/100, Model CSILSTMNet, Train
Loss: 0.0087, Train Acc: 0.9974, Val Loss: 0.0860, Val Acc: 0.9788, Epoch
Duration: 33.49 sec
2025-10-05 01:58:44,971 - root - INFO - Epoch 94/100, Model CSILSTMNet, Train
Loss: 0.0078, Train Acc: 0.9975, Val Loss: 0.0682, Val Acc: 0.9834, Epoch
Duration: 33.49 sec
2025-10-05 01:59:18,257 - root - INFO - Epoch 95/100, Model CSILSTMNet, Train
Loss: 0.0116, Train Acc: 0.9968, Val Loss: 0.0686, Val Acc: 0.9820, Epoch
Duration: 33.29 sec
2025-10-05 01:59:51,768 - root - INFO - Epoch 96/100, Model CSILSTMNet, Train
Loss: 0.0076, Train Acc: 0.9976, Val Loss: 0.0751, Val Acc: 0.9812, Epoch
Duration: 33.51 sec
2025-10-05 02:00:25,211 - root - INFO - Epoch 97/100, Model CSILSTMNet, Train
Loss: 0.0117, Train Acc: 0.9961, Val Loss: 0.0781, Val Acc: 0.9819, Epoch
Duration: 33.44 sec
2025-10-05 02:00:58,520 - root - INFO - Epoch 98/100, Model CSILSTMNet, Train
Loss: 0.0064, Train Acc: 0.9981, Val Loss: 0.0620, Val Acc: 0.9851, Epoch
Duration: 33.31 sec
2025-10-05 02:01:34,211 - root - INFO - Epoch 99/100, Model CSILSTMNet, Train
Loss: 0.0120, Train Acc: 0.9964, Val Loss: 0.0741, Val Acc: 0.9824, Epoch
Duration: 33.50 sec
2025-10-05 02:02:07,521 - root - INFO - Epoch 100/100, Model CSILSTMNet, Train
Loss: 0.0075, Train Acc: 0.9975, Val Loss: 0.0777, Val Acc: 0.9785, Epoch
Duration: 33.31 sec
2025-10-05 02:02:07,521 - root - CRITICAL - Training complete. Best Val Acc:
0.9851 at epoch 98.
2025-10-05 02:02:07,525 - root - CRITICAL - SAVED BEST MODEL
/Users/sanjeev/VNIT/FINAL_PRJ_PHASE2/experiments/20251005_000538/checkpoints/bes
t_model_CSILSTMNet_epoch98.pt USING LEARNING RATE 0.0005
2025-10-05 02:02:07,525 - root - CRITICAL - BEST VALIDATION ACCURACY: 0.9851 at
epoch 98.
2025-10-05 02:02:07,611 - root - INFO - Saved stats plot to
/Users/sanjeev/VNIT/FINAL_PRJ_PHASE2/experiments/20251005_000538/plots/CSILSTMNe
t_{'csi_input_size': 99, 'meta_input_size': 12, 'window_size': 128,
'num_classes': 31, 'lr': 0.0005, 'epochs': 100, 'model_name':
'CSILSTMNet'}_stats.png
2025-10-05 02:02:10,660 - root - INFO - Confusion matrix:
[[274  0  1  0  0  1  0  0  0  1  0  0  0  0  0  0  0  0
  0  0  0  0  0  0  0  0  0  0  0  0]
 [  0 269  0  0  9  2  0  3  0  0  0  0  0  0  0  0  0  0
  0  0  0  0  0  0  0  0  0  0  0  0]
 [  4  0 281  0  1  0  0  0  0  6  0  0  0  0  0  0  0  0
  0  0  0  0  0  0  0  0  0  0  0  0]
 [  0  0  0 285  0  0  0  0  0  0  0  0  0  0  0  0  1  0

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0 0 0 0 0 2 0 0 0 0 0 0]
[ 0 8 1 0 261 0 0 0 0 2 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 0 0 0 0]
[ 2 0 0 0 0 293 0 0 0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 0 0 0 0]
[ 0 0 0 0 0 0 288 0 0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 2 0 0 0 0]
[ 0 0 0 0 0 1 0 266 0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 0 0 0 0]
[ 0 0 0 1 0 0 0 0 264 0 0 0 0 0 0 0 0
0 0 0 0 0 1 1 0 0 0 0 0]
[ 3 0 1 0 0 0 0 0 0 268 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 0 0 0 0]
[ 0 0 0 0 0 0 0 0 0 0 268 0 0 0 0 0 0
0 0 1 0 0 0 0 0 1 0 0 0]
[ 0 0 0 0 0 0 0 0 0 0 0 271 0 0 0 0 0
0 0 0 0 0 0 0 0 0 0 1 0]
[ 0 0 0 0 0 0 0 0 0 0 0 0 313 0 0 0 0
11 0 0 0 0 0 0 8 0 0 0 0]
[ 0 0 0 0 0 0 0 0 0 0 0 0 0 301 0 0 0
0 0 0 0 0 0 0 0 0 0 0 0]
[ 0 0 0 0 0 0 0 0 0 0 0 0 0 0 249 0 4
0 0 0 0 0 0 0 0 0 0 0 0]
[ 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 307 0 0
0 0 0 0 0 0 0 0 0 0 0 0]
[ 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 270 19
0 0 0 0 0 0 0 0 0 0 0 0]
[ 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 2 303
0 0 0 0 0 0 0 0 0 0 0 0]
[ 0 0 0 0 0 0 0 0 0 1 0 0 10 1 0 0 0 0
269 1 0 0 0 0 0 0 0 0 0 0]
[ 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0 282 0 0 0 0 0 0 0 0 2 1]
[ 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0
0 0 287 0 0 0 0 0 0 0 1 0]
[ 0 0 0 0 0 0 0 0 0 0 0 2 0 0 0 0 0 0
0 0 1 296 0 0 0 0 0 0 14 3]
[ 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0 0 0 0 287 0 1 0 0 0 0 0]
[ 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0 0 0 0 3 266 3 0 0 0 0 0]
[ 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 2 0
1 0 0 0 1 0 304 0 0 0 0 0]
[ 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 262 0 0 0 0]
[ 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 287 0 0 0]
[ 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 1 292 0 0]
[ 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0 1 1 4 0 0 0 0 0 0 283 5]
[ 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0 0 4 3 0 0 0 0 0 0 12 246]]

```

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| 1            | 0.97      | 0.99   | 0.98     | 277     |
| 2            | 0.97      | 0.95   | 0.96     | 283     |
| 3            | 0.99      | 0.96   | 0.98     | 292     |
| 4            | 1.00      | 0.99   | 0.99     | 288     |
| 5            | 0.96      | 0.96   | 0.96     | 272     |
| 6            | 0.99      | 0.99   | 0.99     | 295     |
| 7            | 0.99      | 0.99   | 0.99     | 290     |
| 8            | 0.99      | 1.00   | 0.99     | 267     |
| 9            | 1.00      | 0.99   | 0.99     | 267     |
| 10           | 0.96      | 0.99   | 0.97     | 272     |
| 11           | 0.99      | 0.99   | 0.99     | 270     |
| 12           | 1.00      | 1.00   | 1.00     | 272     |
| 13           | 0.97      | 0.94   | 0.96     | 332     |
| 14           | 1.00      | 1.00   | 1.00     | 301     |
| 15           | 1.00      | 0.98   | 0.99     | 253     |
| 16           | 1.00      | 1.00   | 1.00     | 307     |
| 17           | 0.97      | 0.93   | 0.95     | 290     |
| 18           | 0.94      | 0.99   | 0.96     | 306     |
| 19           | 0.96      | 0.95   | 0.96     | 282     |
| 20           | 0.99      | 0.99   | 0.99     | 285     |
| 21           | 0.98      | 0.99   | 0.98     | 289     |
| 22           | 0.98      | 0.94   | 0.96     | 316     |
| 23           | 0.99      | 1.00   | 0.99     | 288     |
| 24           | 0.99      | 0.98   | 0.98     | 272     |
| 25           | 0.98      | 0.99   | 0.99     | 308     |
| 26           | 0.96      | 1.00   | 0.98     | 263     |
| 27           | 0.99      | 1.00   | 1.00     | 287     |
| 28           | 1.00      | 1.00   | 1.00     | 293     |
| 29           | 0.90      | 0.96   | 0.93     | 294     |
| 30           | 0.96      | 0.93   | 0.95     | 265     |
| accuracy     |           |        |          | 0.98    |
| macro avg    |           |        |          | 0.98    |
| weighted avg |           |        |          | 0.98    |

```

2025-10-05 02:02:10,664 - root - CRITICAL - 99. SAVED BEST MODEL IN CSILSTMNet.
BEST VALIDATION ACCURACY: 0.9851 at epoch 98.
2025-10-05 02:02:10,664 - root - CRITICAL - 100. LOGGED BEST MODEL IN MLFLOW
/Users/sanjeev/VNIT/FINAL_PRJ_PHASE2/experiments/20251005_000538/checkpoints/bes
t_model_CSILSTMNet_epoch98.pt. BEST VALIDATION ACCURACY: 0.9851 at epoch 98.
2025-10-05 02:02:10,666 - root - INFO -
2025-10-05 02:02:10,666 - root - INFO - N
2025-10-05 02:02:10,666 - root - INFO - E
2025-10-05 02:02:10,666 - root - INFO - W
2025-10-05 02:02:10,666 - root - INFO - STARTING TRAINING FLOW :
DenseNet1D:('csi_channels', 'meta_feature_dim', 'num_classes', 'lr',
'epochs'):([99], [12], [31], [0.001, 0.0005], [100])
2025-10-05 02:02:10,670 - root - INFO - START RUN MLFLOW :
DenseNet1D:{'csi_channels': 99, 'meta_feature_dim': 12, 'num_classes': 31, 'lr':
0.001, 'epochs': 100, 'model_name': 'DenseNet1D'}
2025-10-05 02:02:10,682 - root - INFO - START T-N-E DenseNet1D
2025-10-05 02:02:10,683 - root - INFO - DenseNet1D USING LEARNING RATE 0.001

```



2025-10-05 02:02:56,171 - root - INFO - Epoch 1/100, Model DenseNet1D, Train Loss: 0.7528, Train Acc: 0.7531, Val Loss: 0.3496, Val Acc: 0.8801, Epoch Duration: 45.49 sec

2025-10-05 02:03:42,784 - root - INFO - Epoch 2/100, Model DenseNet1D, Train Loss: 0.4220, Train Acc: 0.8550, Val Loss: 0.2509, Val Acc: 0.9174, Epoch Duration: 44.54 sec

2025-10-05 02:04:29,265 - root - INFO - Epoch 3/100, Model DenseNet1D, Train Loss: 0.3454, Train Acc: 0.8788, Val Loss: 0.2246, Val Acc: 0.9226, Epoch Duration: 44.36 sec

2025-10-05 02:05:15,349 - root - INFO - Epoch 4/100, Model DenseNet1D, Train Loss: 0.3024, Train Acc: 0.8937, Val Loss: 0.1762, Val Acc: 0.9427, Epoch Duration: 44.08 sec

2025-10-05 02:06:01,746 - root - INFO - Epoch 5/100, Model DenseNet1D, Train Loss: 0.2586, Train Acc: 0.9081, Val Loss: 0.1655, Val Acc: 0.9417, Epoch Duration: 44.44 sec

2025-10-05 02:06:45,921 - root - INFO - Epoch 6/100, Model DenseNet1D, Train Loss: 0.2382, Train Acc: 0.9173, Val Loss: 0.1461, Val Acc: 0.9511, Epoch Duration: 44.17 sec

2025-10-05 02:07:32,144 - root - INFO - Epoch 7/100, Model DenseNet1D, Train Loss: 0.2189, Train Acc: 0.9218, Val Loss: 0.1684, Val Acc: 0.9445, Epoch Duration: 44.34 sec

2025-10-05 02:08:16,259 - root - INFO - Epoch 8/100, Model DenseNet1D, Train Loss: 0.2073, Train Acc: 0.9262, Val Loss: 0.1360, Val Acc: 0.9536, Epoch Duration: 44.11 sec

2025-10-05 02:09:02,749 - root - INFO - Epoch 9/100, Model DenseNet1D, Train Loss: 0.1841, Train Acc: 0.9353, Val Loss: 0.1256, Val Acc: 0.9551, Epoch Duration: 44.44 sec

2025-10-05 02:09:49,629 - root - INFO - Epoch 10/100, Model DenseNet1D, Train Loss: 0.1687, Train Acc: 0.9404, Val Loss: 0.1138, Val Acc: 0.9621, Epoch Duration: 44.91 sec

2025-10-05 02:10:35,832 - root - INFO - Epoch 11/100, Model DenseNet1D, Train Loss: 0.1625, Train Acc: 0.9411, Val Loss: 0.1067, Val Acc: 0.9642, Epoch Duration: 44.30 sec

2025-10-05 02:11:22,743 - root - INFO - Epoch 12/100, Model DenseNet1D, Train Loss: 0.1488, Train Acc: 0.9471, Val Loss: 0.1157, Val Acc: 0.9620, Epoch Duration: 45.05 sec

2025-10-05 02:12:07,398 - root - INFO - Epoch 13/100, Model DenseNet1D, Train Loss: 0.1454, Train Acc: 0.9472, Val Loss: 0.1013, Val Acc: 0.9651, Epoch Duration: 44.65 sec

2025-10-05 02:12:53,874 - root - INFO - Epoch 14/100, Model DenseNet1D, Train Loss: 0.1348, Train Acc: 0.9522, Val Loss: 0.1175, Val Acc: 0.9595, Epoch Duration: 44.30 sec

2025-10-05 02:13:38,148 - root - INFO - Epoch 15/100, Model DenseNet1D, Train Loss: 0.1254, Train Acc: 0.9551, Val Loss: 0.1383, Val Acc: 0.9528, Epoch Duration: 44.27 sec

2025-10-05 02:14:22,355 - root - INFO - Epoch 16/100, Model DenseNet1D, Train Loss: 0.1251, Train Acc: 0.9565, Val Loss: 0.0936, Val Acc: 0.9693, Epoch Duration: 44.21 sec

2025-10-05 02:15:08,522 - root - INFO - Epoch 17/100, Model DenseNet1D, Train Loss: 0.1179, Train Acc: 0.9578, Val Loss: 0.1215, Val Acc: 0.9591, Epoch Duration: 44.25 sec

2025-10-05 02:15:52,866 - root - INFO - Epoch 18/100, Model DenseNet1D, Train Loss: 0.1110, Train Acc: 0.9604, Val Loss: 0.0781, Val Acc: 0.9732, Epoch Duration: 44.34 sec

2025-10-05 02:16:39,118 - root - INFO - Epoch 19/100, Model DenseNet1D, Train Loss: 0.1033, Train Acc: 0.9625, Val Loss: 0.0853, Val Acc: 0.9719, Epoch Duration: 44.30 sec

2025-10-05 02:17:23,308 - root - INFO - Epoch 20/100, Model DenseNet1D, Train Loss: 0.1088, Train Acc: 0.9611, Val Loss: 0.0782, Val Acc: 0.9705, Epoch Duration: 44.19 sec

2025-10-05 02:18:07,265 - root - INFO - Epoch 21/100, Model DenseNet1D, Train Loss: 0.0980, Train Acc: 0.9659, Val Loss: 0.0694, Val Acc: 0.9763, Epoch Duration: 43.96 sec

2025-10-05 02:18:53,816 - root - INFO - Epoch 22/100, Model DenseNet1D, Train Loss: 0.0981, Train Acc: 0.9648, Val Loss: 0.0840, Val Acc: 0.9740, Epoch Duration: 44.52 sec

2025-10-05 02:19:38,129 - root - INFO - Epoch 23/100, Model DenseNet1D, Train Loss: 0.0937, Train Acc: 0.9676, Val Loss: 0.0806, Val Acc: 0.9732, Epoch Duration: 44.31 sec

2025-10-05 02:20:22,225 - root - INFO - Epoch 24/100, Model DenseNet1D, Train Loss: 0.0916, Train Acc: 0.9664, Val Loss: 0.0836, Val Acc: 0.9729, Epoch Duration: 44.10 sec

2025-10-05 02:21:06,387 - root - INFO - Epoch 25/100, Model DenseNet1D, Train Loss: 0.0867, Train Acc: 0.9691, Val Loss: 0.0722, Val Acc: 0.9747, Epoch Duration: 44.16 sec

2025-10-05 02:21:51,051 - root - INFO - Epoch 26/100, Model DenseNet1D, Train Loss: 0.0850, Train Acc: 0.9687, Val Loss: 0.0730, Val Acc: 0.9768, Epoch Duration: 44.66 sec

2025-10-05 02:22:37,890 - root - INFO - Epoch 27/100, Model DenseNet1D, Train Loss: 0.0811, Train Acc: 0.9711, Val Loss: 0.0750, Val Acc: 0.9750, Epoch Duration: 44.78 sec

2025-10-05 02:23:21,827 - root - INFO - Epoch 28/100, Model DenseNet1D, Train Loss: 0.0811, Train Acc: 0.9713, Val Loss: 0.0690, Val Acc: 0.9762, Epoch Duration: 43.94 sec

2025-10-05 02:24:05,900 - root - INFO - Epoch 29/100, Model DenseNet1D, Train Loss: 0.0779, Train Acc: 0.9727, Val Loss: 0.0750, Val Acc: 0.9763, Epoch Duration: 44.07 sec

2025-10-05 02:24:50,693 - root - INFO - Epoch 30/100, Model DenseNet1D, Train Loss: 0.0716, Train Acc: 0.9746, Val Loss: 0.0492, Val Acc: 0.9861, Epoch Duration: 44.79 sec

2025-10-05 02:25:36,817 - root - INFO - Epoch 31/100, Model DenseNet1D, Train Loss: 0.0768, Train Acc: 0.9730, Val Loss: 0.0737, Val Acc: 0.9781, Epoch Duration: 44.19 sec

2025-10-05 02:26:20,946 - root - INFO - Epoch 32/100, Model DenseNet1D, Train Loss: 0.0660, Train Acc: 0.9767, Val Loss: 0.0630, Val Acc: 0.9788, Epoch Duration: 44.13 sec

2025-10-05 02:27:05,416 - root - INFO - Epoch 33/100, Model DenseNet1D, Train Loss: 0.0746, Train Acc: 0.9736, Val Loss: 0.0605, Val Acc: 0.9817, Epoch Duration: 44.47 sec

2025-10-05 02:27:49,789 - root - INFO - Epoch 34/100, Model DenseNet1D, Train Loss: 0.0704, Train Acc: 0.9748, Val Loss: 0.0552, Val Acc: 0.9806, Epoch Duration: 44.37 sec

2025-10-05 02:28:33,993 - root - INFO - Epoch 35/100, Model DenseNet1D, Train Loss: 0.0652, Train Acc: 0.9768, Val Loss: 0.0573, Val Acc: 0.9838, Epoch Duration: 44.20 sec

2025-10-05 02:29:18,410 - root - INFO - Epoch 36/100, Model DenseNet1D, Train Loss: 0.0638, Train Acc: 0.9769, Val Loss: 0.0549, Val Acc: 0.9818, Epoch Duration: 44.41 sec

2025-10-05 02:30:03,124 - root - INFO - Epoch 37/100, Model DenseNet1D, Train Loss: 0.0615, Train Acc: 0.9785, Val Loss: 0.0707, Val Acc: 0.9771, Epoch Duration: 44.71 sec

2025-10-05 02:30:47,923 - root - INFO - Epoch 38/100, Model DenseNet1D, Train Loss: 0.0594, Train Acc: 0.9792, Val Loss: 0.0540, Val Acc: 0.9844, Epoch Duration: 44.80 sec

2025-10-05 02:31:33,109 - root - INFO - Epoch 39/100, Model DenseNet1D, Train Loss: 0.0570, Train Acc: 0.9801, Val Loss: 0.0558, Val Acc: 0.9825, Epoch Duration: 45.18 sec

2025-10-05 02:32:18,511 - root - INFO - Epoch 40/100, Model DenseNet1D, Train Loss: 0.0633, Train Acc: 0.9773, Val Loss: 0.0649, Val Acc: 0.9769, Epoch Duration: 45.40 sec

2025-10-05 02:33:03,419 - root - INFO - Epoch 41/100, Model DenseNet1D, Train Loss: 0.0607, Train Acc: 0.9791, Val Loss: 0.0743, Val Acc: 0.9741, Epoch Duration: 44.91 sec

2025-10-05 02:33:48,267 - root - INFO - Epoch 42/100, Model DenseNet1D, Train Loss: 0.0544, Train Acc: 0.9807, Val Loss: 0.0576, Val Acc: 0.9815, Epoch Duration: 44.85 sec

2025-10-05 02:34:33,148 - root - INFO - Epoch 43/100, Model DenseNet1D, Train Loss: 0.0510, Train Acc: 0.9818, Val Loss: 0.0859, Val Acc: 0.9746, Epoch Duration: 44.88 sec

2025-10-05 02:35:18,027 - root - INFO - Epoch 44/100, Model DenseNet1D, Train Loss: 0.0527, Train Acc: 0.9820, Val Loss: 0.0490, Val Acc: 0.9845, Epoch Duration: 44.88 sec

2025-10-05 02:36:02,607 - root - INFO - Epoch 45/100, Model DenseNet1D, Train Loss: 0.0512, Train Acc: 0.9809, Val Loss: 0.0615, Val Acc: 0.9825, Epoch Duration: 44.58 sec

2025-10-05 02:36:47,468 - root - INFO - Epoch 46/100, Model DenseNet1D, Train Loss: 0.0555, Train Acc: 0.9801, Val Loss: 0.0482, Val Acc: 0.9853, Epoch Duration: 44.86 sec

2025-10-05 02:37:31,959 - root - INFO - Epoch 47/100, Model DenseNet1D, Train Loss: 0.0542, Train Acc: 0.9817, Val Loss: 0.0576, Val Acc: 0.9809, Epoch Duration: 44.49 sec

2025-10-05 02:38:16,461 - root - INFO - Epoch 48/100, Model DenseNet1D, Train Loss: 0.0465, Train Acc: 0.9831, Val Loss: 0.0625, Val Acc: 0.9785, Epoch Duration: 44.50 sec

2025-10-05 02:39:00,898 - root - INFO - Epoch 49/100, Model DenseNet1D, Train Loss: 0.0459, Train Acc: 0.9840, Val Loss: 0.0562, Val Acc: 0.9797, Epoch Duration: 44.43 sec

2025-10-05 02:39:45,669 - root - INFO - Epoch 50/100, Model DenseNet1D, Train Loss: 0.0489, Train Acc: 0.9834, Val Loss: 0.0568, Val Acc: 0.9830, Epoch Duration: 44.77 sec

2025-10-05 02:40:30,150 - root - INFO - Epoch 51/100, Model DenseNet1D, Train Loss: 0.0462, Train Acc: 0.9840, Val Loss: 0.0551, Val Acc: 0.9837, Epoch Duration: 44.48 sec

2025-10-05 02:41:14,476 - root - INFO - Epoch 52/100, Model DenseNet1D, Train Loss: 0.0456, Train Acc: 0.9838, Val Loss: 0.0514, Val Acc: 0.9834, Epoch Duration: 44.32 sec

2025-10-05 02:41:59,570 - root - INFO - Epoch 53/100, Model DenseNet1D, Train Loss: 0.0447, Train Acc: 0.9841, Val Loss: 0.0603, Val Acc: 0.9813, Epoch Duration: 45.09 sec

2025-10-05 02:42:44,060 - root - INFO - Epoch 54/100, Model DenseNet1D, Train Loss: 0.0461, Train Acc: 0.9835, Val Loss: 0.0589, Val Acc: 0.9817, Epoch Duration: 44.49 sec

2025-10-05 02:43:28,113 - root - INFO - Epoch 55/100, Model DenseNet1D, Train Loss: 0.0429, Train Acc: 0.9853, Val Loss: 0.0603, Val Acc: 0.9806, Epoch Duration: 44.05 sec

2025-10-05 02:44:12,303 - root - INFO - Epoch 56/100, Model DenseNet1D, Train Loss: 0.0440, Train Acc: 0.9845, Val Loss: 0.0719, Val Acc: 0.9755, Epoch Duration: 44.19 sec

2025-10-05 02:44:56,588 - root - INFO - Epoch 57/100, Model DenseNet1D, Train Loss: 0.0375, Train Acc: 0.9865, Val Loss: 0.0580, Val Acc: 0.9820, Epoch Duration: 44.28 sec

2025-10-05 02:45:40,873 - root - INFO - Epoch 58/100, Model DenseNet1D, Train Loss: 0.0399, Train Acc: 0.9858, Val Loss: 0.0588, Val Acc: 0.9811, Epoch Duration: 44.28 sec

2025-10-05 02:46:25,173 - root - INFO - Epoch 59/100, Model DenseNet1D, Train Loss: 0.0400, Train Acc: 0.9855, Val Loss: 0.0766, Val Acc: 0.9791, Epoch Duration: 44.30 sec

2025-10-05 02:47:09,500 - root - INFO - Epoch 60/100, Model DenseNet1D, Train Loss: 0.0406, Train Acc: 0.9853, Val Loss: 0.0476, Val Acc: 0.9850, Epoch Duration: 44.33 sec

2025-10-05 02:47:53,892 - root - INFO - Epoch 61/100, Model DenseNet1D, Train Loss: 0.0372, Train Acc: 0.9869, Val Loss: 0.0725, Val Acc: 0.9798, Epoch Duration: 44.39 sec

2025-10-05 02:48:38,315 - root - INFO - Epoch 62/100, Model DenseNet1D, Train Loss: 0.0437, Train Acc: 0.9850, Val Loss: 0.0566, Val Acc: 0.9834, Epoch Duration: 44.42 sec

2025-10-05 02:49:22,704 - root - INFO - Epoch 63/100, Model DenseNet1D, Train Loss: 0.0354, Train Acc: 0.9876, Val Loss: 0.0622, Val Acc: 0.9836, Epoch Duration: 44.39 sec

2025-10-05 02:50:07,591 - root - INFO - Epoch 64/100, Model DenseNet1D, Train Loss: 0.0406, Train Acc: 0.9859, Val Loss: 0.0707, Val Acc: 0.9799, Epoch Duration: 44.89 sec

2025-10-05 02:50:52,022 - root - INFO - Epoch 65/100, Model DenseNet1D, Train Loss: 0.0357, Train Acc: 0.9874, Val Loss: 0.0508, Val Acc: 0.9848, Epoch Duration: 44.43 sec

2025-10-05 02:51:36,495 - root - INFO - Epoch 66/100, Model DenseNet1D, Train Loss: 0.0361, Train Acc: 0.9873, Val Loss: 0.0541, Val Acc: 0.9840, Epoch Duration: 44.47 sec

2025-10-05 02:52:20,782 - root - INFO - Epoch 67/100, Model DenseNet1D, Train Loss: 0.0346, Train Acc: 0.9887, Val Loss: 0.0568, Val Acc: 0.9830, Epoch Duration: 44.28 sec

2025-10-05 02:53:04,920 - root - INFO - Epoch 68/100, Model DenseNet1D, Train Loss: 0.0331, Train Acc: 0.9883, Val Loss: 0.0536, Val Acc: 0.9848, Epoch Duration: 44.14 sec

2025-10-05 02:53:49,022 - root - INFO - Epoch 69/100, Model DenseNet1D, Train Loss: 0.0351, Train Acc: 0.9872, Val Loss: 0.0864, Val Acc: 0.9763, Epoch Duration: 44.10 sec

2025-10-05 02:54:33,497 - root - INFO - Epoch 70/100, Model DenseNet1D, Train Loss: 0.0381, Train Acc: 0.9876, Val Loss: 0.0611, Val Acc: 0.9837, Epoch Duration: 44.47 sec

2025-10-05 02:55:17,802 - root - INFO - Epoch 71/100, Model DenseNet1D, Train Loss: 0.0301, Train Acc: 0.9895, Val Loss: 0.0484, Val Acc: 0.9855, Epoch Duration: 44.30 sec

2025-10-05 02:56:02,359 - root - INFO - Epoch 72/100, Model DenseNet1D, Train Loss: 0.0361, Train Acc: 0.9883, Val Loss: 0.0465, Val Acc: 0.9861, Epoch Duration: 44.55 sec

2025-10-05 02:56:47,060 - root - INFO - Epoch 73/100, Model DenseNet1D, Train Loss: 0.0329, Train Acc: 0.9885, Val Loss: 0.0587, Val Acc: 0.9833, Epoch Duration: 44.70 sec

2025-10-05 02:57:32,075 - root - INFO - Epoch 74/100, Model DenseNet1D, Train Loss: 0.0340, Train Acc: 0.9883, Val Loss: 0.0564, Val Acc: 0.9841, Epoch Duration: 45.01 sec

2025-10-05 02:58:16,818 - root - INFO - Epoch 75/100, Model DenseNet1D, Train Loss: 0.0315, Train Acc: 0.9894, Val Loss: 0.0493, Val Acc: 0.9875, Epoch Duration: 44.74 sec

2025-10-05 02:59:04,133 - root - INFO - Epoch 76/100, Model DenseNet1D, Train Loss: 0.0335, Train Acc: 0.9884, Val Loss: 0.0697, Val Acc: 0.9850, Epoch Duration: 45.14 sec

2025-10-05 02:59:48,994 - root - INFO - Epoch 77/100, Model DenseNet1D, Train Loss: 0.0332, Train Acc: 0.9883, Val Loss: 0.0498, Val Acc: 0.9853, Epoch Duration: 44.86 sec

2025-10-05 03:00:33,749 - root - INFO - Epoch 78/100, Model DenseNet1D, Train Loss: 0.0329, Train Acc: 0.9887, Val Loss: 0.0610, Val Acc: 0.9832, Epoch Duration: 44.75 sec

2025-10-05 03:01:18,584 - root - INFO - Epoch 79/100, Model DenseNet1D, Train Loss: 0.0317, Train Acc: 0.9893, Val Loss: 0.0468, Val Acc: 0.9867, Epoch Duration: 44.83 sec

2025-10-05 03:02:03,292 - root - INFO - Epoch 80/100, Model DenseNet1D, Train Loss: 0.0308, Train Acc: 0.9899, Val Loss: 0.0626, Val Acc: 0.9810, Epoch Duration: 44.71 sec

2025-10-05 03:02:47,874 - root - INFO - Epoch 81/100, Model DenseNet1D, Train Loss: 0.0305, Train Acc: 0.9893, Val Loss: 0.0432, Val Acc: 0.9883, Epoch Duration: 44.58 sec

2025-10-05 03:03:34,867 - root - INFO - Epoch 82/100, Model DenseNet1D, Train Loss: 0.0301, Train Acc: 0.9901, Val Loss: 0.0622, Val Acc: 0.9809, Epoch Duration: 44.82 sec

2025-10-05 03:04:19,243 - root - INFO - Epoch 83/100, Model DenseNet1D, Train Loss: 0.0275, Train Acc: 0.9911, Val Loss: 0.0684, Val Acc: 0.9830, Epoch Duration: 44.37 sec

2025-10-05 03:05:04,094 - root - INFO - Epoch 84/100, Model DenseNet1D, Train Loss: 0.0293, Train Acc: 0.9903, Val Loss: 0.0604, Val Acc: 0.9822, Epoch Duration: 44.85 sec

2025-10-05 03:05:48,420 - root - INFO - Epoch 85/100, Model DenseNet1D, Train Loss: 0.0291, Train Acc: 0.9901, Val Loss: 0.0695, Val Acc: 0.9817, Epoch Duration: 44.32 sec

2025-10-05 03:06:32,935 - root - INFO - Epoch 86/100, Model DenseNet1D, Train Loss: 0.0267, Train Acc: 0.9904, Val Loss: 0.0551, Val Acc: 0.9853, Epoch Duration: 44.51 sec

2025-10-05 03:07:17,441 - root - INFO - Epoch 87/100, Model DenseNet1D, Train Loss: 0.0304, Train Acc: 0.9897, Val Loss: 0.0538, Val Acc: 0.9857, Epoch Duration: 44.50 sec

2025-10-05 03:08:01,979 - root - INFO - Epoch 88/100, Model DenseNet1D, Train Loss: 0.0260, Train Acc: 0.9909, Val Loss: 0.0625, Val Acc: 0.9844, Epoch Duration: 44.54 sec

2025-10-05 03:08:46,411 - root - INFO - Epoch 89/100, Model DenseNet1D, Train Loss: 0.0247, Train Acc: 0.9919, Val Loss: 0.0486, Val Acc: 0.9864, Epoch Duration: 44.43 sec

2025-10-05 03:09:31,234 - root - INFO - Epoch 90/100, Model DenseNet1D, Train Loss: 0.0294, Train Acc: 0.9897, Val Loss: 0.0564, Val Acc: 0.9850, Epoch Duration: 44.82 sec

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2025-10-05 03:10:16,443 - root - INFO - Epoch 91/100, Model DenseNet1D, Train
Loss: 0.0270, Train Acc: 0.9908, Val Loss: 0.0493, Val Acc: 0.9868, Epoch
Duration: 45.21 sec
2025-10-05 03:11:00,874 - root - INFO - Epoch 92/100, Model DenseNet1D, Train
Loss: 0.0272, Train Acc: 0.9908, Val Loss: 0.0617, Val Acc: 0.9830, Epoch
Duration: 44.43 sec
2025-10-05 03:11:46,072 - root - INFO - Epoch 93/100, Model DenseNet1D, Train
Loss: 0.0266, Train Acc: 0.9917, Val Loss: 0.0600, Val Acc: 0.9830, Epoch
Duration: 45.20 sec
2025-10-05 03:12:30,566 - root - INFO - Epoch 94/100, Model DenseNet1D, Train
Loss: 0.0253, Train Acc: 0.9915, Val Loss: 0.0542, Val Acc: 0.9853, Epoch
Duration: 44.49 sec
2025-10-05 03:13:14,540 - root - INFO - Epoch 95/100, Model DenseNet1D, Train
Loss: 0.0270, Train Acc: 0.9901, Val Loss: 0.0672, Val Acc: 0.9817, Epoch
Duration: 43.97 sec
2025-10-05 03:13:56,838 - root - INFO - Epoch 96/100, Model DenseNet1D, Train
Loss: 0.0240, Train Acc: 0.9914, Val Loss: 0.0629, Val Acc: 0.9818, Epoch
Duration: 42.30 sec
2025-10-05 03:14:38,489 - root - INFO - Epoch 97/100, Model DenseNet1D, Train
Loss: 0.0261, Train Acc: 0.9917, Val Loss: 0.0624, Val Acc: 0.9853, Epoch
Duration: 41.65 sec
2025-10-05 03:15:20,245 - root - INFO - Epoch 98/100, Model DenseNet1D, Train
Loss: 0.0250, Train Acc: 0.9916, Val Loss: 0.0548, Val Acc: 0.9846, Epoch
Duration: 41.75 sec
2025-10-05 03:16:01,957 - root - INFO - Epoch 99/100, Model DenseNet1D, Train
Loss: 0.0249, Train Acc: 0.9923, Val Loss: 0.0713, Val Acc: 0.9806, Epoch
Duration: 41.71 sec
2025-10-05 03:16:43,821 - root - INFO - Epoch 100/100, Model DenseNet1D, Train
Loss: 0.0238, Train Acc: 0.9920, Val Loss: 0.0646, Val Acc: 0.9830, Epoch
Duration: 41.86 sec
2025-10-05 03:16:43,821 - root - CRITICAL - Training complete. Best Val Acc:
0.9883 at epoch 81.
2025-10-05 03:16:43,831 - root - CRITICAL - SAVED BEST MODEL
/Users/sanjeev/VNIT/FINAL_PRJ_PHASE2/experiments/20251005_000538/checkpoints/bes
t_model_DenseNet1D_epoch81.pt USING LEARNING RATE 0.001
2025-10-05 03:16:43,831 - root - CRITICAL - BEST VALIDATION ACCURACY: 0.9883 at
epoch 81.
2025-10-05 03:16:43,913 - root - INFO - Saved stats plot to
/Users/sanjeev/VNIT/FINAL_PRJ_PHASE2/experiments/20251005_000538/plots/DenseNet1
D_{'csi_channels': 99, 'meta_feature_dim': 12, 'num_classes': 31, 'lr': 0.001,
'epochs': 100, 'model_name': 'DenseNet1D'}_stats.png
2025-10-05 03:16:47,670 - root - INFO - Confusion matrix:
[[276    0    0    0    0    1    0    0    0    0    0    0    0    0    0    0
   0    0    0    0    0    0    0    0    0    0    0    0]
 [  0 266    0    0    7  10    0    0    0    0    0    0    0    0    0    0
   0    0    0    0    0    0    0    0    0    0    0    0]
 [  1    0 291    0    0    0    0    0    0    0    0    0    0    0    0    0
   0    0    0    0    0    0    0    0    0    0    0    0]
 [  0    0    0 287    0    0    0    0    1    0    0    0    0    0    0    0
   0    0    0    0    0    0    0    0    0    0    0    0]
 [  0    4    2    0 265    1    0    0    0    0    0    0    0    0    0    0
   0    0    0    0    0    0    0    0    0    0    0    0]
 [  5    1    1    0    1 287    0    0    0    0    0    0    0    0    0    0
   0    0    0    0    0    0    0    0    0    0    0    0]

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[ 0 0 0 0 0 0 289 0 0 0 0 0 0 0 0 0 0 0
 0 0 0 0 0 0 0 1 0 0 0 0]
[ 0 0 1 0 0 0 0 262 0 2 0 0 0 0 0 0 0 0
 0 0 1 0 0 0 0 0 0 0 1 0]
[ 0 0 0 0 0 0 0 0 263 0 0 0 0 0 0 0 3 0
 0 0 0 0 0 1 0 0 0 0 0 0]
[ 0 0 2 0 1 0 0 4 0 265 0 0 0 0 0 0 0 0
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[ 0 0 0 0 0 0 0 0 0 0 269 0 0 0 0 0 0 0
 0 1 0 0 0 0 0 0 0 0 0 0]
[ 0 0 0 0 0 0 0 0 0 0 0 272 0 0 0 0 0 0
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[ 0 0 0 1 0 0 0 0 0 0 0 0 330 0 0 0 0 0
 1 0 0 0 0 0 0 0 0 0 0 0]
[ 0 0 0 0 0 0 0 0 0 0 0 0 0 300 1 0 0 0
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 3 0 0 0 0 0 0 0 0 0 0 0]
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 0 0 0 0 0 0 0 263 0 0 0 0]
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 0 0 1 0 0 0 0 0 285 0 1 0]
[ 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
 0 0 0 0 0 0 0 0 1 292 0 0]
[ 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
 0 0 1 3 0 0 0 0 0 0 287 3]
[ 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
 0 0 0 0 0 0 0 0 0 0 2 263]]

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2025-10-05 03:16:47,670 - root - INFO - Classification report:

|   | precision | recall | f1-score | support |
|---|-----------|--------|----------|---------|
| 1 | 0.98      | 1.00   | 0.99     | 277     |
| 2 | 0.98      | 0.94   | 0.96     | 283     |
| 3 | 0.98      | 1.00   | 0.99     | 292     |

|    |      |      |      |     |
|----|------|------|------|-----|
| 4  | 0.99 | 1.00 | 0.99 | 288 |
| 5  | 0.97 | 0.97 | 0.97 | 272 |
| 6  | 0.96 | 0.97 | 0.97 | 295 |
| 7  | 1.00 | 1.00 | 1.00 | 290 |
| 8  | 0.98 | 0.98 | 0.98 | 267 |
| 9  | 0.99 | 0.99 | 0.99 | 267 |
| 10 | 0.99 | 0.97 | 0.98 | 272 |
| 11 | 1.00 | 1.00 | 1.00 | 270 |
| 12 | 1.00 | 1.00 | 1.00 | 272 |
| 13 | 0.96 | 0.99 | 0.97 | 332 |
| 14 | 1.00 | 1.00 | 1.00 | 301 |
| 15 | 0.97 | 1.00 | 0.99 | 253 |
| 16 | 1.00 | 1.00 | 1.00 | 307 |
| 17 | 0.96 | 0.91 | 0.94 | 290 |
| 18 | 0.95 | 0.98 | 0.96 | 306 |
| 19 | 0.98 | 0.94 | 0.96 | 282 |
| 20 | 1.00 | 0.99 | 0.99 | 285 |
| 21 | 0.99 | 0.98 | 0.98 | 289 |
| 22 | 0.99 | 0.97 | 0.98 | 316 |
| 23 | 0.99 | 1.00 | 1.00 | 288 |
| 24 | 0.99 | 0.98 | 0.99 | 272 |
| 25 | 0.99 | 0.98 | 0.99 | 308 |
| 26 | 1.00 | 1.00 | 1.00 | 263 |
| 27 | 1.00 | 0.99 | 0.99 | 287 |
| 28 | 1.00 | 1.00 | 1.00 | 293 |
| 29 | 0.97 | 0.98 | 0.97 | 294 |
| 30 | 0.95 | 0.99 | 0.97 | 265 |

|              |      |      |      |      |
|--------------|------|------|------|------|
| accuracy     |      |      | 0.98 | 8576 |
| macro avg    | 0.98 | 0.98 | 0.98 | 8576 |
| weighted avg | 0.98 | 0.98 | 0.98 | 8576 |

2025-10-05 03:16:47,679 - root - CRITICAL - 99. SAVED BEST MODEL IN DenseNet1D. BEST VALIDATION ACCURACY: 0.9883 at epoch 81.

2025-10-05 03:16:47,679 - root - CRITICAL - 100. LOGGED BEST MODEL IN MLFLOW /Users/sanjeev/VNIT/FINAL\_PRJ\_PHASE2/experiments/20251005\_000538/checkpoints/best\_model\_DenseNet1D\_epoch81.pt. BEST VALIDATION ACCURACY: 0.9883 at epoch 81.

2025-10-05 03:16:47,680 - root - INFO - START RUN MLFLOW :

DenseNet1D: {'csi\_channels': 99, 'meta\_feature\_dim': 12, 'num\_classes': 31, 'lr': 0.0005, 'epochs': 100, 'model\_name': 'DenseNet1D'}

2025-10-05 03:16:47,688 - root - INFO - START T-N-E DenseNet1D

2025-10-05 03:16:47,689 - root - INFO - DenseNet1D USING LEARNING RATE 0.0005

2025-10-05 03:17:29,570 - root - INFO - Epoch 1/100, Model DenseNet1D, Train Loss: 0.7541, Train Acc: 0.7594, Val Loss: 0.3287, Val Acc: 0.8967, Epoch Duration: 41.88 sec

2025-10-05 03:18:13,041 - root - INFO - Epoch 2/100, Model DenseNet1D, Train Loss: 0.4071, Train Acc: 0.8614, Val Loss: 0.2575, Val Acc: 0.9136, Epoch Duration: 41.74 sec

2025-10-05 03:18:56,371 - root - INFO - Epoch 3/100, Model DenseNet1D, Train Loss: 0.3229, Train Acc: 0.8875, Val Loss: 0.2103, Val Acc: 0.9271, Epoch Duration: 41.84 sec

2025-10-05 03:19:39,888 - root - INFO - Epoch 4/100, Model DenseNet1D, Train Loss: 0.2817, Train Acc: 0.9027, Val Loss: 0.1810, Val Acc: 0.9411, Epoch Duration: 42.05 sec



2025-10-05 03:20:22,876 - root - INFO - Epoch 5/100, Model DenseNet1D, Train Loss: 0.2422, Train Acc: 0.9155, Val Loss: 0.1552, Val Acc: 0.9489, Epoch Duration: 41.53 sec

2025-10-05 03:21:05,911 - root - INFO - Epoch 6/100, Model DenseNet1D, Train Loss: 0.2285, Train Acc: 0.9202, Val Loss: 0.1572, Val Acc: 0.9452, Epoch Duration: 41.52 sec

2025-10-05 03:21:47,653 - root - INFO - Epoch 7/100, Model DenseNet1D, Train Loss: 0.2070, Train Acc: 0.9277, Val Loss: 0.1457, Val Acc: 0.9500, Epoch Duration: 41.74 sec

2025-10-05 03:22:31,022 - root - INFO - Epoch 8/100, Model DenseNet1D, Train Loss: 0.1892, Train Acc: 0.9330, Val Loss: 0.1154, Val Acc: 0.9573, Epoch Duration: 41.89 sec

2025-10-05 03:23:14,097 - root - INFO - Epoch 9/100, Model DenseNet1D, Train Loss: 0.1779, Train Acc: 0.9369, Val Loss: 0.1173, Val Acc: 0.9558, Epoch Duration: 41.59 sec

2025-10-05 03:23:55,848 - root - INFO - Epoch 10/100, Model DenseNet1D, Train Loss: 0.1648, Train Acc: 0.9409, Val Loss: 0.1021, Val Acc: 0.9653, Epoch Duration: 41.75 sec

2025-10-05 03:24:39,230 - root - INFO - Epoch 11/100, Model DenseNet1D, Train Loss: 0.1623, Train Acc: 0.9426, Val Loss: 0.1145, Val Acc: 0.9608, Epoch Duration: 41.90 sec

2025-10-05 03:25:20,880 - root - INFO - Epoch 12/100, Model DenseNet1D, Train Loss: 0.1478, Train Acc: 0.9470, Val Loss: 0.1038, Val Acc: 0.9634, Epoch Duration: 41.65 sec

2025-10-05 03:26:02,571 - root - INFO - Epoch 13/100, Model DenseNet1D, Train Loss: 0.1389, Train Acc: 0.9501, Val Loss: 0.1016, Val Acc: 0.9646, Epoch Duration: 41.69 sec

2025-10-05 03:26:44,608 - root - INFO - Epoch 14/100, Model DenseNet1D, Train Loss: 0.1381, Train Acc: 0.9502, Val Loss: 0.0887, Val Acc: 0.9701, Epoch Duration: 42.04 sec

2025-10-05 03:27:27,965 - root - INFO - Epoch 15/100, Model DenseNet1D, Train Loss: 0.1237, Train Acc: 0.9553, Val Loss: 0.0867, Val Acc: 0.9691, Epoch Duration: 41.87 sec

2025-10-05 03:28:09,712 - root - INFO - Epoch 16/100, Model DenseNet1D, Train Loss: 0.1232, Train Acc: 0.9567, Val Loss: 0.0864, Val Acc: 0.9700, Epoch Duration: 41.75 sec

2025-10-05 03:28:51,795 - root - INFO - Epoch 17/100, Model DenseNet1D, Train Loss: 0.1143, Train Acc: 0.9587, Val Loss: 0.0811, Val Acc: 0.9728, Epoch Duration: 42.08 sec

2025-10-05 03:29:35,226 - root - INFO - Epoch 18/100, Model DenseNet1D, Train Loss: 0.1180, Train Acc: 0.9588, Val Loss: 0.0830, Val Acc: 0.9705, Epoch Duration: 41.96 sec

2025-10-05 03:30:16,856 - root - INFO - Epoch 19/100, Model DenseNet1D, Train Loss: 0.1067, Train Acc: 0.9616, Val Loss: 0.0813, Val Acc: 0.9705, Epoch Duration: 41.63 sec

2025-10-05 03:30:58,535 - root - INFO - Epoch 20/100, Model DenseNet1D, Train Loss: 0.1089, Train Acc: 0.9602, Val Loss: 0.0884, Val Acc: 0.9684, Epoch Duration: 41.68 sec

2025-10-05 03:31:40,508 - root - INFO - Epoch 21/100, Model DenseNet1D, Train Loss: 0.0970, Train Acc: 0.9653, Val Loss: 0.0766, Val Acc: 0.9729, Epoch Duration: 41.97 sec

2025-10-05 03:32:23,738 - root - INFO - Epoch 22/100, Model DenseNet1D, Train Loss: 0.0966, Train Acc: 0.9659, Val Loss: 0.0779, Val Acc: 0.9728, Epoch Duration: 41.73 sec

2025-10-05 03:33:05,462 - root - INFO - Epoch 23/100, Model DenseNet1D, Train Loss: 0.0973, Train Acc: 0.9654, Val Loss: 0.0794, Val Acc: 0.9689, Epoch Duration: 41.72 sec

2025-10-05 03:33:47,369 - root - INFO - Epoch 24/100, Model DenseNet1D, Train Loss: 0.0956, Train Acc: 0.9666, Val Loss: 0.0707, Val Acc: 0.9748, Epoch Duration: 41.91 sec

2025-10-05 03:34:30,790 - root - INFO - Epoch 25/100, Model DenseNet1D, Train Loss: 0.0917, Train Acc: 0.9664, Val Loss: 0.0654, Val Acc: 0.9791, Epoch Duration: 41.93 sec

2025-10-05 03:35:14,138 - root - INFO - Epoch 26/100, Model DenseNet1D, Train Loss: 0.0896, Train Acc: 0.9687, Val Loss: 0.0805, Val Acc: 0.9707, Epoch Duration: 41.85 sec

2025-10-05 03:35:56,482 - root - INFO - Epoch 27/100, Model DenseNet1D, Train Loss: 0.0801, Train Acc: 0.9715, Val Loss: 0.0599, Val Acc: 0.9782, Epoch Duration: 42.34 sec

2025-10-05 03:36:38,665 - root - INFO - Epoch 28/100, Model DenseNet1D, Train Loss: 0.0857, Train Acc: 0.9696, Val Loss: 0.0547, Val Acc: 0.9820, Epoch Duration: 42.18 sec

2025-10-05 03:37:22,062 - root - INFO - Epoch 29/100, Model DenseNet1D, Train Loss: 0.0819, Train Acc: 0.9698, Val Loss: 0.0637, Val Acc: 0.9785, Epoch Duration: 41.92 sec

2025-10-05 03:38:04,065 - root - INFO - Epoch 30/100, Model DenseNet1D, Train Loss: 0.0812, Train Acc: 0.9713, Val Loss: 0.0683, Val Acc: 0.9762, Epoch Duration: 42.00 sec

2025-10-05 03:38:45,890 - root - INFO - Epoch 31/100, Model DenseNet1D, Train Loss: 0.0769, Train Acc: 0.9718, Val Loss: 0.0560, Val Acc: 0.9815, Epoch Duration: 41.82 sec

2025-10-05 03:39:27,640 - root - INFO - Epoch 32/100, Model DenseNet1D, Train Loss: 0.0722, Train Acc: 0.9742, Val Loss: 0.0546, Val Acc: 0.9810, Epoch Duration: 41.75 sec

2025-10-05 03:40:09,500 - root - INFO - Epoch 33/100, Model DenseNet1D, Train Loss: 0.0765, Train Acc: 0.9722, Val Loss: 0.0541, Val Acc: 0.9817, Epoch Duration: 41.86 sec

2025-10-05 03:40:51,416 - root - INFO - Epoch 34/100, Model DenseNet1D, Train Loss: 0.0705, Train Acc: 0.9758, Val Loss: 0.0616, Val Acc: 0.9783, Epoch Duration: 41.91 sec

2025-10-05 03:41:33,455 - root - INFO - Epoch 35/100, Model DenseNet1D, Train Loss: 0.0688, Train Acc: 0.9743, Val Loss: 0.0472, Val Acc: 0.9839, Epoch Duration: 42.04 sec

2025-10-05 03:42:16,787 - root - INFO - Epoch 36/100, Model DenseNet1D, Train Loss: 0.0673, Train Acc: 0.9756, Val Loss: 0.0610, Val Acc: 0.9804, Epoch Duration: 41.83 sec

2025-10-05 03:42:58,655 - root - INFO - Epoch 37/100, Model DenseNet1D, Train Loss: 0.0681, Train Acc: 0.9751, Val Loss: 0.0535, Val Acc: 0.9805, Epoch Duration: 41.87 sec

2025-10-05 03:43:40,739 - root - INFO - Epoch 38/100, Model DenseNet1D, Train Loss: 0.0667, Train Acc: 0.9769, Val Loss: 0.0558, Val Acc: 0.9816, Epoch Duration: 42.08 sec

2025-10-05 03:44:22,518 - root - INFO - Epoch 39/100, Model DenseNet1D, Train Loss: 0.0634, Train Acc: 0.9769, Val Loss: 0.0490, Val Acc: 0.9831, Epoch Duration: 41.78 sec

2025-10-05 03:45:04,442 - root - INFO - Epoch 40/100, Model DenseNet1D, Train Loss: 0.0646, Train Acc: 0.9779, Val Loss: 0.0553, Val Acc: 0.9812, Epoch Duration: 41.92 sec

2025-10-05 03:45:46,639 - root - INFO - Epoch 41/100, Model DenseNet1D, Train Loss: 0.0614, Train Acc: 0.9782, Val Loss: 0.0495, Val Acc: 0.9823, Epoch Duration: 42.20 sec

2025-10-05 03:46:28,709 - root - INFO - Epoch 42/100, Model DenseNet1D, Train Loss: 0.0596, Train Acc: 0.9788, Val Loss: 0.0534, Val Acc: 0.9832, Epoch Duration: 42.07 sec

2025-10-05 03:47:10,625 - root - INFO - Epoch 43/100, Model DenseNet1D, Train Loss: 0.0570, Train Acc: 0.9796, Val Loss: 0.0502, Val Acc: 0.9855, Epoch Duration: 41.91 sec

2025-10-05 03:47:53,877 - root - INFO - Epoch 44/100, Model DenseNet1D, Train Loss: 0.0554, Train Acc: 0.9798, Val Loss: 0.0418, Val Acc: 0.9871, Epoch Duration: 41.76 sec

2025-10-05 03:48:37,364 - root - INFO - Epoch 45/100, Model DenseNet1D, Train Loss: 0.0573, Train Acc: 0.9794, Val Loss: 0.0695, Val Acc: 0.9766, Epoch Duration: 41.99 sec

2025-10-05 03:49:19,052 - root - INFO - Epoch 46/100, Model DenseNet1D, Train Loss: 0.0506, Train Acc: 0.9812, Val Loss: 0.0575, Val Acc: 0.9839, Epoch Duration: 41.69 sec

2025-10-05 03:50:00,909 - root - INFO - Epoch 47/100, Model DenseNet1D, Train Loss: 0.0535, Train Acc: 0.9807, Val Loss: 0.0407, Val Acc: 0.9868, Epoch Duration: 41.86 sec

2025-10-05 03:50:42,887 - root - INFO - Epoch 48/100, Model DenseNet1D, Train Loss: 0.0564, Train Acc: 0.9802, Val Loss: 0.0579, Val Acc: 0.9809, Epoch Duration: 41.98 sec

2025-10-05 03:51:24,716 - root - INFO - Epoch 49/100, Model DenseNet1D, Train Loss: 0.0487, Train Acc: 0.9820, Val Loss: 0.0496, Val Acc: 0.9843, Epoch Duration: 41.83 sec

2025-10-05 03:52:06,490 - root - INFO - Epoch 50/100, Model DenseNet1D, Train Loss: 0.0515, Train Acc: 0.9825, Val Loss: 0.0508, Val Acc: 0.9838, Epoch Duration: 41.77 sec

2025-10-05 03:52:48,324 - root - INFO - Epoch 51/100, Model DenseNet1D, Train Loss: 0.0463, Train Acc: 0.9823, Val Loss: 0.0523, Val Acc: 0.9820, Epoch Duration: 41.83 sec

2025-10-05 03:53:30,218 - root - INFO - Epoch 52/100, Model DenseNet1D, Train Loss: 0.0484, Train Acc: 0.9834, Val Loss: 0.0391, Val Acc: 0.9876, Epoch Duration: 41.89 sec

2025-10-05 03:54:13,698 - root - INFO - Epoch 53/100, Model DenseNet1D, Train Loss: 0.0503, Train Acc: 0.9811, Val Loss: 0.0480, Val Acc: 0.9850, Epoch Duration: 41.97 sec

2025-10-05 03:54:55,714 - root - INFO - Epoch 54/100, Model DenseNet1D, Train Loss: 0.0469, Train Acc: 0.9837, Val Loss: 0.0484, Val Acc: 0.9852, Epoch Duration: 42.01 sec

2025-10-05 03:55:37,987 - root - INFO - Epoch 55/100, Model DenseNet1D, Train Loss: 0.0499, Train Acc: 0.9821, Val Loss: 0.0398, Val Acc: 0.9879, Epoch Duration: 42.27 sec

2025-10-05 03:56:21,506 - root - INFO - Epoch 56/100, Model DenseNet1D, Train Loss: 0.0433, Train Acc: 0.9848, Val Loss: 0.0447, Val Acc: 0.9847, Epoch Duration: 42.00 sec

2025-10-05 03:57:03,384 - root - INFO - Epoch 57/100, Model DenseNet1D, Train Loss: 0.0492, Train Acc: 0.9826, Val Loss: 0.0455, Val Acc: 0.9861, Epoch Duration: 41.88 sec

2025-10-05 03:57:45,227 - root - INFO - Epoch 58/100, Model DenseNet1D, Train Loss: 0.0418, Train Acc: 0.9845, Val Loss: 0.0364, Val Acc: 0.9889, Epoch Duration: 41.84 sec

2025-10-05 03:58:28,694 - root - INFO - Epoch 59/100, Model DenseNet1D, Train Loss: 0.0460, Train Acc: 0.9844, Val Loss: 0.0348, Val Acc: 0.9893, Epoch Duration: 41.94 sec

2025-10-05 03:59:12,002 - root - INFO - Epoch 60/100, Model DenseNet1D, Train Loss: 0.0399, Train Acc: 0.9855, Val Loss: 0.0525, Val Acc: 0.9840, Epoch Duration: 41.70 sec

2025-10-05 03:59:53,955 - root - INFO - Epoch 61/100, Model DenseNet1D, Train Loss: 0.0441, Train Acc: 0.9845, Val Loss: 0.0453, Val Acc: 0.9854, Epoch Duration: 41.95 sec

2025-10-05 04:00:35,942 - root - INFO - Epoch 62/100, Model DenseNet1D, Train Loss: 0.0429, Train Acc: 0.9854, Val Loss: 0.0429, Val Acc: 0.9864, Epoch Duration: 41.99 sec

2025-10-05 04:01:17,780 - root - INFO - Epoch 63/100, Model DenseNet1D, Train Loss: 0.0441, Train Acc: 0.9845, Val Loss: 0.0572, Val Acc: 0.9840, Epoch Duration: 41.84 sec

2025-10-05 04:01:59,818 - root - INFO - Epoch 64/100, Model DenseNet1D, Train Loss: 0.0380, Train Acc: 0.9862, Val Loss: 0.0444, Val Acc: 0.9868, Epoch Duration: 42.04 sec

2025-10-05 04:02:41,850 - root - INFO - Epoch 65/100, Model DenseNet1D, Train Loss: 0.0408, Train Acc: 0.9848, Val Loss: 0.0387, Val Acc: 0.9883, Epoch Duration: 42.03 sec

2025-10-05 04:03:23,591 - root - INFO - Epoch 66/100, Model DenseNet1D, Train Loss: 0.0387, Train Acc: 0.9858, Val Loss: 0.0418, Val Acc: 0.9885, Epoch Duration: 41.74 sec

2025-10-05 04:04:05,557 - root - INFO - Epoch 67/100, Model DenseNet1D, Train Loss: 0.0399, Train Acc: 0.9864, Val Loss: 0.0351, Val Acc: 0.9888, Epoch Duration: 41.97 sec

2025-10-05 04:04:47,702 - root - INFO - Epoch 68/100, Model DenseNet1D, Train Loss: 0.0381, Train Acc: 0.9871, Val Loss: 0.0383, Val Acc: 0.9876, Epoch Duration: 42.14 sec

2025-10-05 04:05:29,700 - root - INFO - Epoch 69/100, Model DenseNet1D, Train Loss: 0.0363, Train Acc: 0.9872, Val Loss: 0.0363, Val Acc: 0.9889, Epoch Duration: 42.00 sec

2025-10-05 04:06:11,331 - root - INFO - Epoch 70/100, Model DenseNet1D, Train Loss: 0.0385, Train Acc: 0.9865, Val Loss: 0.0374, Val Acc: 0.9878, Epoch Duration: 41.63 sec

2025-10-05 04:06:53,141 - root - INFO - Epoch 71/100, Model DenseNet1D, Train Loss: 0.0366, Train Acc: 0.9872, Val Loss: 0.0350, Val Acc: 0.9892, Epoch Duration: 41.81 sec

2025-10-05 04:07:35,003 - root - INFO - Epoch 72/100, Model DenseNet1D, Train Loss: 0.0337, Train Acc: 0.9883, Val Loss: 0.0438, Val Acc: 0.9872, Epoch Duration: 41.86 sec

2025-10-05 04:08:16,615 - root - INFO - Epoch 73/100, Model DenseNet1D, Train Loss: 0.0340, Train Acc: 0.9886, Val Loss: 0.0414, Val Acc: 0.9876, Epoch Duration: 41.61 sec

2025-10-05 04:08:58,430 - root - INFO - Epoch 74/100, Model DenseNet1D, Train Loss: 0.0338, Train Acc: 0.9882, Val Loss: 0.0418, Val Acc: 0.9895, Epoch Duration: 41.81 sec

2025-10-05 04:09:42,640 - root - INFO - Epoch 75/100, Model DenseNet1D, Train Loss: 0.0394, Train Acc: 0.9859, Val Loss: 0.0568, Val Acc: 0.9816, Epoch Duration: 42.47 sec

2025-10-05 04:10:24,416 - root - INFO - Epoch 76/100, Model DenseNet1D, Train Loss: 0.0316, Train Acc: 0.9894, Val Loss: 0.0521, Val Acc: 0.9850, Epoch Duration: 41.78 sec

2025-10-05 04:11:06,264 - root - INFO - Epoch 77/100, Model DenseNet1D, Train Loss: 0.0398, Train Acc: 0.9863, Val Loss: 0.0466, Val Acc: 0.9858, Epoch Duration: 41.85 sec

2025-10-05 04:11:48,280 - root - INFO - Epoch 78/100, Model DenseNet1D, Train Loss: 0.0342, Train Acc: 0.9880, Val Loss: 0.0409, Val Acc: 0.9887, Epoch Duration: 42.02 sec

2025-10-05 04:12:30,369 - root - INFO - Epoch 79/100, Model DenseNet1D, Train Loss: 0.0328, Train Acc: 0.9887, Val Loss: 0.0493, Val Acc: 0.9851, Epoch Duration: 42.09 sec

2025-10-05 04:13:12,215 - root - INFO - Epoch 80/100, Model DenseNet1D, Train Loss: 0.0349, Train Acc: 0.9879, Val Loss: 0.0472, Val Acc: 0.9851, Epoch Duration: 41.84 sec

2025-10-05 04:13:54,360 - root - INFO - Epoch 81/100, Model DenseNet1D, Train Loss: 0.0319, Train Acc: 0.9887, Val Loss: 0.0411, Val Acc: 0.9887, Epoch Duration: 42.14 sec

2025-10-05 04:14:36,387 - root - INFO - Epoch 82/100, Model DenseNet1D, Train Loss: 0.0327, Train Acc: 0.9888, Val Loss: 0.0396, Val Acc: 0.9887, Epoch Duration: 42.03 sec

2025-10-05 04:15:18,008 - root - INFO - Epoch 83/100, Model DenseNet1D, Train Loss: 0.0318, Train Acc: 0.9888, Val Loss: 0.0559, Val Acc: 0.9841, Epoch Duration: 41.62 sec

2025-10-05 04:15:59,728 - root - INFO - Epoch 84/100, Model DenseNet1D, Train Loss: 0.0302, Train Acc: 0.9893, Val Loss: 0.0540, Val Acc: 0.9859, Epoch Duration: 41.72 sec

2025-10-05 04:16:41,622 - root - INFO - Epoch 85/100, Model DenseNet1D, Train Loss: 0.0362, Train Acc: 0.9871, Val Loss: 0.0439, Val Acc: 0.9871, Epoch Duration: 41.89 sec

2025-10-05 04:17:23,426 - root - INFO - Epoch 86/100, Model DenseNet1D, Train Loss: 0.0267, Train Acc: 0.9907, Val Loss: 0.0434, Val Acc: 0.9886, Epoch Duration: 41.80 sec

2025-10-05 04:18:05,250 - root - INFO - Epoch 87/100, Model DenseNet1D, Train Loss: 0.0313, Train Acc: 0.9891, Val Loss: 0.0367, Val Acc: 0.9907, Epoch Duration: 41.82 sec

2025-10-05 04:18:48,984 - root - INFO - Epoch 88/100, Model DenseNet1D, Train Loss: 0.0291, Train Acc: 0.9894, Val Loss: 0.0505, Val Acc: 0.9855, Epoch Duration: 41.94 sec

2025-10-05 04:19:30,892 - root - INFO - Epoch 89/100, Model DenseNet1D, Train Loss: 0.0318, Train Acc: 0.9889, Val Loss: 0.0393, Val Acc: 0.9887, Epoch Duration: 41.91 sec

2025-10-05 04:20:12,593 - root - INFO - Epoch 90/100, Model DenseNet1D, Train Loss: 0.0256, Train Acc: 0.9906, Val Loss: 0.0475, Val Acc: 0.9873, Epoch Duration: 41.70 sec

2025-10-05 04:20:54,559 - root - INFO - Epoch 91/100, Model DenseNet1D, Train Loss: 0.0328, Train Acc: 0.9890, Val Loss: 0.0435, Val Acc: 0.9876, Epoch Duration: 41.96 sec

2025-10-05 04:21:36,688 - root - INFO - Epoch 92/100, Model DenseNet1D, Train Loss: 0.0285, Train Acc: 0.9901, Val Loss: 0.0554, Val Acc: 0.9827, Epoch Duration: 42.13 sec

2025-10-05 04:22:18,684 - root - INFO - Epoch 93/100, Model DenseNet1D, Train Loss: 0.0255, Train Acc: 0.9911, Val Loss: 0.0371, Val Acc: 0.9899, Epoch Duration: 42.00 sec

2025-10-05 04:23:00,783 - root - INFO - Epoch 94/100, Model DenseNet1D, Train Loss: 0.0306, Train Acc: 0.9892, Val Loss: 0.0396, Val Acc: 0.9882, Epoch Duration: 42.10 sec

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2025-10-05 04:23:42,836 - root - INFO - Epoch 95/100, Model DenseNet1D, Train
Loss: 0.0254, Train Acc: 0.9911, Val Loss: 0.0534, Val Acc: 0.9844, Epoch
Duration: 42.05 sec
2025-10-05 04:24:24,485 - root - INFO - Epoch 96/100, Model DenseNet1D, Train
Loss: 0.0277, Train Acc: 0.9911, Val Loss: 0.0375, Val Acc: 0.9907, Epoch
Duration: 41.65 sec
2025-10-05 04:25:06,199 - root - INFO - Epoch 97/100, Model DenseNet1D, Train
Loss: 0.0280, Train Acc: 0.9903, Val Loss: 0.0385, Val Acc: 0.9893, Epoch
Duration: 41.71 sec
2025-10-05 04:25:48,018 - root - INFO - Epoch 98/100, Model DenseNet1D, Train
Loss: 0.0257, Train Acc: 0.9914, Val Loss: 0.0496, Val Acc: 0.9864, Epoch
Duration: 41.82 sec
2025-10-05 04:26:29,986 - root - INFO - Epoch 99/100, Model DenseNet1D, Train
Loss: 0.0242, Train Acc: 0.9917, Val Loss: 0.0648, Val Acc: 0.9829, Epoch
Duration: 41.97 sec
2025-10-05 04:27:11,874 - root - INFO - Epoch 100/100, Model DenseNet1D, Train
Loss: 0.0282, Train Acc: 0.9903, Val Loss: 0.0395, Val Acc: 0.9889, Epoch
Duration: 41.89 sec
2025-10-05 04:27:11,874 - root - CRITICAL - Training complete. Best Val Acc:
0.9907 at epoch 87.
2025-10-05 04:27:11,886 - root - CRITICAL - SAVED BEST MODEL
/Users/sanjeev/VNIT/FINAL_PRJ_PHASE2/experiments/20251005_000538/checkpoints/bes
t_model_DenseNet1D_epoch87.pt USING LEARNING RATE 0.0005
2025-10-05 04:27:11,886 - root - CRITICAL - BEST VALIDATION ACCURACY: 0.9907 at
epoch 87.
2025-10-05 04:27:11,967 - root - INFO - Saved stats plot to
/Users/sanjeev/VNIT/FINAL_PRJ_PHASE2/experiments/20251005_000538/plots/DenseNet1
D_{'csi_channels': 99, 'meta_feature_dim': 12, 'num_classes': 31, 'lr': 0.0005,
'epochs': 100, 'model_name': 'DenseNet1D'}_stats.png
2025-10-05 04:27:15,537 - root - INFO - Confusion matrix:
[[277  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
  0  0  0  0  0  0  0  0  0  0  0  0  0]
 [  0 261  0  0 16  6  0  0  0  0  0  0  0  0  0  0  0  0
  0  0  0  0  0  0  0  0  0  0  0  0  0]
 [  0  0 292  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
  0  0  0  0  0  0  0  0  0  0  0  0]
 [  0  0  0 288  0  0  0  0  0  0  0  0  0  0  0  0  0  0
  0  0  0  0  0  0  0  0  0  0  0  0]
 [  0  2  1  0 269  0  0  0  0  0  0  0  0  0  0  0  0  0
  0  0  0  0  0  0  0  0  0  0  0  0]
 [  1  0  1  0  0 293  0  0  0  0  0  0  0  0  0  0  0  0
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 [  0  0  0  0  0  0 289  0  0  0  0  0  0  0  0  0  0  0
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[ 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
 0 0 4 1 0 0 0 0 0 0 0 260]]

```

2025-10-05 04:27:15,537 - root - INFO - Classification report:

|    | precision | recall | f1-score | support |
|----|-----------|--------|----------|---------|
| 1  | 0.99      | 1.00   | 1.00     | 277     |
| 2  | 0.99      | 0.92   | 0.96     | 283     |
| 3  | 0.99      | 1.00   | 1.00     | 292     |
| 4  | 1.00      | 1.00   | 1.00     | 288     |
| 5  | 0.94      | 0.99   | 0.96     | 272     |
| 6  | 0.98      | 0.99   | 0.99     | 295     |
| 7  | 1.00      | 1.00   | 1.00     | 290     |
| 8  | 0.99      | 1.00   | 0.99     | 267     |
| 9  | 1.00      | 0.99   | 1.00     | 267     |
| 10 | 0.99      | 0.99   | 0.99     | 272     |
| 11 | 1.00      | 0.99   | 1.00     | 270     |
| 12 | 1.00      | 1.00   | 1.00     | 272     |
| 13 | 0.99      | 0.95   | 0.97     | 332     |
| 14 | 1.00      | 1.00   | 1.00     | 301     |
| 15 | 1.00      | 0.99   | 0.99     | 253     |

|              |      |      |      |           |
|--------------|------|------|------|-----------|
| 16           | 0.99 | 1.00 | 1.00 | 307       |
| 17           | 0.99 | 0.98 | 0.98 | 290       |
| 18           | 0.98 | 1.00 | 0.99 | 306       |
| 19           | 0.95 | 0.99 | 0.97 | 282       |
| 20           | 0.99 | 1.00 | 0.99 | 285       |
| 21           | 0.97 | 1.00 | 0.98 | 289       |
| 22           | 0.99 | 0.97 | 0.98 | 316       |
| 23           | 0.98 | 1.00 | 0.99 | 288       |
| 24           | 1.00 | 1.00 | 1.00 | 272       |
| 25           | 1.00 | 0.98 | 0.99 | 308       |
| 26           | 1.00 | 1.00 | 1.00 | 263       |
| 27           | 1.00 | 1.00 | 1.00 | 287       |
| 28           | 1.00 | 1.00 | 1.00 | 293       |
| 29           | 1.00 | 0.97 | 0.98 | 294       |
| 30           | 0.97 | 0.98 | 0.98 | 265       |
| accuracy     |      |      |      | 0.99 8576 |
| macro avg    | 0.99 | 0.99 | 0.99 | 8576      |
| weighted avg | 0.99 | 0.99 | 0.99 | 8576      |

2025-10-05 04:27:15,547 - root - CRITICAL - 99. SAVED BEST MODEL IN DenseNet1D. BEST VALIDATION ACCURACY: 0.9907 at epoch 87.

2025-10-05 04:27:15,547 - root - CRITICAL - 100. LOGGED BEST MODEL IN MLFLOW /Users/sanjeev/VNIT/FINAL\_PRJ\_PHASE2/experiments/20251005\_000538/checkpoints/best\_model\_DenseNet1D\_epoch87.pt. BEST VALIDATION ACCURACY: 0.9907 at epoch 87.

2025-10-05 04:27:15,549 - root - INFO -

2025-10-05 04:27:15,549 - root - INFO - N

2025-10-05 04:27:15,549 - root - INFO - E

2025-10-05 04:27:15,549 - root - INFO - W

2025-10-05 04:27:15,549 - root - INFO - STARTING TRAINING FLOW :

EfficientNet1DLSTM:('in\_channels', 'meta\_seq\_len', 'meta\_feature\_dim', 'num\_classes', 'lr', 'epochs'):([99], [128], [12], [31], [0.001, 0.0005], [100])

2025-10-05 04:27:15,553 - root - INFO - START RUN MLFLOW :

EfficientNet1DLSTM:{'in\_channels': 99, 'meta\_seq\_len': 128, 'meta\_feature\_dim': 12, 'num\_classes': 31, 'lr': 0.001, 'epochs': 100, 'model\_name': 'EfficientNet1DLSTM'}

2025-10-05 04:27:15,563 - root - INFO - START T-N-E EfficientNet1DLSTM

2025-10-05 04:27:15,563 - root - INFO - EfficientNet1DLSTM USING LEARNING RATE 0.001

2025-10-05 04:28:13,742 - root - WARNING - ROC-AUC subset computation failed: Input contains NaN.

2025-10-05 04:28:13,745 - root - INFO - Epoch 1/100, Model EfficientNet1DLSTM, Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch Duration: 58.18 sec

2025-10-05 04:29:09,685 - root - WARNING - ROC-AUC subset computation failed: Input contains NaN.

2025-10-05 04:29:09,687 - root - INFO - Epoch 2/100, Model EfficientNet1DLSTM, Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch Duration: 55.94 sec

2025-10-05 04:30:05,812 - root - WARNING - ROC-AUC subset computation failed: Input contains NaN.

2025-10-05 04:30:05,814 - root - INFO - Epoch 3/100, Model EfficientNet1DLSTM, Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch Duration: 56.13 sec



2025-10-05 04:31:01,841 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.

2025-10-05 04:31:01,843 - root - INFO - Epoch 4/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 56.03 sec

2025-10-05 04:31:57,647 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.

2025-10-05 04:31:57,650 - root - INFO - Epoch 5/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 55.80 sec

2025-10-05 04:32:53,580 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.

2025-10-05 04:32:53,582 - root - INFO - Epoch 6/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 55.93 sec

2025-10-05 04:33:49,650 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.

2025-10-05 04:33:49,652 - root - INFO - Epoch 7/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 56.07 sec

2025-10-05 04:34:45,856 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.

2025-10-05 04:34:45,859 - root - INFO - Epoch 8/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 56.21 sec

2025-10-05 04:35:41,980 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.

2025-10-05 04:35:41,982 - root - INFO - Epoch 9/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 56.12 sec

2025-10-05 04:36:39,117 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.

2025-10-05 04:36:39,119 - root - INFO - Epoch 10/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 57.13 sec

2025-10-05 04:37:36,434 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.

2025-10-05 04:37:36,436 - root - INFO - Epoch 11/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 57.32 sec

2025-10-05 04:38:33,759 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.

2025-10-05 04:38:33,760 - root - INFO - Epoch 12/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 57.32 sec

2025-10-05 04:39:30,588 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.

2025-10-05 04:39:30,591 - root - INFO - Epoch 13/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 56.83 sec

2025-10-05 04:40:26,655 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.

2025-10-05 04:40:26,658 - root - INFO - Epoch 14/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch

Duration: 56.07 sec  
2025-10-05 04:41:23,722 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 04:41:23,725 - root - INFO - Epoch 15/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 57.06 sec  
2025-10-05 04:42:21,286 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 04:42:21,288 - root - INFO - Epoch 16/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 57.56 sec  
2025-10-05 04:43:18,854 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 04:43:18,856 - root - INFO - Epoch 17/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 57.57 sec  
2025-10-05 04:44:15,703 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 04:44:15,704 - root - INFO - Epoch 18/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 56.85 sec  
2025-10-05 04:45:11,661 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 04:45:11,663 - root - INFO - Epoch 19/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 55.96 sec  
2025-10-05 04:46:07,941 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 04:46:07,943 - root - INFO - Epoch 20/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 56.28 sec  
2025-10-05 04:47:04,304 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 04:47:04,306 - root - INFO - Epoch 21/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 56.36 sec  
2025-10-05 04:48:00,611 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 04:48:00,613 - root - INFO - Epoch 22/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 56.31 sec  
2025-10-05 04:48:56,570 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 04:48:56,572 - root - INFO - Epoch 23/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 55.96 sec  
2025-10-05 04:49:52,458 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 04:49:52,460 - root - INFO - Epoch 24/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 55.89 sec  
2025-10-05 04:50:48,536 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 04:50:48,538 - root - INFO - Epoch 25/100, Model EfficientNet1DLSTM,

Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 56.08 sec  
2025-10-05 04:51:44,698 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 04:51:44,700 - root - INFO - Epoch 26/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 56.16 sec  
2025-10-05 04:52:40,749 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 04:52:40,751 - root - INFO - Epoch 27/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 56.05 sec  
2025-10-05 04:53:36,562 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 04:53:36,564 - root - INFO - Epoch 28/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 55.81 sec  
2025-10-05 04:54:32,418 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 04:54:32,420 - root - INFO - Epoch 29/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 55.86 sec  
2025-10-05 04:55:28,557 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 04:55:28,560 - root - INFO - Epoch 30/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 56.14 sec  
2025-10-05 04:56:25,027 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 04:56:25,029 - root - INFO - Epoch 31/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 56.47 sec  
2025-10-05 04:57:20,656 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 04:57:20,658 - root - INFO - Epoch 32/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 55.63 sec  
2025-10-05 04:58:16,386 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 04:58:16,388 - root - INFO - Epoch 33/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 55.73 sec  
2025-10-05 04:59:12,454 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 04:59:12,456 - root - INFO - Epoch 34/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 56.07 sec  
2025-10-05 05:00:08,468 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 05:00:08,470 - root - INFO - Epoch 35/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 56.01 sec  
2025-10-05 05:01:04,383 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.

2025-10-05 05:01:04,385 - root - INFO - Epoch 36/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 55.91 sec  
2025-10-05 05:02:00,058 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 05:02:00,060 - root - INFO - Epoch 37/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 55.67 sec  
2025-10-05 05:02:55,872 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 05:02:55,874 - root - INFO - Epoch 38/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 55.81 sec  
2025-10-05 05:03:51,732 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 05:03:51,734 - root - INFO - Epoch 39/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 55.86 sec  
2025-10-05 05:04:48,115 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 05:04:48,118 - root - INFO - Epoch 40/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 56.38 sec  
2025-10-05 05:05:44,333 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 05:05:44,335 - root - INFO - Epoch 41/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 56.22 sec  
2025-10-05 05:06:40,787 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 05:06:40,788 - root - INFO - Epoch 42/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 56.45 sec  
2025-10-05 05:07:37,053 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 05:07:37,055 - root - INFO - Epoch 43/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 56.27 sec  
2025-10-05 05:08:33,721 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 05:08:33,723 - root - INFO - Epoch 44/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 56.67 sec  
2025-10-05 05:09:30,320 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 05:09:30,322 - root - INFO - Epoch 45/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 56.60 sec  
2025-10-05 05:10:26,606 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 05:10:26,608 - root - INFO - Epoch 46/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 56.28 sec  
2025-10-05 05:11:22,571 - root - WARNING - ROC-AUC subset computation failed:

Input contains NaN.  
2025-10-05 05:11:22,573 - root - INFO - Epoch 47/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 55.96 sec  
2025-10-05 05:12:18,694 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 05:12:18,695 - root - INFO - Epoch 48/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 56.12 sec  
2025-10-05 05:13:15,186 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 05:13:15,188 - root - INFO - Epoch 49/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 56.49 sec  
2025-10-05 05:14:11,137 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 05:14:11,139 - root - INFO - Epoch 50/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 55.95 sec  
2025-10-05 05:15:07,223 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 05:15:07,224 - root - INFO - Epoch 51/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 56.08 sec  
2025-10-05 05:16:03,551 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 05:16:03,553 - root - INFO - Epoch 52/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 56.33 sec  
2025-10-05 05:16:59,843 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 05:16:59,844 - root - INFO - Epoch 53/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 56.29 sec  
2025-10-05 05:17:56,031 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 05:17:56,032 - root - INFO - Epoch 54/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 56.19 sec  
2025-10-05 05:18:51,877 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 05:18:51,878 - root - INFO - Epoch 55/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 55.84 sec  
2025-10-05 05:19:48,018 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 05:19:48,020 - root - INFO - Epoch 56/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 56.14 sec  
2025-10-05 05:20:44,258 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 05:20:44,261 - root - INFO - Epoch 57/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 56.24 sec

2025-10-05 05:21:40,716 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.

2025-10-05 05:21:40,718 - root - INFO - Epoch 58/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 56.46 sec

2025-10-05 05:22:36,821 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.

2025-10-05 05:22:36,822 - root - INFO - Epoch 59/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 56.10 sec

2025-10-05 05:23:32,907 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.

2025-10-05 05:23:32,909 - root - INFO - Epoch 60/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 56.09 sec

2025-10-05 05:24:28,939 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.

2025-10-05 05:24:28,940 - root - INFO - Epoch 61/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 56.03 sec

2025-10-05 05:25:25,171 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.

2025-10-05 05:25:25,173 - root - INFO - Epoch 62/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 56.23 sec

2025-10-05 05:26:21,650 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.

2025-10-05 05:26:21,652 - root - INFO - Epoch 63/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 56.48 sec

2025-10-05 05:27:17,748 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.

2025-10-05 05:27:17,750 - root - INFO - Epoch 64/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 56.10 sec

2025-10-05 05:28:13,934 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.

2025-10-05 05:28:13,936 - root - INFO - Epoch 65/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 56.18 sec

2025-10-05 05:29:10,303 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.

2025-10-05 05:29:10,305 - root - INFO - Epoch 66/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 56.37 sec

2025-10-05 05:30:06,932 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.

2025-10-05 05:30:06,934 - root - INFO - Epoch 67/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 56.63 sec

2025-10-05 05:31:03,406 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.

2025-10-05 05:31:03,409 - root - INFO - Epoch 68/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch

Duration: 56.47 sec  
2025-10-05 05:31:59,801 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 05:31:59,803 - root - INFO - Epoch 69/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 56.39 sec  
2025-10-05 05:32:56,417 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 05:32:56,419 - root - INFO - Epoch 70/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 56.61 sec  
2025-10-05 05:33:53,084 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 05:33:53,087 - root - INFO - Epoch 71/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 56.67 sec  
2025-10-05 05:34:49,780 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 05:34:49,783 - root - INFO - Epoch 72/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 56.69 sec  
2025-10-05 05:35:46,063 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 05:35:46,065 - root - INFO - Epoch 73/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 56.28 sec  
2025-10-05 05:36:42,424 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 05:36:42,427 - root - INFO - Epoch 74/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 56.36 sec  
2025-10-05 05:37:38,721 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 05:37:38,724 - root - INFO - Epoch 75/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 56.29 sec  
2025-10-05 05:38:35,302 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 05:38:35,304 - root - INFO - Epoch 76/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 56.58 sec  
2025-10-05 05:39:32,140 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 05:39:32,143 - root - INFO - Epoch 77/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 56.84 sec  
2025-10-05 05:40:28,360 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 05:40:28,362 - root - INFO - Epoch 78/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 56.22 sec  
2025-10-05 05:41:24,679 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 05:41:24,682 - root - INFO - Epoch 79/100, Model EfficientNet1DLSTM,

Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 56.32 sec  
2025-10-05 05:42:21,090 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 05:42:21,093 - root - INFO - Epoch 80/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 56.41 sec  
2025-10-05 05:43:17,775 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 05:43:17,777 - root - INFO - Epoch 81/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 56.68 sec  
2025-10-05 05:44:13,992 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 05:44:13,995 - root - INFO - Epoch 82/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 56.22 sec  
2025-10-05 05:45:11,284 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 05:45:11,287 - root - INFO - Epoch 83/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 57.29 sec  
2025-10-05 05:46:08,725 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 05:46:08,726 - root - INFO - Epoch 84/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 57.44 sec  
2025-10-05 05:47:05,774 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 05:47:05,776 - root - INFO - Epoch 85/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 57.05 sec  
2025-10-05 05:48:01,859 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 05:48:01,860 - root - INFO - Epoch 86/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 56.08 sec  
2025-10-05 05:48:57,821 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 05:48:57,823 - root - INFO - Epoch 87/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 55.96 sec  
2025-10-05 05:49:53,950 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 05:49:53,952 - root - INFO - Epoch 88/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 56.13 sec  
2025-10-05 05:50:50,254 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 05:50:50,256 - root - INFO - Epoch 89/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 56.30 sec  
2025-10-05 05:51:46,746 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.



2025-10-05 05:51:46,748 - root - INFO - Epoch 90/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 56.49 sec  
2025-10-05 05:52:42,827 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 05:52:42,829 - root - INFO - Epoch 91/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 56.08 sec  
2025-10-05 05:53:39,096 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 05:53:39,098 - root - INFO - Epoch 92/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 56.27 sec  
2025-10-05 05:54:35,484 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 05:54:35,486 - root - INFO - Epoch 93/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 56.39 sec  
2025-10-05 05:55:32,100 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 05:55:32,102 - root - INFO - Epoch 94/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 56.61 sec  
2025-10-05 05:56:28,616 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 05:56:28,619 - root - INFO - Epoch 95/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 56.52 sec  
2025-10-05 05:57:24,514 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 05:57:24,516 - root - INFO - Epoch 96/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 55.90 sec  
2025-10-05 05:58:20,432 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 05:58:20,434 - root - INFO - Epoch 97/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 55.92 sec  
2025-10-05 05:59:16,773 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 05:59:16,775 - root - INFO - Epoch 98/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 56.34 sec  
2025-10-05 06:00:13,317 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 06:00:13,319 - root - INFO - Epoch 99/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 56.54 sec  
2025-10-05 06:01:09,266 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 06:01:09,267 - root - INFO - Epoch 100/100, Model EfficientNet1DLSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 55.95 sec  
2025-10-05 06:01:09,267 - root - CRITICAL - Training complete. Best Val Acc:

0.0000 at epoch 0.  
2025-10-05 06:01:09,344 - root - INFO - Saved stats plot to  
/Users/sanjeev/VNIT/FINAL\_PRJ\_PHASE2/experiments/20251005\_000538/plots/Efficient  
Net1DLSTM\_{'in\_channels': 99, 'meta\_seq\_len': 128, 'meta\_feature\_dim': 12,  
'num\_classes': 31, 'lr': 0.001, 'epochs': 100, 'model\_name':  
'EfficientNet1DLSTM'}\_stats.png  
2025-10-05 06:01:09,344 - root - WARNING - No best model state found, skipping  
test evaluation.  
2025-10-05 06:01:09,344 - root - INFO - START RUN MLFLOW :  
EfficientNet1DLSTM: {'in\_channels': 99, 'meta\_seq\_len': 128, 'meta\_feature\_dim':  
12, 'num\_classes': 31, 'lr': 0.0005, 'epochs': 100, 'model\_name':  
'EfficientNet1DLSTM'}  
2025-10-05 06:01:09,357 - root - INFO - START T-N-E EfficientNet1DLSTM  
2025-10-05 06:01:09,357 - root - INFO - EfficientNet1DLSTM USING LEARNING RATE  
0.0005  
2025-10-05 06:02:05,663 - root - INFO - Epoch 1/100, Model EfficientNet1DLSTM,  
Train Loss: 0.8245, Train Acc: 0.7479, Val Loss: 0.3380, Val Acc: 0.8900, Epoch  
Duration: 56.30 sec  
2025-10-05 06:03:04,849 - root - INFO - Epoch 2/100, Model EfficientNet1DLSTM,  
Train Loss: 0.3848, Train Acc: 0.8693, Val Loss: 0.2411, Val Acc: 0.9120, Epoch  
Duration: 56.90 sec  
2025-10-05 06:04:03,063 - root - INFO - Epoch 3/100, Model EfficientNet1DLSTM,  
Train Loss: 0.2920, Train Acc: 0.8992, Val Loss: 0.1693, Val Acc: 0.9447, Epoch  
Duration: 56.53 sec  
2025-10-05 06:05:01,108 - root - INFO - Epoch 4/100, Model EfficientNet1DLSTM,  
Train Loss: 0.2503, Train Acc: 0.9117, Val Loss: 0.1741, Val Acc: 0.9374, Epoch  
Duration: 56.35 sec  
2025-10-05 06:05:57,306 - root - INFO - Epoch 5/100, Model EfficientNet1DLSTM,  
Train Loss: 0.2092, Train Acc: 0.9283, Val Loss: 0.1236, Val Acc: 0.9590, Epoch  
Duration: 56.20 sec  
2025-10-05 06:06:55,330 - root - INFO - Epoch 6/100, Model EfficientNet1DLSTM,  
Train Loss: 0.1904, Train Acc: 0.9346, Val Loss: 0.1408, Val Acc: 0.9493, Epoch  
Duration: 56.32 sec  
2025-10-05 06:07:51,589 - root - INFO - Epoch 7/100, Model EfficientNet1DLSTM,  
Train Loss: 0.1720, Train Acc: 0.9393, Val Loss: 0.1133, Val Acc: 0.9607, Epoch  
Duration: 56.26 sec  
2025-10-05 06:08:49,892 - root - INFO - Epoch 8/100, Model EfficientNet1DLSTM,  
Train Loss: 0.1581, Train Acc: 0.9466, Val Loss: 0.0943, Val Acc: 0.9674, Epoch  
Duration: 56.60 sec  
2025-10-05 06:09:47,896 - root - INFO - Epoch 9/100, Model EfficientNet1DLSTM,  
Train Loss: 0.1425, Train Acc: 0.9504, Val Loss: 0.0757, Val Acc: 0.9745, Epoch  
Duration: 56.28 sec  
2025-10-05 06:10:45,824 - root - INFO - Epoch 10/100, Model EfficientNet1DLSTM,  
Train Loss: 0.1306, Train Acc: 0.9547, Val Loss: 0.0845, Val Acc: 0.9694, Epoch  
Duration: 56.18 sec  
2025-10-05 06:11:41,948 - root - INFO - Epoch 11/100, Model EfficientNet1DLSTM,  
Train Loss: 0.1222, Train Acc: 0.9562, Val Loss: 0.0996, Val Acc: 0.9625, Epoch  
Duration: 56.12 sec  
2025-10-05 06:12:38,395 - root - INFO - Epoch 12/100, Model EfficientNet1DLSTM,  
Train Loss: 0.1135, Train Acc: 0.9610, Val Loss: 0.0800, Val Acc: 0.9714, Epoch  
Duration: 56.45 sec  
2025-10-05 06:13:34,950 - root - INFO - Epoch 13/100, Model EfficientNet1DLSTM,  
Train Loss: 0.1089, Train Acc: 0.9612, Val Loss: 0.0812, Val Acc: 0.9693, Epoch  
Duration: 56.55 sec

2025-10-05 06:14:31,219 - root - INFO - Epoch 14/100, Model EfficientNet1DLSTM, Train Loss: 0.1001, Train Acc: 0.9655, Val Loss: 0.0634, Val Acc: 0.9766, Epoch Duration: 56.27 sec

2025-10-05 06:15:29,283 - root - INFO - Epoch 15/100, Model EfficientNet1DLSTM, Train Loss: 0.0924, Train Acc: 0.9674, Val Loss: 0.0689, Val Acc: 0.9767, Epoch Duration: 56.27 sec

2025-10-05 06:16:27,512 - root - INFO - Epoch 16/100, Model EfficientNet1DLSTM, Train Loss: 0.0860, Train Acc: 0.9692, Val Loss: 0.0661, Val Acc: 0.9764, Epoch Duration: 56.53 sec

2025-10-05 06:17:24,222 - root - INFO - Epoch 17/100, Model EfficientNet1DLSTM, Train Loss: 0.0862, Train Acc: 0.9700, Val Loss: 0.0789, Val Acc: 0.9700, Epoch Duration: 56.71 sec

2025-10-05 06:18:20,499 - root - INFO - Epoch 18/100, Model EfficientNet1DLSTM, Train Loss: 0.0729, Train Acc: 0.9742, Val Loss: 0.0621, Val Acc: 0.9804, Epoch Duration: 56.28 sec

2025-10-05 06:19:19,642 - root - INFO - Epoch 19/100, Model EfficientNet1DLSTM, Train Loss: 0.0785, Train Acc: 0.9733, Val Loss: 0.0610, Val Acc: 0.9803, Epoch Duration: 57.40 sec

2025-10-05 06:20:17,306 - root - INFO - Epoch 20/100, Model EfficientNet1DLSTM, Train Loss: 0.0721, Train Acc: 0.9748, Val Loss: 0.0498, Val Acc: 0.9839, Epoch Duration: 57.66 sec

2025-10-05 06:21:16,816 - root - INFO - Epoch 21/100, Model EfficientNet1DLSTM, Train Loss: 0.0700, Train Acc: 0.9755, Val Loss: 0.0580, Val Acc: 0.9812, Epoch Duration: 57.70 sec

2025-10-05 06:22:15,643 - root - INFO - Epoch 22/100, Model EfficientNet1DLSTM, Train Loss: 0.0667, Train Acc: 0.9770, Val Loss: 0.0709, Val Acc: 0.9762, Epoch Duration: 58.83 sec

2025-10-05 06:23:14,194 - root - INFO - Epoch 23/100, Model EfficientNet1DLSTM, Train Loss: 0.0617, Train Acc: 0.9784, Val Loss: 0.0621, Val Acc: 0.9781, Epoch Duration: 58.55 sec

2025-10-05 06:24:12,476 - root - INFO - Epoch 24/100, Model EfficientNet1DLSTM, Train Loss: 0.0586, Train Acc: 0.9791, Val Loss: 0.0552, Val Acc: 0.9824, Epoch Duration: 58.28 sec

2025-10-05 06:25:10,627 - root - INFO - Epoch 25/100, Model EfficientNet1DLSTM, Train Loss: 0.0564, Train Acc: 0.9803, Val Loss: 0.0456, Val Acc: 0.9848, Epoch Duration: 58.15 sec

2025-10-05 06:26:10,586 - root - INFO - Epoch 26/100, Model EfficientNet1DLSTM, Train Loss: 0.0578, Train Acc: 0.9800, Val Loss: 0.0565, Val Acc: 0.9824, Epoch Duration: 58.13 sec

2025-10-05 06:27:08,400 - root - INFO - Epoch 27/100, Model EfficientNet1DLSTM, Train Loss: 0.0549, Train Acc: 0.9810, Val Loss: 0.0420, Val Acc: 0.9874, Epoch Duration: 57.81 sec

2025-10-05 06:28:08,676 - root - INFO - Epoch 28/100, Model EfficientNet1DLSTM, Train Loss: 0.0502, Train Acc: 0.9826, Val Loss: 0.0580, Val Acc: 0.9808, Epoch Duration: 58.42 sec

2025-10-05 06:29:06,710 - root - INFO - Epoch 29/100, Model EfficientNet1DLSTM, Train Loss: 0.0500, Train Acc: 0.9830, Val Loss: 0.0434, Val Acc: 0.9851, Epoch Duration: 58.03 sec

2025-10-05 06:30:04,733 - root - INFO - Epoch 30/100, Model EfficientNet1DLSTM, Train Loss: 0.0485, Train Acc: 0.9840, Val Loss: 0.0439, Val Acc: 0.9844, Epoch Duration: 58.02 sec

2025-10-05 06:31:02,855 - root - INFO - Epoch 31/100, Model EfficientNet1DLSTM, Train Loss: 0.0451, Train Acc: 0.9838, Val Loss: 0.0476, Val Acc: 0.9851, Epoch Duration: 58.12 sec

2025-10-05 06:32:00,514 - root - INFO - Epoch 32/100, Model EfficientNet1DLSTM, Train Loss: 0.0450, Train Acc: 0.9848, Val Loss: 0.0461, Val Acc: 0.9862, Epoch Duration: 57.66 sec

2025-10-05 06:32:57,742 - root - INFO - Epoch 33/100, Model EfficientNet1DLSTM, Train Loss: 0.0440, Train Acc: 0.9851, Val Loss: 0.0575, Val Acc: 0.9816, Epoch Duration: 57.23 sec

2025-10-05 06:33:54,550 - root - INFO - Epoch 34/100, Model EfficientNet1DLSTM, Train Loss: 0.0455, Train Acc: 0.9846, Val Loss: 0.0595, Val Acc: 0.9816, Epoch Duration: 56.80 sec

2025-10-05 06:34:51,655 - root - INFO - Epoch 35/100, Model EfficientNet1DLSTM, Train Loss: 0.0390, Train Acc: 0.9869, Val Loss: 0.0452, Val Acc: 0.9872, Epoch Duration: 57.10 sec

2025-10-05 06:35:48,315 - root - INFO - Epoch 36/100, Model EfficientNet1DLSTM, Train Loss: 0.0406, Train Acc: 0.9870, Val Loss: 0.0418, Val Acc: 0.9867, Epoch Duration: 56.66 sec

2025-10-05 06:36:45,615 - root - INFO - Epoch 37/100, Model EfficientNet1DLSTM, Train Loss: 0.0409, Train Acc: 0.9855, Val Loss: 0.0481, Val Acc: 0.9845, Epoch Duration: 57.30 sec

2025-10-05 06:37:42,544 - root - INFO - Epoch 38/100, Model EfficientNet1DLSTM, Train Loss: 0.0389, Train Acc: 0.9861, Val Loss: 0.0458, Val Acc: 0.9853, Epoch Duration: 56.93 sec

2025-10-05 06:38:39,095 - root - INFO - Epoch 39/100, Model EfficientNet1DLSTM, Train Loss: 0.0390, Train Acc: 0.9870, Val Loss: 0.0408, Val Acc: 0.9883, Epoch Duration: 56.55 sec

2025-10-05 06:39:38,016 - root - INFO - Epoch 40/100, Model EfficientNet1DLSTM, Train Loss: 0.0346, Train Acc: 0.9881, Val Loss: 0.0555, Val Acc: 0.9845, Epoch Duration: 57.12 sec

2025-10-05 06:40:34,382 - root - INFO - Epoch 41/100, Model EfficientNet1DLSTM, Train Loss: 0.0332, Train Acc: 0.9882, Val Loss: 0.0484, Val Acc: 0.9854, Epoch Duration: 56.36 sec

2025-10-05 06:41:30,970 - root - INFO - Epoch 42/100, Model EfficientNet1DLSTM, Train Loss: 0.0325, Train Acc: 0.9879, Val Loss: 0.0406, Val Acc: 0.9866, Epoch Duration: 56.59 sec

2025-10-05 06:42:27,601 - root - INFO - Epoch 43/100, Model EfficientNet1DLSTM, Train Loss: 0.0322, Train Acc: 0.9892, Val Loss: 0.0424, Val Acc: 0.9872, Epoch Duration: 56.63 sec

2025-10-05 06:43:24,320 - root - INFO - Epoch 44/100, Model EfficientNet1DLSTM, Train Loss: 0.0319, Train Acc: 0.9894, Val Loss: 0.0381, Val Acc: 0.9883, Epoch Duration: 56.72 sec

2025-10-05 06:44:20,696 - root - INFO - Epoch 45/100, Model EfficientNet1DLSTM, Train Loss: 0.0328, Train Acc: 0.9884, Val Loss: 0.0431, Val Acc: 0.9869, Epoch Duration: 56.37 sec

2025-10-05 06:45:16,923 - root - INFO - Epoch 46/100, Model EfficientNet1DLSTM, Train Loss: 0.0336, Train Acc: 0.9887, Val Loss: 0.0383, Val Acc: 0.9867, Epoch Duration: 56.22 sec

2025-10-05 06:46:13,286 - root - INFO - Epoch 47/100, Model EfficientNet1DLSTM, Train Loss: 0.0297, Train Acc: 0.9899, Val Loss: 0.0379, Val Acc: 0.9890, Epoch Duration: 56.36 sec

2025-10-05 06:47:11,610 - root - INFO - Epoch 48/100, Model EfficientNet1DLSTM, Train Loss: 0.0277, Train Acc: 0.9905, Val Loss: 0.0466, Val Acc: 0.9848, Epoch Duration: 56.61 sec

2025-10-05 06:48:08,026 - root - INFO - Epoch 49/100, Model EfficientNet1DLSTM, Train Loss: 0.0308, Train Acc: 0.9892, Val Loss: 0.0388, Val Acc: 0.9887, Epoch Duration: 56.41 sec

2025-10-05 06:49:03,964 - root - INFO - Epoch 50/100, Model EfficientNet1DLSTM, Train Loss: 0.0279, Train Acc: 0.9904, Val Loss: 0.0378, Val Acc: 0.9888, Epoch Duration: 55.94 sec

2025-10-05 06:50:00,216 - root - INFO - Epoch 51/100, Model EfficientNet1DLSTM, Train Loss: 0.0269, Train Acc: 0.9905, Val Loss: 0.0326, Val Acc: 0.9902, Epoch Duration: 56.25 sec

2025-10-05 06:50:58,359 - root - INFO - Epoch 52/100, Model EfficientNet1DLSTM, Train Loss: 0.0285, Train Acc: 0.9902, Val Loss: 0.0488, Val Acc: 0.9847, Epoch Duration: 56.44 sec

2025-10-05 06:51:54,799 - root - INFO - Epoch 53/100, Model EfficientNet1DLSTM, Train Loss: 0.0259, Train Acc: 0.9913, Val Loss: 0.0509, Val Acc: 0.9853, Epoch Duration: 56.44 sec

2025-10-05 06:52:51,655 - root - INFO - Epoch 54/100, Model EfficientNet1DLSTM, Train Loss: 0.0262, Train Acc: 0.9905, Val Loss: 0.0565, Val Acc: 0.9841, Epoch Duration: 56.85 sec

2025-10-05 06:53:49,613 - root - INFO - Epoch 55/100, Model EfficientNet1DLSTM, Train Loss: 0.0247, Train Acc: 0.9919, Val Loss: 0.0362, Val Acc: 0.9885, Epoch Duration: 57.96 sec

2025-10-05 06:54:48,745 - root - INFO - Epoch 56/100, Model EfficientNet1DLSTM, Train Loss: 0.0243, Train Acc: 0.9919, Val Loss: 0.0370, Val Acc: 0.9880, Epoch Duration: 59.13 sec

2025-10-05 06:55:46,690 - root - INFO - Epoch 57/100, Model EfficientNet1DLSTM, Train Loss: 0.0255, Train Acc: 0.9916, Val Loss: 0.0506, Val Acc: 0.9854, Epoch Duration: 57.94 sec

2025-10-05 06:56:43,870 - root - INFO - Epoch 58/100, Model EfficientNet1DLSTM, Train Loss: 0.0256, Train Acc: 0.9911, Val Loss: 0.0321, Val Acc: 0.9902, Epoch Duration: 57.18 sec

2025-10-05 06:57:40,603 - root - INFO - Epoch 59/100, Model EfficientNet1DLSTM, Train Loss: 0.0256, Train Acc: 0.9913, Val Loss: 0.0419, Val Acc: 0.9866, Epoch Duration: 56.73 sec

2025-10-05 06:58:36,913 - root - INFO - Epoch 60/100, Model EfficientNet1DLSTM, Train Loss: 0.0227, Train Acc: 0.9920, Val Loss: 0.0336, Val Acc: 0.9895, Epoch Duration: 56.31 sec

2025-10-05 06:59:34,084 - root - INFO - Epoch 61/100, Model EfficientNet1DLSTM, Train Loss: 0.0250, Train Acc: 0.9915, Val Loss: 0.0377, Val Acc: 0.9889, Epoch Duration: 57.17 sec

2025-10-05 07:00:30,799 - root - INFO - Epoch 62/100, Model EfficientNet1DLSTM, Train Loss: 0.0212, Train Acc: 0.9929, Val Loss: 0.0430, Val Acc: 0.9874, Epoch Duration: 56.71 sec

2025-10-05 07:01:27,490 - root - INFO - Epoch 63/100, Model EfficientNet1DLSTM, Train Loss: 0.0213, Train Acc: 0.9926, Val Loss: 0.0418, Val Acc: 0.9880, Epoch Duration: 56.69 sec

2025-10-05 07:02:23,913 - root - INFO - Epoch 64/100, Model EfficientNet1DLSTM, Train Loss: 0.0247, Train Acc: 0.9919, Val Loss: 0.0394, Val Acc: 0.9879, Epoch Duration: 56.42 sec

2025-10-05 07:03:20,698 - root - INFO - Epoch 65/100, Model EfficientNet1DLSTM, Train Loss: 0.0212, Train Acc: 0.9928, Val Loss: 0.0398, Val Acc: 0.9885, Epoch Duration: 56.78 sec

2025-10-05 07:04:17,453 - root - INFO - Epoch 66/100, Model EfficientNet1DLSTM, Train Loss: 0.0245, Train Acc: 0.9917, Val Loss: 0.0381, Val Acc: 0.9883, Epoch Duration: 56.75 sec

2025-10-05 07:05:14,342 - root - INFO - Epoch 67/100, Model EfficientNet1DLSTM, Train Loss: 0.0212, Train Acc: 0.9927, Val Loss: 0.0510, Val Acc: 0.9861, Epoch Duration: 56.89 sec

2025-10-05 07:06:10,313 - root - INFO - Epoch 68/100, Model EfficientNet1DLSTM, Train Loss: 0.0207, Train Acc: 0.9930, Val Loss: 0.0421, Val Acc: 0.9874, Epoch Duration: 55.97 sec

2025-10-05 07:07:06,414 - root - INFO - Epoch 69/100, Model EfficientNet1DLSTM, Train Loss: 0.0224, Train Acc: 0.9923, Val Loss: 0.0361, Val Acc: 0.9892, Epoch Duration: 56.10 sec

2025-10-05 07:08:02,802 - root - INFO - Epoch 70/100, Model EfficientNet1DLSTM, Train Loss: 0.0199, Train Acc: 0.9929, Val Loss: 0.0355, Val Acc: 0.9886, Epoch Duration: 56.39 sec

2025-10-05 07:08:59,414 - root - INFO - Epoch 71/100, Model EfficientNet1DLSTM, Train Loss: 0.0193, Train Acc: 0.9935, Val Loss: 0.0449, Val Acc: 0.9875, Epoch Duration: 56.61 sec

2025-10-05 07:09:55,747 - root - INFO - Epoch 72/100, Model EfficientNet1DLSTM, Train Loss: 0.0218, Train Acc: 0.9924, Val Loss: 0.0347, Val Acc: 0.9904, Epoch Duration: 56.33 sec

2025-10-05 07:10:54,020 - root - INFO - Epoch 73/100, Model EfficientNet1DLSTM, Train Loss: 0.0175, Train Acc: 0.9941, Val Loss: 0.0579, Val Acc: 0.9851, Epoch Duration: 56.23 sec

2025-10-05 07:11:50,130 - root - INFO - Epoch 74/100, Model EfficientNet1DLSTM, Train Loss: 0.0212, Train Acc: 0.9925, Val Loss: 0.0449, Val Acc: 0.9865, Epoch Duration: 56.11 sec

2025-10-05 07:12:46,329 - root - INFO - Epoch 75/100, Model EfficientNet1DLSTM, Train Loss: 0.0182, Train Acc: 0.9937, Val Loss: 0.0336, Val Acc: 0.9893, Epoch Duration: 56.20 sec

2025-10-05 07:13:42,651 - root - INFO - Epoch 76/100, Model EfficientNet1DLSTM, Train Loss: 0.0161, Train Acc: 0.9947, Val Loss: 0.0334, Val Acc: 0.9895, Epoch Duration: 56.32 sec

2025-10-05 07:14:38,454 - root - INFO - Epoch 77/100, Model EfficientNet1DLSTM, Train Loss: 0.0173, Train Acc: 0.9938, Val Loss: 0.0335, Val Acc: 0.9906, Epoch Duration: 55.80 sec

2025-10-05 07:15:36,042 - root - INFO - Epoch 78/100, Model EfficientNet1DLSTM, Train Loss: 0.0187, Train Acc: 0.9941, Val Loss: 0.0407, Val Acc: 0.9881, Epoch Duration: 55.73 sec

2025-10-05 07:16:32,060 - root - INFO - Epoch 79/100, Model EfficientNet1DLSTM, Train Loss: 0.0195, Train Acc: 0.9933, Val Loss: 0.0476, Val Acc: 0.9867, Epoch Duration: 56.02 sec

2025-10-05 07:17:28,361 - root - INFO - Epoch 80/100, Model EfficientNet1DLSTM, Train Loss: 0.0137, Train Acc: 0.9956, Val Loss: 0.0340, Val Acc: 0.9901, Epoch Duration: 56.30 sec

2025-10-05 07:18:24,501 - root - INFO - Epoch 81/100, Model EfficientNet1DLSTM, Train Loss: 0.0214, Train Acc: 0.9928, Val Loss: 0.0381, Val Acc: 0.9883, Epoch Duration: 56.14 sec

2025-10-05 07:19:20,222 - root - INFO - Epoch 82/100, Model EfficientNet1DLSTM, Train Loss: 0.0185, Train Acc: 0.9940, Val Loss: 0.0652, Val Acc: 0.9830, Epoch Duration: 55.72 sec

2025-10-05 07:20:16,138 - root - INFO - Epoch 83/100, Model EfficientNet1DLSTM, Train Loss: 0.0167, Train Acc: 0.9945, Val Loss: 0.0409, Val Acc: 0.9880, Epoch Duration: 55.91 sec

2025-10-05 07:21:12,374 - root - INFO - Epoch 84/100, Model EfficientNet1DLSTM, Train Loss: 0.0180, Train Acc: 0.9934, Val Loss: 0.0564, Val Acc: 0.9837, Epoch Duration: 56.23 sec

2025-10-05 07:22:08,771 - root - INFO - Epoch 85/100, Model EfficientNet1DLSTM, Train Loss: 0.0149, Train Acc: 0.9951, Val Loss: 0.0379, Val Acc: 0.9888, Epoch Duration: 56.39 sec

2025-10-05 07:23:04,600 - root - INFO - Epoch 86/100, Model EfficientNet1DLSTM, Train Loss: 0.0150, Train Acc: 0.9950, Val Loss: 0.0378, Val Acc: 0.9893, Epoch Duration: 55.83 sec

2025-10-05 07:24:00,656 - root - INFO - Epoch 87/100, Model EfficientNet1DLSTM, Train Loss: 0.0167, Train Acc: 0.9942, Val Loss: 0.0361, Val Acc: 0.9895, Epoch Duration: 56.05 sec

2025-10-05 07:24:56,934 - root - INFO - Epoch 88/100, Model EfficientNet1DLSTM, Train Loss: 0.0143, Train Acc: 0.9951, Val Loss: 0.0279, Val Acc: 0.9925, Epoch Duration: 56.28 sec

2025-10-05 07:25:55,286 - root - INFO - Epoch 89/100, Model EfficientNet1DLSTM, Train Loss: 0.0152, Train Acc: 0.9950, Val Loss: 0.0480, Val Acc: 0.9858, Epoch Duration: 56.45 sec

2025-10-05 07:26:51,636 - root - INFO - Epoch 90/100, Model EfficientNet1DLSTM, Train Loss: 0.0160, Train Acc: 0.9947, Val Loss: 0.0443, Val Acc: 0.9885, Epoch Duration: 56.35 sec

2025-10-05 07:27:47,741 - root - INFO - Epoch 91/100, Model EfficientNet1DLSTM, Train Loss: 0.0181, Train Acc: 0.9939, Val Loss: 0.0391, Val Acc: 0.9906, Epoch Duration: 56.10 sec

2025-10-05 07:28:43,739 - root - INFO - Epoch 92/100, Model EfficientNet1DLSTM, Train Loss: 0.0152, Train Acc: 0.9949, Val Loss: 0.0434, Val Acc: 0.9881, Epoch Duration: 56.00 sec

2025-10-05 07:29:40,001 - root - INFO - Epoch 93/100, Model EfficientNet1DLSTM, Train Loss: 0.0168, Train Acc: 0.9943, Val Loss: 0.0586, Val Acc: 0.9853, Epoch Duration: 56.26 sec

2025-10-05 07:30:36,436 - root - INFO - Epoch 94/100, Model EfficientNet1DLSTM, Train Loss: 0.0150, Train Acc: 0.9945, Val Loss: 0.0409, Val Acc: 0.9876, Epoch Duration: 56.43 sec

2025-10-05 07:31:32,588 - root - INFO - Epoch 95/100, Model EfficientNet1DLSTM, Train Loss: 0.0137, Train Acc: 0.9955, Val Loss: 0.0330, Val Acc: 0.9914, Epoch Duration: 56.15 sec

2025-10-05 07:32:28,595 - root - INFO - Epoch 96/100, Model EfficientNet1DLSTM, Train Loss: 0.0149, Train Acc: 0.9950, Val Loss: 0.0421, Val Acc: 0.9878, Epoch Duration: 56.00 sec

2025-10-05 07:33:24,687 - root - INFO - Epoch 97/100, Model EfficientNet1DLSTM, Train Loss: 0.0148, Train Acc: 0.9955, Val Loss: 0.0443, Val Acc: 0.9903, Epoch Duration: 56.09 sec

2025-10-05 07:34:21,020 - root - INFO - Epoch 98/100, Model EfficientNet1DLSTM, Train Loss: 0.0115, Train Acc: 0.9960, Val Loss: 0.0453, Val Acc: 0.9883, Epoch Duration: 56.33 sec

2025-10-05 07:35:17,421 - root - INFO - Epoch 99/100, Model EfficientNet1DLSTM, Train Loss: 0.0125, Train Acc: 0.9958, Val Loss: 0.0429, Val Acc: 0.9892, Epoch Duration: 56.40 sec

2025-10-05 07:36:13,217 - root - INFO - Epoch 100/100, Model EfficientNet1DLSTM, Train Loss: 0.0145, Train Acc: 0.9952, Val Loss: 0.0572, Val Acc: 0.9852, Epoch Duration: 55.79 sec

2025-10-05 07:36:13,217 - root - CRITICAL - Training complete. Best Val Acc: 0.9925 at epoch 88.

2025-10-05 07:36:13,234 - root - CRITICAL - SAVED BEST MODEL  
/Users/sanjeev/VNIT/FINAL\_PRJ\_PHASE2/experiments/20251005\_000538/checkpoints/best\_model\_EfficientNet1DLSTM\_epoch88.pt USING LEARNING RATE 0.0005

2025-10-05 07:36:13,234 - root - CRITICAL - BEST VALIDATION ACCURACY: 0.9925 at epoch 88.

2025-10-05 07:36:13,319 - root - INFO - Saved stats plot to  
/Users/sanjeev/VNIT/FINAL\_PRJ\_PHASE2/experiments/20251005\_000538/plots/Efficient

Net1DLSTM\_{'in\_channels': 99, 'meta\_seq\_len': 128, 'meta\_feature\_dim': 12, 'num\_classes': 31, 'lr': 0.0005, 'epochs': 100, 'model\_name': 'EfficientNet1DLSTM'}\_stats.png

2025-10-05 07:36:17,292 - root - INFO - Confusion matrix:

```
[[276  0  0  0  0  0  0  0  0  0  0  1  0  0  0  0  0
  0  0  0  0  0  0  0  0  0  0  0  0]
 [  0 282  0  0  0  1  0  0  0  0  0  0  0  0  0  0  0
  0  0  0  0  0  0  0  0  0  0  0  0]
 [  0  0 289  0  0  0  0  2  0  1  0  0  0  0  0  0  0
  0  0  0  0  0  0  0  0  0  0  0  0]
 [  0  0  0 288  0  0  0  0  0  0  0  0  0  0  0  0  0
  0  0  0  0  0  0  0  0  0  0  0  0]
 [  0  4  0  0 267  1  0  0  0  0  0  0  0  0  0  0  0
  0  0  0  0  0  0  0  0  0  0  0  0]
 [  0  3  0  0  0 290  0  1  1  0  0  0  0  0  0  0  0
  0  0  0  0  0  0  0  0  0  0  0  0]
 [  0  0  0  0  0  0 289  0  0  0  0  0  0  0  0  0  0
  0  0  0  0  0  0  0  0  0  0  0  1]
 [  0  0  0  0  0  0  0 264  0  0  0  0  0  0  0  0  0
  0  1  0  1  0  0  0  0  0  0  1  0]
 [  0  0  0  2  0  0  0  0 259  0  0  1  0  0  1  0  2
  0  0  0  0  0  0  0  0  0  0  0  0]
 [  0  0  1  0  0  0  0  0  0 270  0  0  0  1  0  0  0
  0  0  0  0  0  0  0  0  0  0  0  0]
 [  0  0  0  0  0  0  0  0  0  0 270  0  0  0  0  0  0
  0  0  0  0  0  0  0  0  0  0  0  0]
 [  0  0  0  0  0  0  0  0  0  0  0 271  0  0  0  1  0
  0  0  0  0  0  0  0  0  0  0  0  0]
 [  0  0  0  0  0  0  0  0  0  0  0  1  0 316  0  0  0
 15 0  0  0  0  0  0  0  0  0  0  0  0]
 [  0  0  0  0  0  0  0  0  0  0  0  0  0  0 301  0  0
  0  0  0  0  0  0  0  0  0  0  0  0]
 [  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0 251  0
  0  0  0  0  2  0  0  0  0  0  0  0]
 [  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0 305
  0  0  0  0  0  0  0  0  0  0  0]
 [  1  0  0  1  0  0  0  0  0  0  0  0  0  0  1 252 35
  0  0  0  0  0  0  0  0  0  0  0  0]
 [  0  0  0  0  0  0  0  0  0  0  0  0  0  0  1  1 304
  0  0  0  0  0  0  0  0  0  0  0  0]
 [  0  0  0  0  0  0  0  0  0  0  0  0  0  3  1  0  0
277 1  0  0  0  0  0  0  0  0  0  0]
 [  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
  0 285  0  0  0  0  0  0  0  0  0  0]
 [  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
  0  0 288  0  0  0  0  0  0  0  0  1]
 [  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
  0  0  5 308  0  0  0  0  1  0  1  1]
 [  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
  0  0  0  1 287  0  0  0  0  0  0]
 [  0  0  0  0  0  0  0  0  0  0  0  0  1  0  0  0  0
  0  0  0  0  0 269  2  0  0  0  0]
 [  0  1  0  0  0  0  0  0  0  0  0  0  0  0  0  1  0
  2  0  0  0  0  0 304  0  0  0  0]
```



```

[ 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
 0 0 0 0 0 0 0 263 0 0 0 0]
[ 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
 0 0 0 0 0 0 0 0 287 0 0 0]
[ 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
 0 0 0 0 0 0 0 0 0 293 0 0]
[ 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
 0 0 1 5 0 0 0 0 1 0 285 2]
[ 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
 0 0 2 3 0 0 0 0 0 0 1 259]]

```

2025-10-05 07:36:17,292 - root - INFO - Classification report:  
precision recall f1-score support

|              |      |      |      |                     |
|--------------|------|------|------|---------------------|
| 1            | 1.00 | 1.00 | 1.00 | 277                 |
| 2            | 0.97 | 1.00 | 0.98 | 283                 |
| 3            | 1.00 | 0.99 | 0.99 | 292                 |
| 4            | 0.99 | 1.00 | 0.99 | 288                 |
| 5            | 1.00 | 0.98 | 0.99 | 272                 |
| 6            | 0.99 | 0.98 | 0.99 | 295                 |
| 7            | 1.00 | 1.00 | 1.00 | 290                 |
| 8            | 0.99 | 0.99 | 0.99 | 267                 |
| 9            | 1.00 | 0.97 | 0.98 | 267                 |
| 10           | 1.00 | 0.99 | 0.99 | 272                 |
| 11           | 1.00 | 1.00 | 1.00 | 270                 |
| 12           | 0.99 | 1.00 | 0.99 | 272                 |
| 13           | 0.99 | 0.95 | 0.97 | 332                 |
| 14           | 1.00 | 1.00 | 1.00 | 301                 |
| 15           | 0.99 | 0.99 | 0.99 | 253                 |
| 16           | 0.99 | 0.99 | 0.99 | 307                 |
| 17           | 0.98 | 0.87 | 0.92 | 290                 |
| 18           | 0.89 | 0.99 | 0.94 | 306                 |
| 19           | 0.94 | 0.98 | 0.96 | 282                 |
| 20           | 0.99 | 1.00 | 1.00 | 285                 |
| 21           | 0.97 | 1.00 | 0.98 | 289                 |
| 22           | 0.97 | 0.97 | 0.97 | 316                 |
| 23           | 0.99 | 1.00 | 0.99 | 288                 |
| 24           | 1.00 | 0.99 | 0.99 | 272                 |
| 25           | 0.99 | 0.99 | 0.99 | 308                 |
| 26           | 1.00 | 1.00 | 1.00 | 263                 |
| 27           | 0.99 | 1.00 | 1.00 | 287                 |
| 28           | 1.00 | 1.00 | 1.00 | 293                 |
| 29           | 0.99 | 0.97 | 0.98 | 294                 |
| 30           | 0.98 | 0.98 | 0.98 | 265                 |
| accuracy     |      |      |      | 0.99 8576           |
| macro avg    |      |      |      | 0.99 0.99 0.99 8576 |
| weighted avg |      |      |      | 0.99 0.99 0.99 8576 |

2025-10-05 07:36:17,307 - root - CRITICAL - 99. SAVED BEST MODEL IN EfficientNet1DLSTM. BEST VALIDATION ACCURACY: 0.9925 at epoch 88.  
2025-10-05 07:36:17,308 - root - CRITICAL - 100. LOGGED BEST MODEL IN MLFLOW /Users/sanjeev/VNIT/FINAL\_PRJ\_PHASE2/experiments/20251005\_000538/checkpoints/best\_model\_EfficientNet1DLSTM\_epoch88.pt. BEST VALIDATION ACCURACY: 0.9925 at epoch 88.

2025-10-05 07:36:17,309 - root - INFO -  
2025-10-05 07:36:17,309 - root - INFO - N  
2025-10-05 07:36:17,309 - root - INFO - E  
2025-10-05 07:36:17,309 - root - INFO - W  
2025-10-05 07:36:17,309 - root - INFO - STARTING TRAINING FLOW :  
MobileNetV3\_1D\_LSTM:('csi\_channels', 'meta\_feature\_dim', 'num\_classes', 'lr',  
'epochs'):([99], [12], [31], [0.001, 0.0005], [100])  
2025-10-05 07:36:17,313 - root - INFO - START RUN MLFLOW :  
MobileNetV3\_1D\_LSTM:{'csi\_channels': 99, 'meta\_feature\_dim': 12, 'num\_classes':  
31, 'lr': 0.001, 'epochs': 100, 'model\_name': 'MobileNetV3\_1D\_LSTM'}  
2025-10-05 07:36:17,321 - root - INFO - START T-N-E MobileNetV3\_1D\_LSTM  
2025-10-05 07:36:17,322 - root - INFO - MobileNetV3\_1D\_LSTM USING LEARNING RATE  
0.001  
2025-10-05 07:37:05,457 - root - INFO - Epoch 1/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: 2.0173, Train Acc: 0.3586, Val Loss: 1.7440, Val Acc: 0.4177, Epoch  
Duration: 48.13 sec  
2025-10-05 07:37:54,309 - root - INFO - Epoch 2/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: 1.6713, Train Acc: 0.4419, Val Loss: 1.6389, Val Acc: 0.4546, Epoch  
Duration: 47.03 sec  
2025-10-05 07:38:43,196 - root - INFO - Epoch 3/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: 1.5142, Train Acc: 0.4847, Val Loss: 1.4632, Val Acc: 0.5075, Epoch  
Duration: 47.16 sec  
2025-10-05 07:39:32,231 - root - INFO - Epoch 4/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: 1.4496, Train Acc: 0.5111, Val Loss: 1.4582, Val Acc: 0.5083, Epoch  
Duration: 47.37 sec  
2025-10-05 07:40:21,540 - root - INFO - Epoch 5/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: 1.3744, Train Acc: 0.5298, Val Loss: 1.3613, Val Acc: 0.5333, Epoch  
Duration: 47.55 sec  
2025-10-05 07:41:10,923 - root - INFO - Epoch 6/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: 1.3287, Train Acc: 0.5455, Val Loss: 1.3322, Val Acc: 0.5597, Epoch  
Duration: 47.66 sec  
2025-10-05 07:42:00,050 - root - INFO - Epoch 7/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: 1.2714, Train Acc: 0.5636, Val Loss: 1.2727, Val Acc: 0.5622, Epoch  
Duration: 47.49 sec  
2025-10-05 07:42:48,865 - root - INFO - Epoch 8/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: 1.2224, Train Acc: 0.5766, Val Loss: 1.2728, Val Acc: 0.5710, Epoch  
Duration: 47.15 sec  
2025-10-05 07:43:37,460 - root - INFO - Epoch 9/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: 1.1953, Train Acc: 0.5880, Val Loss: 1.1871, Val Acc: 0.5935, Epoch  
Duration: 46.92 sec  
2025-10-05 07:44:26,297 - root - INFO - Epoch 10/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: 1.1411, Train Acc: 0.6040, Val Loss: 1.1493, Val Acc: 0.6090, Epoch  
Duration: 47.17 sec  
2025-10-05 07:45:15,089 - root - INFO - Epoch 11/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: 1.1079, Train Acc: 0.6183, Val Loss: 1.2207, Val Acc: 0.5841, Epoch  
Duration: 47.14 sec  
2025-10-05 07:46:02,251 - root - INFO - Epoch 12/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: 1.0796, Train Acc: 0.6278, Val Loss: 1.1340, Val Acc: 0.6137, Epoch  
Duration: 47.16 sec  
2025-10-05 07:46:51,443 - root - INFO - Epoch 13/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: 1.0501, Train Acc: 0.6371, Val Loss: 1.0638, Val Acc: 0.6360, Epoch  
Duration: 47.50 sec  
2025-10-05 07:47:40,645 - root - INFO - Epoch 14/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: 1.0122, Train Acc: 0.6524, Val Loss: 1.0487, Val Acc: 0.6353, Epoch

Duration: 47.54 sec  
2025-10-05 07:48:27,515 - root - INFO - Epoch 15/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.9669, Train Acc: 0.6634, Val Loss: 1.0672, Val Acc: 0.6372, Epoch Duration: 46.87 sec  
2025-10-05 07:49:16,580 - root - INFO - Epoch 16/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.9297, Train Acc: 0.6743, Val Loss: 1.0256, Val Acc: 0.6552, Epoch Duration: 47.38 sec  
2025-10-05 07:50:05,437 - root - INFO - Epoch 17/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.8960, Train Acc: 0.6862, Val Loss: 1.0029, Val Acc: 0.6587, Epoch Duration: 47.21 sec  
2025-10-05 07:50:54,557 - root - INFO - Epoch 18/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.8615, Train Acc: 0.6980, Val Loss: 1.0061, Val Acc: 0.6606, Epoch Duration: 47.48 sec  
2025-10-05 07:51:43,950 - root - INFO - Epoch 19/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.8255, Train Acc: 0.7106, Val Loss: 1.0016, Val Acc: 0.6572, Epoch Duration: 47.71 sec  
2025-10-05 07:52:31,434 - root - INFO - Epoch 20/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.7948, Train Acc: 0.7227, Val Loss: 0.9577, Val Acc: 0.6698, Epoch Duration: 47.48 sec  
2025-10-05 07:53:20,672 - root - INFO - Epoch 21/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.7773, Train Acc: 0.7275, Val Loss: 0.9132, Val Acc: 0.6838, Epoch Duration: 47.57 sec  
2025-10-05 07:54:09,627 - root - INFO - Epoch 22/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.7249, Train Acc: 0.7415, Val Loss: 0.9074, Val Acc: 0.6889, Epoch Duration: 47.26 sec  
2025-10-05 07:54:58,609 - root - INFO - Epoch 23/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.7026, Train Acc: 0.7516, Val Loss: 0.8861, Val Acc: 0.6954, Epoch Duration: 47.33 sec  
2025-10-05 07:55:47,577 - root - INFO - Epoch 24/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.6682, Train Acc: 0.7608, Val Loss: 0.8842, Val Acc: 0.6979, Epoch Duration: 47.28 sec  
2025-10-05 07:56:36,740 - root - INFO - Epoch 25/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.6331, Train Acc: 0.7725, Val Loss: 0.9909, Val Acc: 0.6652, Epoch Duration: 47.50 sec  
2025-10-05 07:57:24,158 - root - INFO - Epoch 26/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.6309, Train Acc: 0.7741, Val Loss: 0.8440, Val Acc: 0.7130, Epoch Duration: 47.42 sec  
2025-10-05 07:58:13,520 - root - INFO - Epoch 27/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.5895, Train Acc: 0.7903, Val Loss: 0.8670, Val Acc: 0.7057, Epoch Duration: 47.69 sec  
2025-10-05 07:59:00,780 - root - INFO - Epoch 28/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.5598, Train Acc: 0.7996, Val Loss: 0.8881, Val Acc: 0.7038, Epoch Duration: 47.26 sec  
2025-10-05 07:59:47,953 - root - INFO - Epoch 29/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.5403, Train Acc: 0.8062, Val Loss: 0.8195, Val Acc: 0.7214, Epoch Duration: 47.17 sec  
2025-10-05 08:00:36,936 - root - INFO - Epoch 30/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.5038, Train Acc: 0.8197, Val Loss: 0.8189, Val Acc: 0.7310, Epoch Duration: 47.31 sec  
2025-10-05 08:01:26,084 - root - INFO - Epoch 31/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.4842, Train Acc: 0.8256, Val Loss: 0.8232, Val Acc: 0.7240, Epoch Duration: 47.41 sec  
2025-10-05 08:02:13,409 - root - INFO - Epoch 32/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.4561, Train Acc: 0.8353, Val Loss: 0.8003, Val Acc: 0.7353, Epoch

Duration: 47.32 sec  
2025-10-05 08:03:02,639 - root - INFO - Epoch 33/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.4398, Train Acc: 0.8413, Val Loss: 0.8358, Val Acc: 0.7253, Epoch Duration: 47.52 sec  
2025-10-05 08:03:50,153 - root - INFO - Epoch 34/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.4229, Train Acc: 0.8461, Val Loss: 0.8324, Val Acc: 0.7243, Epoch Duration: 47.51 sec  
2025-10-05 08:04:38,358 - root - INFO - Epoch 35/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.3908, Train Acc: 0.8616, Val Loss: 0.7932, Val Acc: 0.7422, Epoch Duration: 48.20 sec  
2025-10-05 08:05:28,184 - root - INFO - Epoch 36/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.3773, Train Acc: 0.8656, Val Loss: 0.8048, Val Acc: 0.7361, Epoch Duration: 48.10 sec  
2025-10-05 08:06:15,798 - root - INFO - Epoch 37/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.3562, Train Acc: 0.8728, Val Loss: 0.8202, Val Acc: 0.7442, Epoch Duration: 47.61 sec  
2025-10-05 08:07:04,879 - root - INFO - Epoch 38/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.3396, Train Acc: 0.8786, Val Loss: 0.8258, Val Acc: 0.7409, Epoch Duration: 47.46 sec  
2025-10-05 08:07:52,034 - root - INFO - Epoch 39/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.3275, Train Acc: 0.8818, Val Loss: 0.8242, Val Acc: 0.7420, Epoch Duration: 47.15 sec  
2025-10-05 08:08:39,626 - root - INFO - Epoch 40/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.3056, Train Acc: 0.8909, Val Loss: 0.8057, Val Acc: 0.7459, Epoch Duration: 47.59 sec  
2025-10-05 08:09:29,038 - root - INFO - Epoch 41/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.2889, Train Acc: 0.8971, Val Loss: 0.8405, Val Acc: 0.7499, Epoch Duration: 47.63 sec  
2025-10-05 08:10:18,285 - root - INFO - Epoch 42/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.2880, Train Acc: 0.8973, Val Loss: 0.8284, Val Acc: 0.7486, Epoch Duration: 47.51 sec  
2025-10-05 08:11:05,342 - root - INFO - Epoch 43/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.2903, Train Acc: 0.8976, Val Loss: 0.8381, Val Acc: 0.7448, Epoch Duration: 47.05 sec  
2025-10-05 08:11:52,414 - root - INFO - Epoch 44/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.2557, Train Acc: 0.9096, Val Loss: 0.8499, Val Acc: 0.7438, Epoch Duration: 47.07 sec  
2025-10-05 08:12:39,647 - root - INFO - Epoch 45/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.2490, Train Acc: 0.9126, Val Loss: 0.8537, Val Acc: 0.7481, Epoch Duration: 47.23 sec  
2025-10-05 08:13:26,844 - root - INFO - Epoch 46/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.2465, Train Acc: 0.9130, Val Loss: 0.8457, Val Acc: 0.7477, Epoch Duration: 47.19 sec  
2025-10-05 08:14:14,055 - root - INFO - Epoch 47/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.2228, Train Acc: 0.9228, Val Loss: 0.8354, Val Acc: 0.7586, Epoch Duration: 47.21 sec  
2025-10-05 08:15:03,398 - root - INFO - Epoch 48/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.2238, Train Acc: 0.9211, Val Loss: 0.9113, Val Acc: 0.7435, Epoch Duration: 47.56 sec  
2025-10-05 08:15:50,461 - root - INFO - Epoch 49/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.2025, Train Acc: 0.9301, Val Loss: 0.8728, Val Acc: 0.7516, Epoch Duration: 47.06 sec  
2025-10-05 08:16:37,585 - root - INFO - Epoch 50/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.2022, Train Acc: 0.9311, Val Loss: 0.8819, Val Acc: 0.7512, Epoch

Duration: 47.12 sec  
2025-10-05 08:17:24,782 - root - INFO - Epoch 51/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.1849, Train Acc: 0.9365, Val Loss: 0.8756, Val Acc: 0.7564, Epoch Duration: 47.19 sec  
2025-10-05 08:18:11,721 - root - INFO - Epoch 52/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.1881, Train Acc: 0.9355, Val Loss: 0.8898, Val Acc: 0.7615, Epoch Duration: 46.94 sec  
2025-10-05 08:19:01,000 - root - INFO - Epoch 53/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.1740, Train Acc: 0.9402, Val Loss: 0.8959, Val Acc: 0.7598, Epoch Duration: 47.60 sec  
2025-10-05 08:19:48,130 - root - INFO - Epoch 54/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.1981, Train Acc: 0.9335, Val Loss: 1.0023, Val Acc: 0.7401, Epoch Duration: 47.13 sec  
2025-10-05 08:20:35,454 - root - INFO - Epoch 55/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.1574, Train Acc: 0.9474, Val Loss: 0.9251, Val Acc: 0.7598, Epoch Duration: 47.32 sec  
2025-10-05 08:21:22,453 - root - INFO - Epoch 56/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.1502, Train Acc: 0.9497, Val Loss: 0.9230, Val Acc: 0.7526, Epoch Duration: 47.00 sec  
2025-10-05 08:22:09,452 - root - INFO - Epoch 57/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.1666, Train Acc: 0.9440, Val Loss: 0.9309, Val Acc: 0.7575, Epoch Duration: 47.00 sec  
2025-10-05 08:22:56,590 - root - INFO - Epoch 58/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.1324, Train Acc: 0.9577, Val Loss: 0.9394, Val Acc: 0.7607, Epoch Duration: 47.14 sec  
2025-10-05 08:23:43,761 - root - INFO - Epoch 59/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.1482, Train Acc: 0.9496, Val Loss: 1.0243, Val Acc: 0.7423, Epoch Duration: 47.17 sec  
2025-10-05 08:24:31,020 - root - INFO - Epoch 60/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.1487, Train Acc: 0.9505, Val Loss: 0.9517, Val Acc: 0.7543, Epoch Duration: 47.26 sec  
2025-10-05 08:25:18,369 - root - INFO - Epoch 61/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.1298, Train Acc: 0.9570, Val Loss: 0.9442, Val Acc: 0.7610, Epoch Duration: 47.35 sec  
2025-10-05 08:26:06,028 - root - INFO - Epoch 62/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.1208, Train Acc: 0.9601, Val Loss: 0.9856, Val Acc: 0.7561, Epoch Duration: 47.66 sec  
2025-10-05 08:26:53,138 - root - INFO - Epoch 63/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.1251, Train Acc: 0.9585, Val Loss: 0.9542, Val Acc: 0.7681, Epoch Duration: 47.11 sec  
2025-10-05 08:27:42,198 - root - INFO - Epoch 64/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.1440, Train Acc: 0.9526, Val Loss: 0.9730, Val Acc: 0.7600, Epoch Duration: 47.41 sec  
2025-10-05 08:28:29,264 - root - INFO - Epoch 65/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.1122, Train Acc: 0.9632, Val Loss: 1.0392, Val Acc: 0.7548, Epoch Duration: 47.06 sec  
2025-10-05 08:29:16,596 - root - INFO - Epoch 66/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.1208, Train Acc: 0.9594, Val Loss: 0.9924, Val Acc: 0.7596, Epoch Duration: 47.33 sec  
2025-10-05 08:30:03,839 - root - INFO - Epoch 67/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.1104, Train Acc: 0.9639, Val Loss: 0.9645, Val Acc: 0.7622, Epoch Duration: 47.24 sec  
2025-10-05 08:30:51,215 - root - INFO - Epoch 68/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.1050, Train Acc: 0.9657, Val Loss: 1.0092, Val Acc: 0.7614, Epoch

Duration: 47.37 sec  
2025-10-05 08:31:38,675 - root - INFO - Epoch 69/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.1113, Train Acc: 0.9626, Val Loss: 1.0548, Val Acc: 0.7565, Epoch Duration: 47.46 sec  
2025-10-05 08:32:25,747 - root - INFO - Epoch 70/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.1163, Train Acc: 0.9623, Val Loss: 1.0620, Val Acc: 0.7572, Epoch Duration: 47.07 sec  
2025-10-05 08:33:12,765 - root - INFO - Epoch 71/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.0986, Train Acc: 0.9680, Val Loss: 1.0461, Val Acc: 0.7589, Epoch Duration: 47.02 sec  
2025-10-05 08:33:59,950 - root - INFO - Epoch 72/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.1231, Train Acc: 0.9603, Val Loss: 1.1412, Val Acc: 0.7372, Epoch Duration: 47.18 sec  
2025-10-05 08:34:47,257 - root - INFO - Epoch 73/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.1071, Train Acc: 0.9648, Val Loss: 1.0281, Val Acc: 0.7534, Epoch Duration: 47.30 sec  
2025-10-05 08:35:34,481 - root - INFO - Epoch 74/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.0887, Train Acc: 0.9706, Val Loss: 1.0324, Val Acc: 0.7689, Epoch Duration: 47.22 sec  
2025-10-05 08:36:24,001 - root - INFO - Epoch 75/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.0964, Train Acc: 0.9680, Val Loss: 1.0829, Val Acc: 0.7516, Epoch Duration: 47.73 sec  
2025-10-05 08:37:11,083 - root - INFO - Epoch 76/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.1054, Train Acc: 0.9655, Val Loss: 1.0797, Val Acc: 0.7578, Epoch Duration: 47.08 sec  
2025-10-05 08:37:58,183 - root - INFO - Epoch 77/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.0768, Train Acc: 0.9761, Val Loss: 1.0616, Val Acc: 0.7587, Epoch Duration: 47.10 sec  
2025-10-05 08:38:45,295 - root - INFO - Epoch 78/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.0952, Train Acc: 0.9690, Val Loss: 1.0692, Val Acc: 0.7604, Epoch Duration: 47.11 sec  
2025-10-05 08:39:32,895 - root - INFO - Epoch 79/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.0907, Train Acc: 0.9713, Val Loss: 1.1425, Val Acc: 0.7469, Epoch Duration: 47.60 sec  
2025-10-05 08:40:20,397 - root - INFO - Epoch 80/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.0961, Train Acc: 0.9676, Val Loss: 1.0802, Val Acc: 0.7591, Epoch Duration: 47.50 sec  
2025-10-05 08:41:07,895 - root - INFO - Epoch 81/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.0912, Train Acc: 0.9695, Val Loss: 1.1132, Val Acc: 0.7513, Epoch Duration: 47.50 sec  
2025-10-05 08:41:55,586 - root - INFO - Epoch 82/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.0832, Train Acc: 0.9726, Val Loss: 1.0911, Val Acc: 0.7633, Epoch Duration: 47.69 sec  
2025-10-05 08:42:42,680 - root - INFO - Epoch 83/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.0779, Train Acc: 0.9749, Val Loss: 1.0878, Val Acc: 0.7615, Epoch Duration: 47.09 sec  
2025-10-05 08:43:29,751 - root - INFO - Epoch 84/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.0770, Train Acc: 0.9755, Val Loss: 1.3083, Val Acc: 0.7298, Epoch Duration: 47.07 sec  
2025-10-05 08:44:16,757 - root - INFO - Epoch 85/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.0896, Train Acc: 0.9698, Val Loss: 1.1022, Val Acc: 0.7620, Epoch Duration: 47.00 sec  
2025-10-05 08:45:04,032 - root - INFO - Epoch 86/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.0730, Train Acc: 0.9773, Val Loss: 1.1537, Val Acc: 0.7565, Epoch

Duration: 47.27 sec  
2025-10-05 08:45:51,224 - root - INFO - Epoch 87/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.0886, Train Acc: 0.9722, Val Loss: 1.1738, Val Acc: 0.7514, Epoch Duration: 47.19 sec  
2025-10-05 08:46:38,735 - root - INFO - Epoch 88/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.0750, Train Acc: 0.9770, Val Loss: 1.0901, Val Acc: 0.7684, Epoch Duration: 47.51 sec  
2025-10-05 08:47:25,985 - root - INFO - Epoch 89/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.0764, Train Acc: 0.9754, Val Loss: 1.1248, Val Acc: 0.7611, Epoch Duration: 47.25 sec  
2025-10-05 08:48:12,942 - root - INFO - Epoch 90/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.0738, Train Acc: 0.9761, Val Loss: 1.1784, Val Acc: 0.7566, Epoch Duration: 46.95 sec  
2025-10-05 08:48:59,910 - root - INFO - Epoch 91/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.0753, Train Acc: 0.9760, Val Loss: 1.1274, Val Acc: 0.7663, Epoch Duration: 46.97 sec  
2025-10-05 08:49:47,046 - root - INFO - Epoch 92/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.0877, Train Acc: 0.9712, Val Loss: 1.1033, Val Acc: 0.7695, Epoch Duration: 47.13 sec  
2025-10-05 08:50:36,394 - root - INFO - Epoch 93/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.0585, Train Acc: 0.9821, Val Loss: 1.1689, Val Acc: 0.7579, Epoch Duration: 47.58 sec  
2025-10-05 08:51:23,609 - root - INFO - Epoch 94/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.0697, Train Acc: 0.9770, Val Loss: 1.1191, Val Acc: 0.7628, Epoch Duration: 47.21 sec  
2025-10-05 08:52:10,977 - root - INFO - Epoch 95/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.0773, Train Acc: 0.9745, Val Loss: 1.1587, Val Acc: 0.7636, Epoch Duration: 47.37 sec  
2025-10-05 08:52:58,463 - root - INFO - Epoch 96/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.0687, Train Acc: 0.9769, Val Loss: 1.2111, Val Acc: 0.7558, Epoch Duration: 47.48 sec  
2025-10-05 08:53:45,394 - root - INFO - Epoch 97/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.0625, Train Acc: 0.9801, Val Loss: 1.1985, Val Acc: 0.7615, Epoch Duration: 46.93 sec  
2025-10-05 08:54:32,398 - root - INFO - Epoch 98/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.0748, Train Acc: 0.9757, Val Loss: 1.1580, Val Acc: 0.7645, Epoch Duration: 47.00 sec  
2025-10-05 08:55:19,508 - root - INFO - Epoch 99/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.0676, Train Acc: 0.9776, Val Loss: 1.1618, Val Acc: 0.7610, Epoch Duration: 47.11 sec  
2025-10-05 08:56:06,672 - root - INFO - Epoch 100/100, Model MobileNetV3\_1D\_LSTM, Train Loss: 0.0686, Train Acc: 0.9774, Val Loss: 1.1483, Val Acc: 0.7668, Epoch Duration: 47.16 sec  
2025-10-05 08:56:06,672 - root - CRITICAL - Training complete. Best Val Acc: 0.7695 at epoch 92.  
2025-10-05 08:56:06,683 - root - CRITICAL - SAVED BEST MODEL  
/Users/sanjeev/VNIT/FINAL\_PRJ\_PHASE2/experiments/20251005\_000538/checkpoints/best\_model\_MobileNetV3\_1D\_LSTM\_epoch92.pt USING LEARNING RATE 0.001  
2025-10-05 08:56:06,683 - root - CRITICAL - BEST VALIDATION ACCURACY: 0.7695 at epoch 92.  
2025-10-05 08:56:06,770 - root - INFO - Saved stats plot to  
/Users/sanjeev/VNIT/FINAL\_PRJ\_PHASE2/experiments/20251005\_000538/plots/MobileNetV3\_1D\_LSTM\_{'csi\_channels': 99, 'meta\_feature\_dim': 12, 'num\_classes': 31, 'lr': 0.001, 'epochs': 100, 'model\_name': 'MobileNetV3\_1D\_LSTM'}\_stats.png

2025-10-05 08:56:10,298 - root - INFO - Confusion matrix:

```
[[188  0  1  0  0  3  5  3 18  2  1 17  0  0  0 17  6  2
  0  3  0  0  0  4  7  0  0  0  0  0]
 [ 0 202  1  0 41 32  0  1  1  0  0  0  0  3  0  2  0  0
  0  0  0  0  0  0  0  0  0  0  0  0]
 [ 0  0 246  0  4  3  4 11  2  6  6  0  0  1  0  2  0  0
  0  2  0  1  0  1  0  1  0  2  0  0]
 [ 0  0  0 279  0  0  0  0  0  1  0  0  0  0  0  0  1  0
  0  0  1  0  0  2  4  0  0  0  0  0]
 [ 1 19  0  0 233 13  2  0  1  1  0  1  0  1  0  0  0  0
  0  0  0  0  0  0  0  0  0  0  0  0]
 [ 5 26  1  1  41 182  2  6  9  4  1  4  0  1  0  0  0  0
  0  1  1  0  0  5  1  0  0  3  1  0]
 [ 3  0  2  0  1  2 248 21  1  6  0  1  0  0  0  0  0  0
  0  0  0  0  1  0  1  1  0  0  0  2]
 [ 3  0  4  1  4 10 21 159 15 31  4  3  0  3  0  0  0  0
  1  1  0  1  2  1  0  2  0  0  0  1]
 [26  1  2  5  0  5  3  8 168 18  3  3  0  0  1  1  6  2
  1  3  0  1  1  2  5  0  0  0  2  0]
 [12  0  8  2  0  2 14 34 15 172  2  1  0  0  1  2  0  0
  0  1  3  1  0  0  1  0  0  1  0]
 [ 0  0  2  0  0  0  0  2  1  0 258  0  0  0  0  4  0  3
  0  0  0  0  0  0  0  0  0  0  0]
 [ 3  0  0  1  0  2  3  0  1  0  0 238  0  0  1  0  6  2
  5  0  0  0  0  5  3  1  0  0  1  0]
 [ 0  0  0  0  0  0  0  0  0  0  0  1  1 284  0  5  0  1  5
 25  0  0  2  1  0  0  5  0  1  0  1]
 [ 0  0  3  2  1  0  0  0  1  0  0  0  0 280  3  5  1  1
  0  0  0  0  0  1  2  0  1  0  0  0]
 [ 0  0  0  0  0  0  0  0  0  0  0  0  1  0  0 217  0  8  2
 12  0  0  0  1  0  1  2  0  0  3  6]
 [ 9  0  1  0  0  1  0  0  3  0  4  0  0  1  0 257  2  9
  0  9  0  0  0  8  2  0  0  0  1  0]
 [ 2  0  0  1  0  0  0  0  1  0  2  5  2  1  9  1 208 29
 10 10  0  0  0  2  7  0  0  0  0  0]
 [ 2  0  0  0  0  0  0  0  2  0  1  1  0  0  1  7 18 219
  8 43  0  1  0  1  1  0  0  0  0  1]
 [ 1  0  0  0  0  0  1  0  0  0  0  0  1 14  0  5  0 11 12
 228 2  1  0  0  0  3  3  0  0  0  0]
 [ 2  0  0  1  0  3  1  0  0  1  0  2  0  1  0  5 14 41
  2 210 0  0  0  2  0  0  0  0  0  0]
 [ 0  0  0  0  0  0  0  0  0  1  0  0  0  0  4  0  0  0
  5  0 203 10 12  1  0 10  4  0 16 23]
 [ 0  0  3  1  0  0  0  1  0  1  2  0  3  0  2  0  0  0
  1  0  9 218  8 10  0 38  1  0 15  3]
 [ 0  0  0  5  0  1  1  1  0  1  1  0  0  0  4  0  2  0
  1  0 12  1 167  1  3  4  7  0 45 31]
 [ 3  0  2  0  0  0  0  0  0  0  1  2  1  0  0  2  2  0
  0  3  1  0  1 229 22  0  0  0  2  1]
 [ 2  0  0  5  0  3  0  1  2  1  0  0  0  1  0  0  3  2
  1  2  1  0  3  41 230 0  0  4  6  0]
 [ 0  0  0  0  0  0  0  0  0  0  0  0  2  0  2  0  0  0
  1  0  2 15  3  1  1 217  4  0  3 12]
 [ 0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0]
```



```

    3  0  1  2  2  2  2  21 250  1  3  0]
[  0  0  0  1  0  0  0  0  0  0  0  0  0  0  0  0  0
  0  0  0  0  0  1  2  1  0 288  0  0]
[  0  0  1  2  0  1  0  1  1  0  1  0  0  0  4  0  2  0
  1  0 22 12 62  3  0  8  2  0 140 31]
[  0  0  0  0  0  0  0  0  0  1  0  0  1  1  9  1  0  0
  0  0 16  3 32  0  0 14  0  0 29 158]]

```

2025-10-05 08:56:10,298 - root - INFO - Classification report:

|  | precision | recall | f1-score | support |
|--|-----------|--------|----------|---------|
|--|-----------|--------|----------|---------|

|    |      |      |      |     |
|----|------|------|------|-----|
| 1  | 0.72 | 0.68 | 0.70 | 277 |
| 2  | 0.81 | 0.71 | 0.76 | 283 |
| 3  | 0.89 | 0.84 | 0.86 | 292 |
| 4  | 0.91 | 0.97 | 0.94 | 288 |
| 5  | 0.72 | 0.86 | 0.78 | 272 |
| 6  | 0.69 | 0.62 | 0.65 | 295 |
| 7  | 0.81 | 0.86 | 0.83 | 290 |
| 8  | 0.64 | 0.60 | 0.62 | 267 |
| 9  | 0.69 | 0.63 | 0.66 | 267 |
| 10 | 0.70 | 0.63 | 0.66 | 272 |
| 11 | 0.90 | 0.96 | 0.92 | 270 |
| 12 | 0.85 | 0.88 | 0.86 | 272 |
| 13 | 0.93 | 0.86 | 0.89 | 332 |
| 14 | 0.95 | 0.93 | 0.94 | 301 |
| 15 | 0.81 | 0.86 | 0.83 | 253 |
| 16 | 0.84 | 0.84 | 0.84 | 307 |
| 17 | 0.71 | 0.72 | 0.72 | 290 |
| 18 | 0.67 | 0.72 | 0.69 | 306 |
| 19 | 0.75 | 0.81 | 0.78 | 282 |
| 20 | 0.72 | 0.74 | 0.73 | 285 |
| 21 | 0.74 | 0.70 | 0.72 | 289 |
| 22 | 0.81 | 0.69 | 0.75 | 316 |
| 23 | 0.56 | 0.58 | 0.57 | 288 |
| 24 | 0.71 | 0.84 | 0.77 | 272 |
| 25 | 0.77 | 0.75 | 0.76 | 308 |
| 26 | 0.66 | 0.83 | 0.73 | 263 |
| 27 | 0.93 | 0.87 | 0.90 | 287 |
| 28 | 0.96 | 0.98 | 0.97 | 293 |
| 29 | 0.52 | 0.48 | 0.50 | 294 |
| 30 | 0.59 | 0.60 | 0.59 | 265 |

|              |      |      |      |      |
|--------------|------|------|------|------|
| accuracy     |      |      | 0.77 | 8576 |
| macro avg    | 0.77 | 0.77 | 0.76 | 8576 |
| weighted avg | 0.77 | 0.77 | 0.77 | 8576 |

2025-10-05 08:56:10,299 - root - INFO - START RUN MLFLOW :

MobileNetV3\_1D\_LSTM: {'csi\_channels': 99, 'meta\_feature\_dim': 12, 'num\_classes': 31, 'lr': 0.0005, 'epochs': 100, 'model\_name': 'MobileNetV3\_1D\_LSTM'}

2025-10-05 08:56:10,308 - root - INFO - START T-N-E MobileNetV3\_1D\_LSTM

2025-10-05 08:56:10,309 - root - INFO - MobileNetV3\_1D\_LSTM USING LEARNING RATE 0.0005

2025-10-05 08:56:57,720 - root - WARNING - ROC-AUC subset computation failed: Input contains NaN.

2025-10-05 08:56:57,722 - root - INFO - Epoch 1/100, Model MobileNetV3\_1D\_LSTM,

Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.41 sec  
2025-10-05 08:57:45,306 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 08:57:45,309 - root - INFO - Epoch 2/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.58 sec  
2025-10-05 08:58:32,582 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 08:58:32,585 - root - INFO - Epoch 3/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.27 sec  
2025-10-05 08:59:19,565 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 08:59:19,568 - root - INFO - Epoch 4/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 46.98 sec  
2025-10-05 09:00:06,774 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 09:00:06,776 - root - INFO - Epoch 5/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.21 sec  
2025-10-05 09:00:54,231 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 09:00:54,234 - root - INFO - Epoch 6/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.46 sec  
2025-10-05 09:01:41,452 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 09:01:41,454 - root - INFO - Epoch 7/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.22 sec  
2025-10-05 09:02:28,821 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 09:02:28,824 - root - INFO - Epoch 8/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.37 sec  
2025-10-05 09:03:16,670 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 09:03:16,673 - root - INFO - Epoch 9/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.85 sec  
2025-10-05 09:04:03,807 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 09:04:03,809 - root - INFO - Epoch 10/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.13 sec  
2025-10-05 09:04:50,855 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 09:04:50,856 - root - INFO - Epoch 11/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.05 sec  
2025-10-05 09:05:38,223 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.

2025-10-05 09:05:38,224 - root - INFO - Epoch 12/100, Model MobileNetV3\_1D\_LSTM, Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch Duration: 47.37 sec

2025-10-05 09:06:25,436 - root - WARNING - ROC-AUC subset computation failed: Input contains NaN.

2025-10-05 09:06:25,437 - root - INFO - Epoch 13/100, Model MobileNetV3\_1D\_LSTM, Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch Duration: 47.21 sec

2025-10-05 09:07:13,655 - root - WARNING - ROC-AUC subset computation failed: Input contains NaN.

2025-10-05 09:07:13,657 - root - INFO - Epoch 14/100, Model MobileNetV3\_1D\_LSTM, Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch Duration: 48.22 sec

2025-10-05 09:08:02,036 - root - WARNING - ROC-AUC subset computation failed: Input contains NaN.

2025-10-05 09:08:02,037 - root - INFO - Epoch 15/100, Model MobileNetV3\_1D\_LSTM, Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch Duration: 48.38 sec

2025-10-05 09:08:50,386 - root - WARNING - ROC-AUC subset computation failed: Input contains NaN.

2025-10-05 09:08:50,388 - root - INFO - Epoch 16/100, Model MobileNetV3\_1D\_LSTM, Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch Duration: 48.35 sec

2025-10-05 09:09:38,829 - root - WARNING - ROC-AUC subset computation failed: Input contains NaN.

2025-10-05 09:09:38,832 - root - INFO - Epoch 17/100, Model MobileNetV3\_1D\_LSTM, Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch Duration: 48.44 sec

2025-10-05 09:10:26,208 - root - WARNING - ROC-AUC subset computation failed: Input contains NaN.

2025-10-05 09:10:26,211 - root - INFO - Epoch 18/100, Model MobileNetV3\_1D\_LSTM, Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch Duration: 47.38 sec

2025-10-05 09:11:13,154 - root - WARNING - ROC-AUC subset computation failed: Input contains NaN.

2025-10-05 09:11:13,157 - root - INFO - Epoch 19/100, Model MobileNetV3\_1D\_LSTM, Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch Duration: 46.94 sec

2025-10-05 09:12:00,364 - root - WARNING - ROC-AUC subset computation failed: Input contains NaN.

2025-10-05 09:12:00,367 - root - INFO - Epoch 20/100, Model MobileNetV3\_1D\_LSTM, Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch Duration: 47.21 sec

2025-10-05 09:12:47,826 - root - WARNING - ROC-AUC subset computation failed: Input contains NaN.

2025-10-05 09:12:47,829 - root - INFO - Epoch 21/100, Model MobileNetV3\_1D\_LSTM, Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch Duration: 47.46 sec

2025-10-05 09:13:35,343 - root - WARNING - ROC-AUC subset computation failed: Input contains NaN.

2025-10-05 09:13:35,346 - root - INFO - Epoch 22/100, Model MobileNetV3\_1D\_LSTM, Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch Duration: 47.52 sec

2025-10-05 09:14:22,667 - root - WARNING - ROC-AUC subset computation failed:

Input contains NaN.  
2025-10-05 09:14:22,669 - root - INFO - Epoch 23/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.32 sec  
2025-10-05 09:15:09,400 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 09:15:09,401 - root - INFO - Epoch 24/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 46.73 sec  
2025-10-05 09:15:56,529 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 09:15:56,530 - root - INFO - Epoch 25/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.13 sec  
2025-10-05 09:16:43,767 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 09:16:43,769 - root - INFO - Epoch 26/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.24 sec  
2025-10-05 09:17:30,920 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 09:17:30,922 - root - INFO - Epoch 27/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.15 sec  
2025-10-05 09:18:18,183 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 09:18:18,185 - root - INFO - Epoch 28/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.26 sec  
2025-10-05 09:19:05,509 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 09:19:05,511 - root - INFO - Epoch 29/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.32 sec  
2025-10-05 09:19:52,697 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 09:19:52,700 - root - INFO - Epoch 30/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.19 sec  
2025-10-05 09:20:39,629 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 09:20:39,631 - root - INFO - Epoch 31/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 46.93 sec  
2025-10-05 09:21:26,669 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 09:21:26,671 - root - INFO - Epoch 32/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.04 sec  
2025-10-05 09:22:13,689 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 09:22:13,691 - root - INFO - Epoch 33/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.02 sec

2025-10-05 09:23:00,966 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.

2025-10-05 09:23:00,967 - root - INFO - Epoch 34/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.28 sec

2025-10-05 09:23:48,170 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.

2025-10-05 09:23:48,173 - root - INFO - Epoch 35/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.20 sec

2025-10-05 09:24:35,622 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.

2025-10-05 09:24:35,624 - root - INFO - Epoch 36/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.45 sec

2025-10-05 09:25:22,748 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.

2025-10-05 09:25:22,750 - root - INFO - Epoch 37/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.12 sec

2025-10-05 09:26:09,783 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.

2025-10-05 09:26:09,786 - root - INFO - Epoch 38/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.03 sec

2025-10-05 09:26:56,947 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.

2025-10-05 09:26:56,949 - root - INFO - Epoch 39/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.16 sec

2025-10-05 09:27:44,075 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.

2025-10-05 09:27:44,077 - root - INFO - Epoch 40/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.13 sec

2025-10-05 09:28:31,262 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.

2025-10-05 09:28:31,264 - root - INFO - Epoch 41/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.19 sec

2025-10-05 09:29:18,446 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.

2025-10-05 09:29:18,448 - root - INFO - Epoch 42/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.18 sec

2025-10-05 09:30:05,713 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.

2025-10-05 09:30:05,715 - root - INFO - Epoch 43/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.27 sec

2025-10-05 09:30:52,783 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.

2025-10-05 09:30:52,786 - root - INFO - Epoch 44/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch

Duration: 47.07 sec  
2025-10-05 09:31:40,101 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 09:31:40,103 - root - INFO - Epoch 45/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.32 sec  
2025-10-05 09:32:27,175 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 09:32:27,177 - root - INFO - Epoch 46/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.07 sec  
2025-10-05 09:33:14,362 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 09:33:14,363 - root - INFO - Epoch 47/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.18 sec  
2025-10-05 09:34:01,525 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 09:34:01,528 - root - INFO - Epoch 48/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.16 sec  
2025-10-05 09:34:49,019 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 09:34:49,022 - root - INFO - Epoch 49/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.49 sec  
2025-10-05 09:35:36,432 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 09:35:36,434 - root - INFO - Epoch 50/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.41 sec  
2025-10-05 09:36:23,488 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 09:36:23,491 - root - INFO - Epoch 51/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.05 sec  
2025-10-05 09:37:10,761 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 09:37:10,763 - root - INFO - Epoch 52/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.27 sec  
2025-10-05 09:37:57,831 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 09:37:57,833 - root - INFO - Epoch 53/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.07 sec  
2025-10-05 09:38:45,106 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 09:38:45,108 - root - INFO - Epoch 54/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.27 sec  
2025-10-05 09:39:32,635 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 09:39:32,637 - root - INFO - Epoch 55/100, Model MobileNetV3\_1D\_LSTM,

Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.53 sec  
2025-10-05 09:40:20,107 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 09:40:20,109 - root - INFO - Epoch 56/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.47 sec  
2025-10-05 09:41:07,483 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 09:41:07,486 - root - INFO - Epoch 57/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.37 sec  
2025-10-05 09:41:54,572 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 09:41:54,574 - root - INFO - Epoch 58/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.09 sec  
2025-10-05 09:42:41,456 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 09:42:41,458 - root - INFO - Epoch 59/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 46.88 sec  
2025-10-05 09:43:28,842 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 09:43:28,844 - root - INFO - Epoch 60/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.38 sec  
2025-10-05 09:44:15,891 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 09:44:15,894 - root - INFO - Epoch 61/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.05 sec  
2025-10-05 09:45:03,188 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 09:45:03,190 - root - INFO - Epoch 62/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.30 sec  
2025-10-05 09:45:50,637 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 09:45:50,640 - root - INFO - Epoch 63/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.45 sec  
2025-10-05 09:46:38,157 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 09:46:38,159 - root - INFO - Epoch 64/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.52 sec  
2025-10-05 09:47:25,230 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 09:47:25,232 - root - INFO - Epoch 65/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.07 sec  
2025-10-05 09:48:12,298 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.

2025-10-05 09:48:12,300 - root - INFO - Epoch 66/100, Model MobileNetV3\_1D\_LSTM, Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch Duration: 47.07 sec

2025-10-05 09:48:59,375 - root - WARNING - ROC-AUC subset computation failed: Input contains NaN.

2025-10-05 09:48:59,378 - root - INFO - Epoch 67/100, Model MobileNetV3\_1D\_LSTM, Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch Duration: 47.08 sec

2025-10-05 09:49:46,446 - root - WARNING - ROC-AUC subset computation failed: Input contains NaN.

2025-10-05 09:49:46,448 - root - INFO - Epoch 68/100, Model MobileNetV3\_1D\_LSTM, Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch Duration: 47.07 sec

2025-10-05 09:50:33,901 - root - WARNING - ROC-AUC subset computation failed: Input contains NaN.

2025-10-05 09:50:33,903 - root - INFO - Epoch 69/100, Model MobileNetV3\_1D\_LSTM, Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch Duration: 47.45 sec

2025-10-05 09:51:21,222 - root - WARNING - ROC-AUC subset computation failed: Input contains NaN.

2025-10-05 09:51:21,224 - root - INFO - Epoch 70/100, Model MobileNetV3\_1D\_LSTM, Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch Duration: 47.32 sec

2025-10-05 09:52:08,564 - root - WARNING - ROC-AUC subset computation failed: Input contains NaN.

2025-10-05 09:52:08,566 - root - INFO - Epoch 71/100, Model MobileNetV3\_1D\_LSTM, Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch Duration: 47.34 sec

2025-10-05 09:52:55,486 - root - WARNING - ROC-AUC subset computation failed: Input contains NaN.

2025-10-05 09:52:55,488 - root - INFO - Epoch 72/100, Model MobileNetV3\_1D\_LSTM, Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch Duration: 46.92 sec

2025-10-05 09:53:42,708 - root - WARNING - ROC-AUC subset computation failed: Input contains NaN.

2025-10-05 09:53:42,711 - root - INFO - Epoch 73/100, Model MobileNetV3\_1D\_LSTM, Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch Duration: 47.22 sec

2025-10-05 09:54:29,909 - root - WARNING - ROC-AUC subset computation failed: Input contains NaN.

2025-10-05 09:54:29,911 - root - INFO - Epoch 74/100, Model MobileNetV3\_1D\_LSTM, Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch Duration: 47.20 sec

2025-10-05 09:55:17,109 - root - WARNING - ROC-AUC subset computation failed: Input contains NaN.

2025-10-05 09:55:17,112 - root - INFO - Epoch 75/100, Model MobileNetV3\_1D\_LSTM, Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch Duration: 47.20 sec

2025-10-05 09:56:04,507 - root - WARNING - ROC-AUC subset computation failed: Input contains NaN.

2025-10-05 09:56:04,508 - root - INFO - Epoch 76/100, Model MobileNetV3\_1D\_LSTM, Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch Duration: 47.40 sec

2025-10-05 09:56:51,983 - root - WARNING - ROC-AUC subset computation failed:



Input contains NaN.  
2025-10-05 09:56:51,986 - root - INFO - Epoch 77/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.48 sec  
2025-10-05 09:57:39,083 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 09:57:39,086 - root - INFO - Epoch 78/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.10 sec  
2025-10-05 09:58:25,966 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 09:58:25,969 - root - INFO - Epoch 79/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 46.88 sec  
2025-10-05 09:59:13,126 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 09:59:13,128 - root - INFO - Epoch 80/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.16 sec  
2025-10-05 10:00:00,244 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 10:00:00,247 - root - INFO - Epoch 81/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.12 sec  
2025-10-05 10:00:47,581 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 10:00:47,583 - root - INFO - Epoch 82/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.33 sec  
2025-10-05 10:01:35,159 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 10:01:35,162 - root - INFO - Epoch 83/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.58 sec  
2025-10-05 10:02:22,817 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 10:02:22,819 - root - INFO - Epoch 84/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.66 sec  
2025-10-05 10:03:09,979 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 10:03:09,982 - root - INFO - Epoch 85/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.16 sec  
2025-10-05 10:03:57,261 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 10:03:57,264 - root - INFO - Epoch 86/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.28 sec  
2025-10-05 10:04:44,559 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 10:04:44,561 - root - INFO - Epoch 87/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.30 sec

2025-10-05 10:05:31,841 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.

2025-10-05 10:05:31,843 - root - INFO - Epoch 88/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.28 sec

2025-10-05 10:06:19,526 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.

2025-10-05 10:06:19,529 - root - INFO - Epoch 89/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.68 sec

2025-10-05 10:07:07,171 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.

2025-10-05 10:07:07,173 - root - INFO - Epoch 90/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.64 sec

2025-10-05 10:07:54,967 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.

2025-10-05 10:07:54,969 - root - INFO - Epoch 91/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.79 sec

2025-10-05 10:08:42,154 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.

2025-10-05 10:08:42,156 - root - INFO - Epoch 92/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.19 sec

2025-10-05 10:09:29,529 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.

2025-10-05 10:09:29,530 - root - INFO - Epoch 93/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.37 sec

2025-10-05 10:10:17,533 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.

2025-10-05 10:10:17,536 - root - INFO - Epoch 94/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 48.00 sec

2025-10-05 10:11:05,342 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.

2025-10-05 10:11:05,344 - root - INFO - Epoch 95/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.81 sec

2025-10-05 10:11:52,731 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.

2025-10-05 10:11:52,734 - root - INFO - Epoch 96/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.39 sec

2025-10-05 10:12:40,349 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.

2025-10-05 10:12:40,352 - root - INFO - Epoch 97/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.62 sec

2025-10-05 10:13:27,912 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.

2025-10-05 10:13:27,914 - root - INFO - Epoch 98/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch

Duration: 47.56 sec  
2025-10-05 10:14:15,192 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 10:14:15,194 - root - INFO - Epoch 99/100, Model MobileNetV3\_1D\_LSTM,  
Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc: 0.0000, Epoch  
Duration: 47.28 sec  
2025-10-05 10:15:02,852 - root - WARNING - ROC-AUC subset computation failed:  
Input contains NaN.  
2025-10-05 10:15:02,854 - root - INFO - Epoch 100/100, Model  
MobileNetV3\_1D\_LSTM, Train Loss: nan, Train Acc: 0.0000, Val Loss: nan, Val Acc:  
0.0000, Epoch Duration: 47.66 sec  
2025-10-05 10:15:02,855 - root - CRITICAL - Training complete. Best Val Acc:  
0.0000 at epoch 0.  
2025-10-05 10:15:02,953 - root - INFO - Saved stats plot to  
/Users/sanjeev/VNIT/FINAL\_PRJ\_PHASE2/experiments/20251005\_000538/plots/MobileNet  
V3\_1D\_LSTM\_{'csi\_channels': 99, 'meta\_feature\_dim': 12, 'num\_classes': 31, 'lr':  
0.0005, 'epochs': 100, 'model\_name': 'MobileNetV3\_1D\_LSTM'}\_stats.png  
2025-10-05 10:15:02,954 - root - WARNING - No best model state found, skipping  
test evaluation.  
2025-10-05 10:15:02,961 - root - INFO - Logged best model path to MLflow  
/Users/sanjeev/VNIT/FINAL\_PRJ\_PHASE2/experiments/20251005\_000538/checkpoints/bes  
t\_model\_EfficientNet1DLSTM\_epoch88.pt  
2025-10-05 10:15:02,961 - root - CRITICAL - best\_overall : {'model':  
'EfficientNet1DLSTM', 'params': {'in\_channels': 99, 'meta\_seq\_len': 128,  
'meta\_feature\_dim': 12, 'num\_classes': 31, 'lr': 0.0005, 'epochs': 100,  
'model\_name': 'EfficientNet1DLSTM'}, 'val\_acc': 0.9925373134328358, 'epoch': 88,  
'lr': 0.0005}